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ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القريشي

أساسيات القيادة الكهربائية

درايف

طلاب النور والإتصالات _ البوليتكنك

Spark Team

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بسم الله الرحمن الرحيم

تحية طيبة وبعد ...

اخواننا الطلبة الكرام ...

فأننا في فريق #سبارك نضع بين ايديكم مجموعة من الاوراق "تلخيص" التي نأمل من الله عز وجل أن تكون في ميزان حسنات من قام بها وتعب عليها ...

حيث أننا نفيدكم انه من قام بعمل هذه التلاخيص هم ثلة رائعة من طلاب تخصص هندسة الطاقة الكهربائية في كلية الهندسة التكنولوجية / البوليتكنك ...

نود اعلامكم أن هذه الاوراق المختصرة لا تغني بأي شكل من الاشكال لأي مادة تخصصية ومرجعك الاول والأخير ألا وهو #الكتاب ...

اخوانكم في فريق #سبارك

لا تنسونا من صالح الدعاء

قام بالتلخيص

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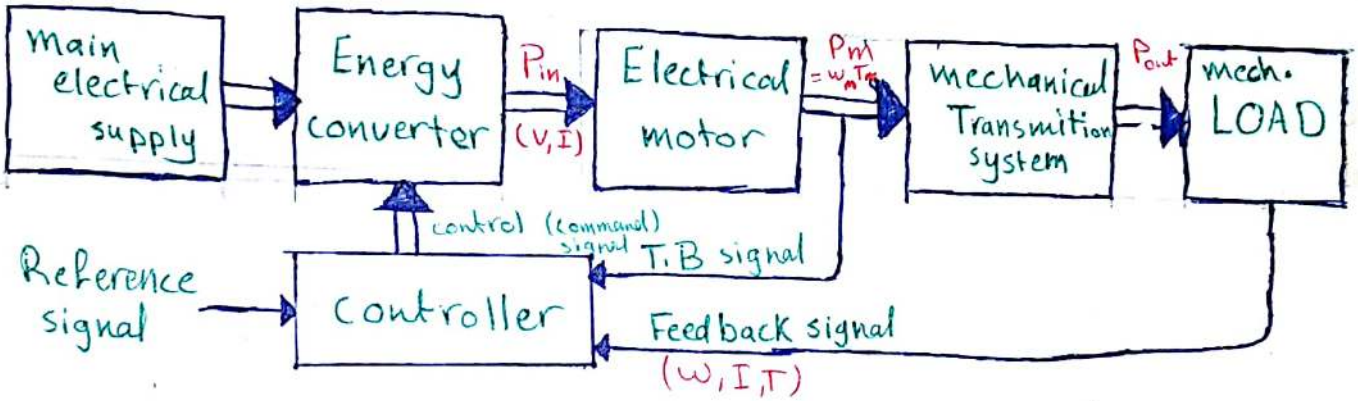
[سؤال: عرف] ⇒ **Drives**- is a system used to produce or generate motion and translate this motion to the Load "The motion is controlled"

⇒ electrical drive system:-

It's a system use electrical motor as a source of motion

* structure of electrical Drive system:- [سؤال: ارسم]



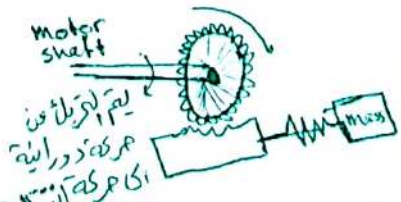
⇒ Block diagram to drive system → closed loop (حسب الاستخدام)
→ open loop

[سؤال: اذكر مثال] * **Energy converter**:- To condition or modify the electrical power to another form

① Rectifier (AC → DC)
② chopper (DC → Dc)
③ Inverter (DC → Ac)
④ transformer (AC → AC)

[سؤال: اذكر وظيفة] [سؤال: اذكر مثال] * **Mech. Transmition system**:-

used to ① to speed control
② to change the direction of motion
③ to convert the motion from rotational to another type (such as → حركة انتقالية)



ex) Gear تسنان Belt حزام cheave بكرات

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* Load :- For the drive system (It's mech load) (moving Part in the sys.) [الحمل]

The motion will be

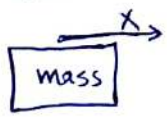
rotational دورانية

$P_{LR} = T \cdot w$

$E_R = \frac{1}{2} y \cdot w^2$

y = moment of inatia

Translational انتقالية



$P_{LT} = F \cdot x = F \cdot R$

$E_T = \frac{1}{2} m v^2$

Power

Kinatic energy

* Function of drive system :- [س: اذكر وظيفة]

① Perform transient operation [start & automatic braking]

② automatic speed control

If the drive system is varible speed control

③ The mentain the speed constant with in aspecified erorr

الحفاظ على السرعة مع خطأ مقبول.

④ protective function الحماية

⑤ programing control . "معرفة كيفية عمل النظام مسبقاً"

⑥ optimal .control (adaptive control) نظام متكيف

تغيير النظام في وظيفته بما يتناسب مع التغير الحاصل حتى يوطين التحكم الأمثل ويلبي ما يحتاجه النظام مثل (كفاءة عالية، خسائر اقل، .....الخ)

⑦ Truking (servo drive) position drive

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*classification of electrical drive system:- [سؤال: اذكر اقسام نظام القيادة حسب .....؟]

① according of the type of motor
   Ⓐ AC drive sys. Ⓑ dc drive sys.

② according to speed control
   Ⓐ Varible speed Drive sys. Ⓑ Constant speed control (fixed)

③ according to [excesting of gear] or [mechanical drive system]
   Ⓐ gear drive sys. Ⓑ gearless drive system

④ according to the Function
   Ⓐ open loop/closed loop Ⓑ programmable electrical drive مبرمج
   Ⓒ Adaptive electrical drive تكيفي Ⓓ servo/position drive sys. متابعة والتحكم بالموقع
   Ⓔ motor-load relation علاقة المحرك بالحمل

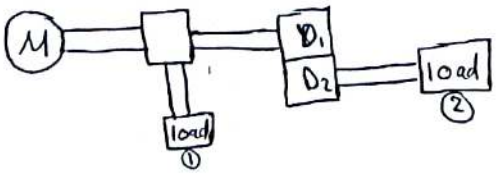
⑤ according to relation between the motor and load.
   Ⓐ single motor for each individual load [individual drive sys.]

كل حمل لديه محرك خاص فيه

   Ⓑ multi motor for one load [group drive]

اكثر من محرك مستخدم لنفس الحمل

   Ⓒ one motor-multi load [Line shaft drive system]



محرك واحد لأكثر من حمل.

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للفهم

* Characteristic of electrical drive [خواص نظام القيادة]

① static characteristic

⇐ تمثل حالات الـ steady-state وتقسم إلى :-

Ⓐ Natural characteristic [الخواص الطبيعية]

⇐ هي الحالة المثالية للنظام من غير احداث اي تغيير [كل القيم nominal]

1- electromechanical characteristic

$I = f(w)$ * هي علاقة بين جزء ميكانيكي وجزء كهربائي

2- mechanical characteristic

$T = f(w)$ * هي علاقة بين جزئين ميكانيكيين

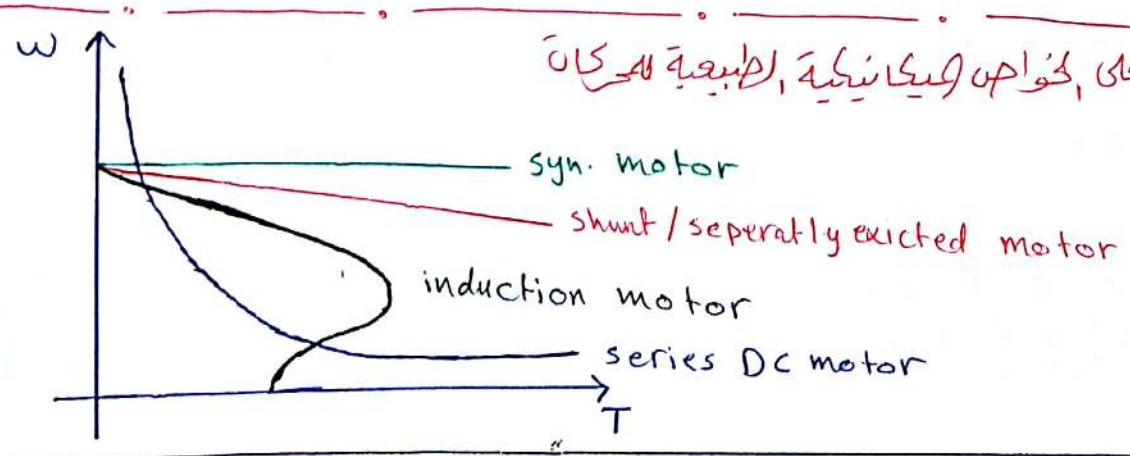
Ⓑ Regulated characteristic [الخواص التنظيمية]

⇐ هي الحالة التي يتم فيها تغيير [اضافة] اي شيء على النظام وبذلك تتغير القيم عن القيم المقررة [nominal value]

② Dynamic characteristic

* تمثل الحالات العابرة [Transient-state]

$I = f(t)$, $T = f(t)$, $w = f(t)$ * وهي علاقة اي قيمة مع الزمن



* امثلة على الخواص الميكانيكية الطبيعية للمحركات

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*mechanical characteristic of industrial load:-

* الخواص الميكانيكية للاحمال الصناعية

$$T_L = T_o + C(T_n - T_o)\left(\frac{w}{w_n}\right)^K$$ العلاقة العامة للاحمال الصناعية

T_o :- intial Torque before starting

T_n :- rated Torque [constant]

w_n :- rated speed [nominal] [constant]

at T_o = Zero $\longrightarrow T_L = C^* w^K$

① $K=0 \Rightarrow T_L$ = constant [constant load drive]

* called unidirection load or active load
* Load due to gravity force
* example :- Creer الرافعة elevator وصاعد
  escalator سلم كهربائي transportation معدات نقل البضائع والركاب
* independent on the speed

* الاتجاه لا يتغير
* عزم الحمل يعتمد على الجاذبية الارضية

② $K=1 \Rightarrow T_L \propto w$

* Torque due to fraction source يعتمد على الاحتكاك
* Example :- Traction & repulsion machine معدات الجر والسحب
  - Trains القطارات

③ $K=-1 \Rightarrow T_L \propto w^{-1}$

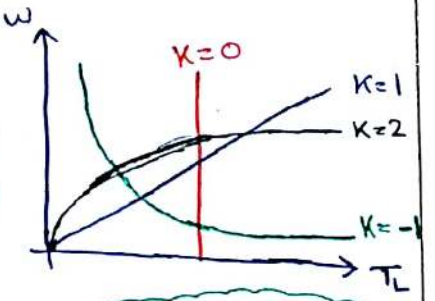
* called constant power drive

$P = T \cdot w = C^* w^{K+1}$ = constant

* example :- Drilling and cuting machine [معدات ثقب وقطع المعادن]

④ $K=2 \Rightarrow T_L \propto w^2$

* called fan characterstic
* example :- fan مروحة compressor الضاغط
  pump مضخة



ملاحظات مهمة:
- uni-directonal load → K=0
- Bi-directional load → K=-1, 1, 2
  - Passive load خيبة السؤال
- بعطيك اسرعة ويطلب اسرعة ومثال عليها وقيمة K
- بعطيك اسرعة ويطلب الباقي 😊
- ممكن يضع اسم العلاقة في سؤال حل وانت تستنتج علاقة w مع T

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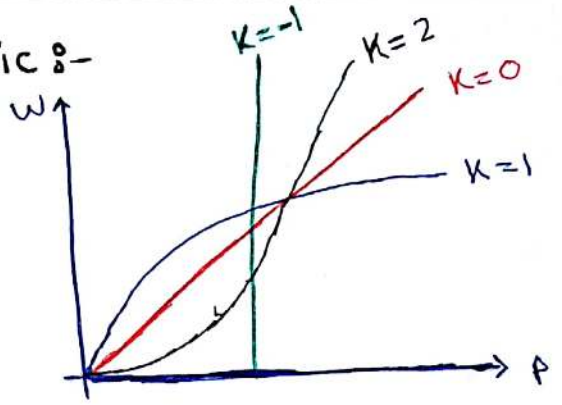
* Power & speed characteristic :-

$P = T\omega = C^* \omega^{K+1}$



- $K=0 \rightarrow P \propto \omega$
- $K=1 \rightarrow P \propto \omega^2$
- $K=2 \rightarrow P \propto \omega^3$
- $K=-1 \rightarrow P = \text{constant}$

* Hardness of mechanical characteristic (β) :- الصلابة هي نسبة تغير السرعة مع تغير العزم ويختلف من نوع لآخر.

$\beta = \frac{dT}{d\omega}$ ⟶ for linear characteristic $\beta = \frac{\Delta T}{\Delta \omega}$

مهم ⟶ Type of hardness [سؤال :- اذكر مثال على .....]

① Hard mechanical characteristic :-
* β is high value ⟶ Large change in Torque cause small change in speed $\Delta T \uparrow$, $\Delta \omega \downarrow$
* example :- DC shunt / separetly exicted motor

② Absolutely hard characteristic :-
* $\beta = \infty$ no change of speed due to change of load (Torque) $\Delta T \uparrow, \Delta \omega = 0$
* example : syncrounous motor

③ soft hard characteristic
* β is small value → low change of torque cause high change of speed $\Delta T \downarrow$, $\Delta \omega \uparrow$
* example :- (induction machine (non-linear part)), DC series motor
DC compound motor

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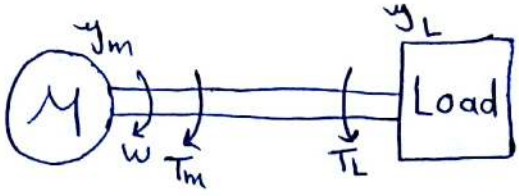
* Equation of motion:-

$$T_m - T_L = J_\Sigma \frac{dw}{dt} \qquad \text{where } J_\Sigma = J_L + J_m$$

T_m :- mechanical torque
T_L :- Load torque
J_Σ :- moment of intia
$T_{acc} = J_\Sigma \frac{dw}{dt}$:- acceleration torque
$\frac{dw}{dt}$:- acceleration



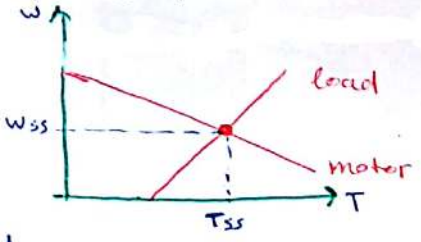
* Application of mechanical charesterstic:-

① To defined The steady state point of the system

⇒ For steady state operating mode

$w = w_{ss} = \text{constant} \rightarrow \frac{dw}{dt} = 0 \rightarrow T_{m_{ss}} = T_{L_{ss}}$

طريقة إيجاد نقطة S.S operating point ← يتم بالمساواة بين معادلة العزم للحرك والعزم للحمل
وتعويض $w = w_{ss}$ ومن ثم إيجاد قيمة (قيم) w_{ss}
← يتم تعويض قيمة w_{ss} في احدى معادلتي العزم وإيجاد T_{Lss}



② To check The steady state stability

* if the system can return to its intial steady state operating point
The system is stable
* if the system go far away to its intial steady state point
The system is unstable.

يتم التحقق من الاستقرارية بطريقتين

① بأستخدام معادلة الحركة eq. of motion للأحمال
$T_m - T_L = J_\Sigma \frac{dw}{dt}$

② بأستخدام المساواة للمنحنيات للمعادلات
$B_m < B_L$ condition of stability

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* check the stability using equation of motion.

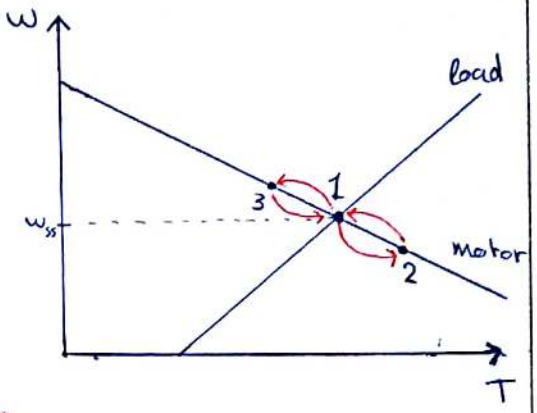
⇒ Example 1

From 1 → 2 $w\downarrow$, $T_m\uparrow$, $T_L\downarrow$

$T_m - T_L = J_\Sigma \frac{dw}{dt} > 0 \rightarrow \frac{dw}{dt} > 0$

→ $w\uparrow$ increase of speed

→ The system return to the steady state operating point



* The steady state operating point is stable

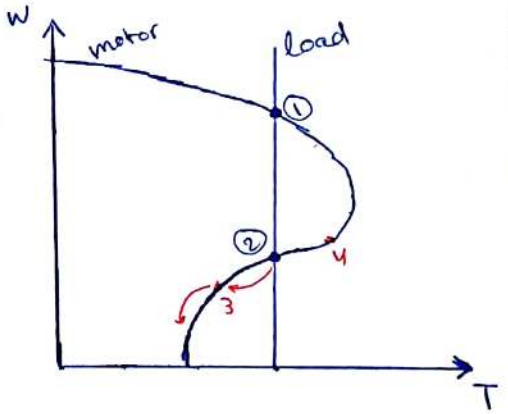
From 1 → 3 $w\uparrow$, $T_m\downarrow$, $T_L\uparrow$, $T_m - T_L = J_\Sigma \frac{dw}{dt} < 0 \rightarrow \frac{dw}{dt} < 0$

* $w\downarrow$ decrease of speed

→ The system return to the steady state operating point

⇒ Example 2

* النقطة ① كما في المثال السابق



Point ②

From 2 → 3 $w\downarrow$, $T_m\downarrow$; T_L - constant

$T_m - T_L = J \frac{dw}{dt} < 0 \rightarrow \frac{dw}{dt} < 0$

$w\downarrow$ decrease the speed

→ The system goes far away from S.S operating point

→ The point ② is un stable operating point

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Example (1) :- $T_m = 100 - 0.5w$ $T_L = 10 + 1.5w$

ⓐ Find The steady state operating point

$T_m(w_{ss}) = T_L(w_{ss})$

$100 - 0.5w_{ss} = 10 + 1.5w_{ss} \rightarrow w_{ss} = 45 \text{ rad/s}$

$T_m(w_{ss}) = T_L(w_{ss}) = 10 + 1.5(45) = 77.5 \text{ N.m}$

The S.S operating point (45 rad/s, 77.5 N.m)

ملاحظات مهمة
① لاتنسى تعويض w_{ss} اذا عوضت لا تحسب خطأ
② ضع وحدات لكل ناتج تجده لان الكتور يهتم بالوحدات

ⓑ cheak The stability

$B_m(w_{ss}) = \left.\frac{dT_m}{dw}\right|_{w=45} = -0.5$ $B_L(w_{ss}) = \left.\frac{dT_L}{dw}\right|_{w=45} = 1.5$

※ $B_m(w_{ss}) < B_L(w_{ss}) \Rightarrow$ The S.S operating point is stable

Example (2) :- $T_m = 1 + 2w$ $T_L = 3\sqrt{w}$

ⓐ Find The steady state operating point.

$T_m(w_{ss}) = T_L(w_{ss})$

$1 + 2w_{ss} = 3\sqrt{w_{ss}} \rightarrow 2w_{ss} - 3\sqrt{w_{ss}} + 1 = 0$

ملاحظة
هنا بعتبر $x = \sqrt{w_{ss}}$
$x^2 = w_{ss}$
وفي آخر بجد كل من الدالة الكاسبة و بأخذ تربيع الاجوبة $w_{ss} = x^2$

$w_{ss_1} = 1 \text{ rad/s} \rightarrow T(w_{ss_1}) = 3 \text{ N.m}$

$w_{ss2} = 0.25 \text{ rad/s} \rightarrow T(w_{ss2}) = 1.5 \text{ N.m}$

Point ① → (1 rad/s, 3 N.m) Point ② → (0.25 rad/s, 3 N.m)

ⓑ cheak The stability

Point ① $B_m(w_{ss_1}) = 2 > B_L(w_{ss_1}) = 3/2 \rightarrow$ Point ① is unstable

Point ② $B_m(w_{ss_2}) = 2 < B_L(w_{ss_2}) = 3 \rightarrow$ Point ② is stable

$\Rightarrow B_m = \frac{dT_m}{dw} = 2$ $B_L = \frac{dT_L}{dw} = \frac{3}{2\sqrt{w}}$

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Example (3) :- $T_m = 500 - 0.1n$ $T_L = 20\,n^2$

ⓐ Find steady state operating point.

$T_m(n_{ss}) = T_L(n_{ss})$

$(500 - 1.0 n_{ss}) = 20 n_{ss}^2$

$n_{ss1} = -5.002500625$ rpm $\rightarrow T(n_{ss1}) \simeq 500.5$ N.m

$n_{ss2} = 4.997500625$ rpm $\rightarrow T(n_{ss2}) \simeq 499.5$ N.m

ⓑ Cheak the stability

$B_m = \frac{dT_m}{dw} = -0.1$ $B_L = 40n$

Point ① $B_m = -0.1 > B_L = -200.1$ Point ① is unstable

Point ② $B_m = -0.1 < B_L = 199.9$ Point ② is stable

ملاحظة مهمة :- اذا كان $T_m = f_1(n)$ $T_L = f_2(n)$

فلا داعي للتحويل من n إلى w

إلا اذا طلب في السؤال (اهرب سؤال ممكن تجيبه على هذا الموضوع)

ويتم التحويل عن طريق القانون $w = \frac{2\pi n}{60}$

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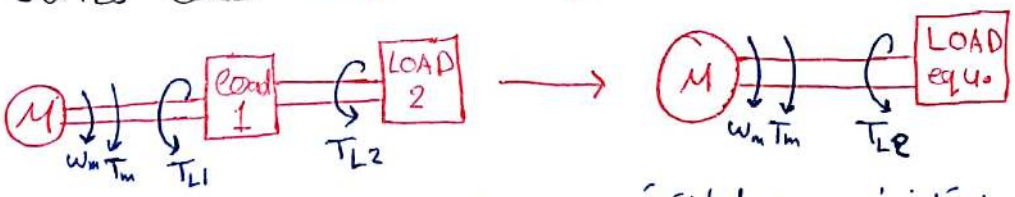
* Claculation of drive system, mechanical part حسابات النظام الميكانيكي في نظام القيادة

⇒ How to calculate(J_Σ) Total moment of intia & load Torque equivillent.?

← المطلوب هو حساب الجزء الميكانيكي لنظام القيادة ← حيث يتم احتساب عزم الحمل المنسوب لعمود المحرك وعزم القصور الذاتي المنسوب لعمود المحرك مع الأخذ بعين الاعتبار هل الحركة خطية أم دورانية.

← يتم تحويل النظام الموجود في نظام محوري على حمل ميكانيكي منسوب لعمود المحرك

① series load أو محتوى حمل على عامود المحرك



※ From saving energy rule من قانون حفظ الطاقة

$K_{\Sigma\,before} = K_{\Sigma\,after}$ → Kinatic energy

$$\frac{1}{2} J_m w_m^2 + \frac{1}{2} J_{L1} w_m^2 + \frac{1}{2} J_{L2} w_m^2 = \frac{1}{2} J_m w_m^2 + \frac{1}{2} J_{LE} w_m^2$$

$J_{LE} = J_{L1} + J_{L2}$ → equivillent moment of intia of load.

$J_\Sigma = J_{LE} + J_m$ → Total moment of intia of system.

* Load torque:-

$$P_{2\,before} = P_{2\,after}$$

$$T_{L1} w_m + T_{L2} w_m = T_{LE} \cdot w_m$$

$$T_{LE} = T_{L1} + T_{L2}$$

ملاحظة
← استخدمنا w_m في جميع المعادلات لأن جميع الأحمال على نفس العمود وهو عمود المحرك

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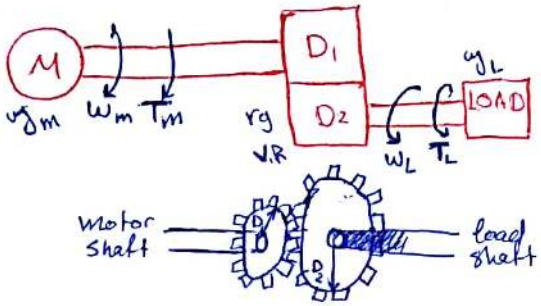
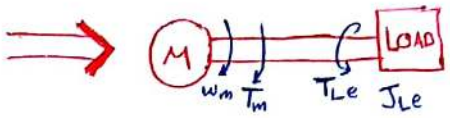
② gear box





gear ratio $(rg) = \frac{D_2}{D_1} = \frac{\omega_m}{\omega_L}$

velocity ratio $(V.R) = \frac{D_1}{D_2} = \frac{1}{rg} = \frac{\omega_L}{\omega_m}$

* eqⁿ. moment of inertia of load (J_{Le})

$$K_{\Sigma\,before} = K_{\Sigma\,after}$$

$$\frac{1}{2} J_m \omega_m^2 + \frac{1}{2} J_L \omega_L^2 = \frac{1}{2} J_m \omega_m^2 + \frac{1}{2} J_{Le} \omega_m^2 \eta$$

but $\frac{\omega_L}{\omega_m} = V.R = \frac{1}{rg}$

$$J_{Le} = J_L \frac{\omega_L^2}{\omega_m^2} \cdot \frac{1}{\eta} = J_L \left(\frac{1}{rg}\right)^2 \cdot \frac{1}{\eta}$$

* Load Torque.

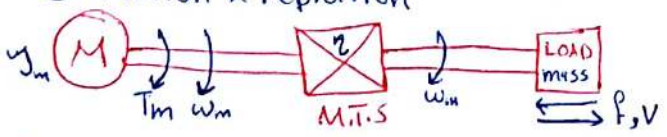
$$P_{2\,before} = P_{2\,after}$$

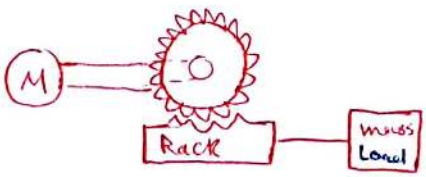
$$\omega_L \cdot T_L = T_{L\Sigma} \cdot \omega_m \cdot \eta$$

$$T_{L\Sigma} = T_L \cdot \frac{\omega_L}{\omega_m} \cdot \frac{1}{\eta} = T_L \cdot \frac{1}{rg} \cdot \frac{1}{\eta}$$

③ transational load

ⓐ traction & repultion





J_{Le} →

$$K_{\Sigma\,before} = K_{\Sigma\,after}$$

$$\frac{1}{2} m V^2 = \frac{1}{2} J_{Le} \omega_m^2 \cdot \eta$$

$$J_{Le} = m \left(\frac{V}{\omega_m}\right)^2 \cdot \frac{1}{\eta}$$

T_{Le} →

$$P_{2\,before} = P_{2\,after}$$

$$F \cdot V = T_{Le} \cdot \omega_m \cdot \eta$$

$$T_{Le} = F \left(\frac{V}{\omega_m}\right) \cdot \frac{1}{\eta}$$

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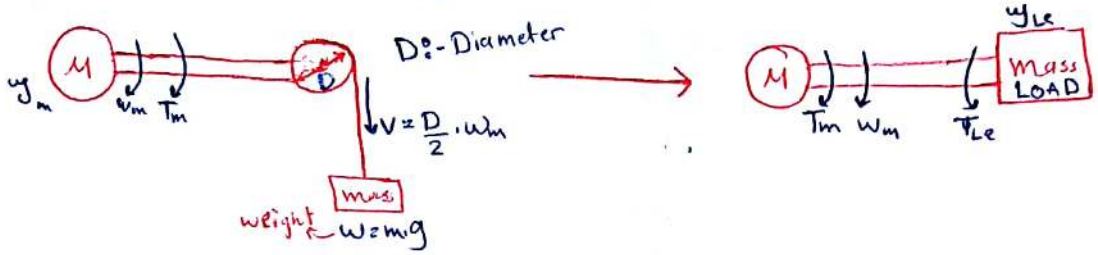
② Load depend on gravity



J_{Le} ⇒ $K.E_{before} = K.E_{after}$

$\frac{1}{2} m v^2 = \frac{1}{2} J_{Le} \omega_m^2 \frac{1}{\eta}$

$J_{Le} = m \left(\frac{V}{\omega_m}\right)^2 \frac{1}{\eta} = m \left(\frac{D}{2}\right)^2 \cdot \frac{1}{\eta}$

T_{Le} ⇒ $P_{2\,before} = P_{2\,after}$

$\omega \cdot V = T_{Le} \cdot \omega_m \cdot \eta$

$T_{Le} = m \cdot g \left(\frac{V}{\omega_m}\right) \cdot \frac{1}{\eta} = m \cdot g \left(\frac{D}{2}\right) \cdot \frac{1}{\eta}$

هذا ما ظهر على هذا الموضوع

$P_2 = T_m \cdot \omega_m$ → Power delivered by the motor ①

$P_{2L} = T_{Le} \cdot \omega_m$ → Power to the load

② نحن لاحظنا أن T_{Le} هو عن طريق الكتلة لكن

① اذا كان على محور المحرك فإنه يجمع كما هو

② اذا كان على محور الحمل بعد gear box فإنه يجمع مع الحمل ومن ثم يحول

③ غالباً السؤال على هذا الموضوع يكون مقالياً وأنت تستنتج الرسمة لذلك عليك النظر الى اسئلة سنوات يكون فيها سؤال مقالياً على هذا الموضوع

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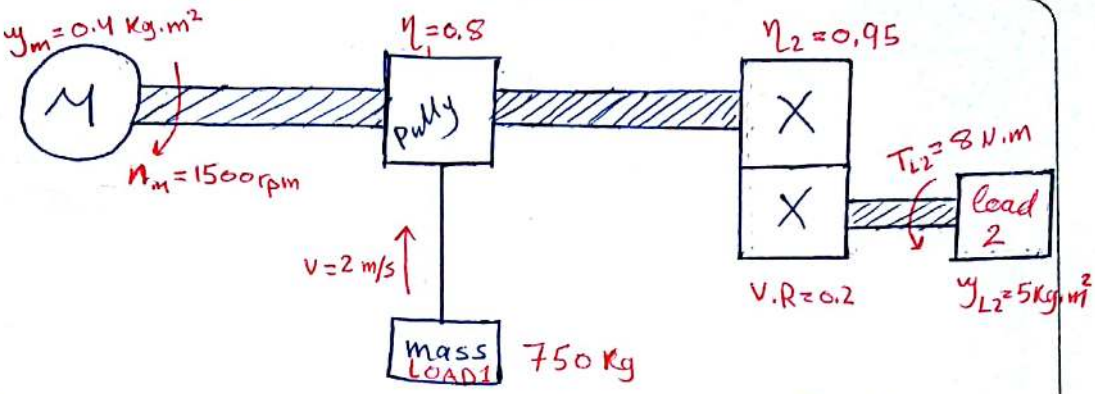
Example :-



① Find The total moment of intia of the system $[J_\epsilon]$

$$J_\epsilon = J_m + \frac{m}{\eta_1}\left(\frac{V}{\omega_m}\right)^2 + J_{L_2}\cdot(V.R)^2\cdot\frac{1}{\eta_2}$$

$$= 0.4 + 0.152 + 0.21$$

$$= 0.762 \text{ Kg.m}^2$$

$$\omega_m = \frac{2\pi n_m}{60} = 157.08 \text{ rad/s}$$

② The developed mechanical power in the shaft??

$$T_{Le} = T_{Le_1} + T_{Le_2}$$

$$= mg\left(\frac{V}{\omega_m}\right)\frac{1}{\eta_1} + T_{L2}(V.R)\cdot\frac{1}{\eta_2}$$

$$= 117.09 + 1.684$$

$$= 118.782 \text{ N.m}$$

$g = 9.81 \text{ m/s}^2$

$$P_2 = T_m \cdot \omega_m$$

$$= 118.782 * 157.08$$

$$= 18.64 \text{ KW}$$

but at steady state $T_m = T_{Le}$

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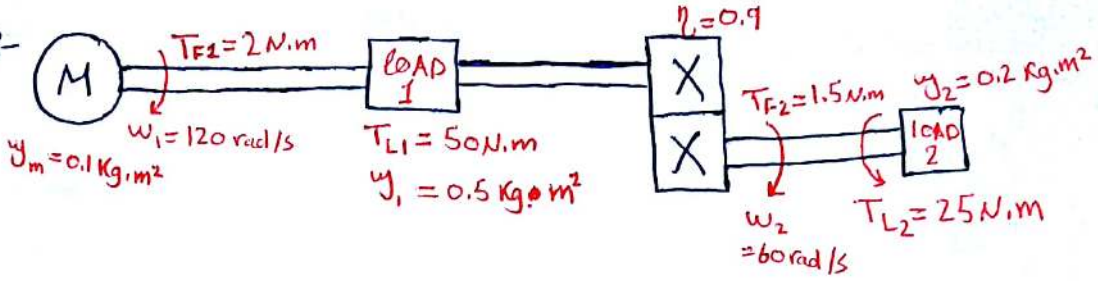
Example(2):-



① Find The total moment of intia of the system (J_Σ)

$$J_\Sigma = J_m + J_{L_1} + J_{L2e}$$

$$= J_m + J_{L_1} + J_{L2}\left(\frac{1}{r_g}\right)^2 \frac{1}{\eta}$$

$$= 0.1 + 0.5 + 0.056 = 0.656 \text{ Kg.m}^2$$

$$r_g = \frac{\omega_1}{\omega_2} = 2$$

تذكير
لا تنسى كتابة وحدة كل ناتج تخرجه

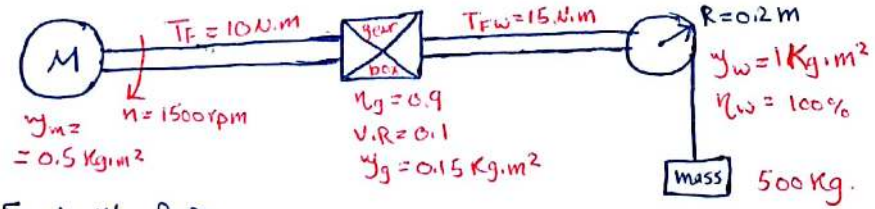
② develop mechanical power

$$T_{Le\Sigma} = T_{F1} + T_{L_1} + (T_{F2} + T_{L2}) \cdot \frac{1}{r_g} \cdot \frac{1}{\eta}$$

$$= 2 + 50 + (1.5 + 25) * \frac{1}{2} * \frac{1}{0.9} = 68.72 \text{ N.m}$$

$$P_2 = T_{Le\Sigma} \cdot \omega_1 = 8006.6 \text{ watt}$$

Example (3):-



⇒ Find J_Σ & P_2

$$J_\Sigma = J_m + J_g + \left(J_w + \underbrace{m\left(\frac{V}{\omega_m}\right)^2}_{J \text{ for mass}}\right) \cdot \left(\frac{1}{r_g}\right)^2 * \frac{1}{\eta_g}$$

$$= 0.883 \text{ Kg.m}^2$$

$$r_g = \frac{1}{V.R} = 10$$

$$\frac{V}{\omega_m} = R = 0.2 \text{ m}$$

$$T_m = T_{Le} = T_F + \left(T_{F.w} + m.g\left(\frac{V}{\omega_m}\right)\right) \frac{1}{r_g} \cdot \frac{1}{\eta} = 120.67 \text{ N.m}$$

$$P_2 = T_m \cdot \omega_m = 18.95 \text{ Kw}$$

but $\omega_m = \frac{2\pi n}{60} = 157.08$ rad/s

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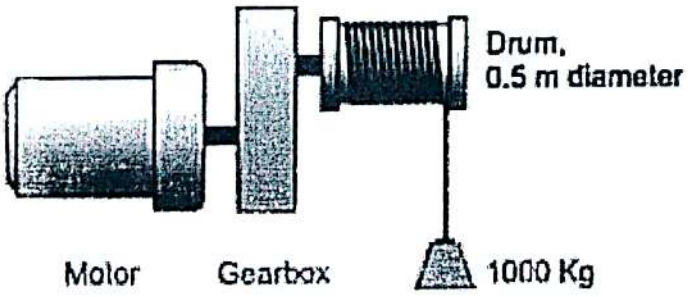
Question 3

The following drive system is used to raise a load at constant velocity. The motor rotational speed is 1000 rpm, and the rotational speed of the drum is 200 rpm. The efficiency of the gear box is 0.85. The efficiency of the drum is 100%. The moment of inertia of the motor (input shaft) is 0.5 kg.m^2, and the moment of inertia of the drum (output shaft) is 0.75 kg.m^2. There is a friction on the motor shaft equivalent to 10 N.m. Calculate:

(1) The total (equivalent) moment of inertia of the drive system referred to the motor shaft.

(2) The developed mechanical power on the motor shaft.



(سؤال وقالي)

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* Transient operation

$$T_m - T_L = J_\epsilon \frac{dw}{dt}$$

$$\Rightarrow \int dt = J_\epsilon \int_{w_1}^{w_2} \frac{1}{T_m - T_L} dw$$

$$\Rightarrow t_{transient} = J_\epsilon \int_{w_1}^{w_2} \frac{1}{T_m - T_L} dw$$

ملاحظة
* الموجة تصبح مقبولة اذا وصل الخطأ لـ 2.5 % من القيمة النهائية (steady state)

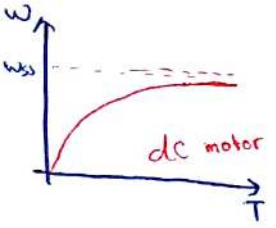
* Type of transient operation

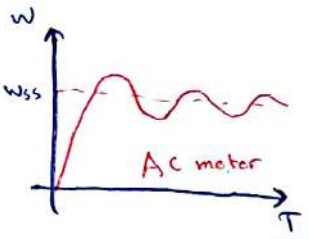
① starting

$w_1 = 0$

$w_2 = 97.5\, w_{ss}$

$$t_{st} = J_\epsilon \int_0^{97.5\% w_{ss}} \frac{1}{T_m - T_L} dw$$





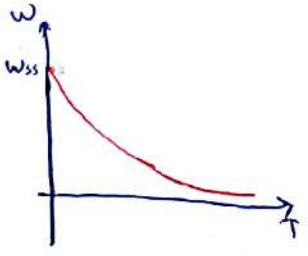
ملاحظة
هذا الموضوع يأتي جهته سؤال ايجاد s.s operating point
يعني لأي منفرداً

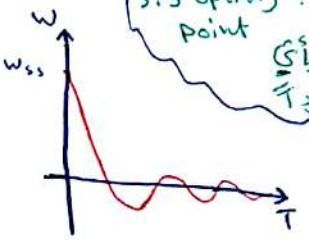
② Braking (stopping)

$w_1 = w_{ss}$

$w_2 = 2.5\% w_{ss}$

$$t_{br} = J_\epsilon \int_{w_{ss}}^{0.025 w_{ss}} \frac{1}{T_m - T_L} dw$$



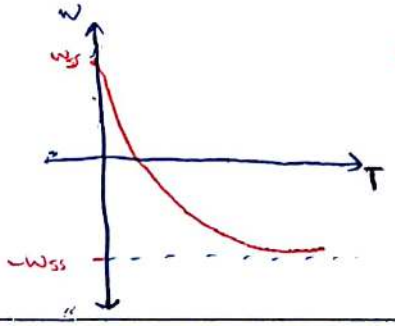


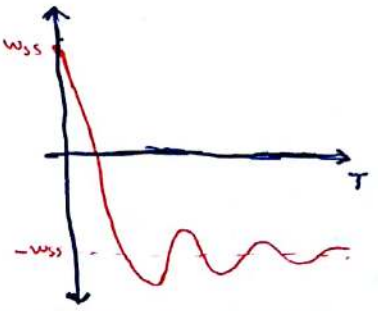
③ reverse

$w_1 = w_{ss}$

$w_2 = 0.975\, w_{ss}$

$$t_{rev} = \int_{w_{ss}}^{-0.975 w_{ss}} \frac{1}{T_m - T_L} dw$$





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Example :- $J_\Sigma = 1\ Kg.m^2$ $T_m = 30 + 0.2w$ $T_L = 10 + 0.6w$

سؤال سؤال 2018 صيفي

Find ① The acceleration of the drive system at starting?

عطلوب $\left.\frac{dw}{dt}\right|_{w=0}$ ⇒ $\frac{dw}{dt} = \frac{30+0.2w-10-0.6w}{J_\Sigma} = \frac{T_m - T_L}{J_\Sigma}$

$= 20 - 0.4w$

$\left.\frac{dw}{dt}\right|_{w=0} = 20\ rad/s^2$

② Find The S.S operating point & cheak the stability.

$T_{mss} = T_{Lss}$

$30 + 0.2\,w_{ss} = 10 + 0.6\,w_{ss}$ ⇒ $w_{ss} = 50\ rad/s$

↳ $T_{mss} = T_{Lss} = 40\ N.m$

$\beta_m = 0.2$ $\beta_L = 0.6$ ⇒ $\beta_m < \beta_L$

stable S.S operating point.

③ calculate the starting time of the drive system.

$w_1 =$ zero $w_2 = 97.5\ w_{ss} = 48.75\ rad/s$

$$\int_{w_1}^{w_2} \frac{J_\Sigma}{T_m - T_L} dw = t_{st} = \int_0^{48.75} \frac{1}{20 - 0.4w} dw = 9.223\ s$$

④ calculate the braking time if the motor is disconnected

$T_m = 0$ $w_1 = w_{ss} = 50\ rad/s$ $w_2 = 2.5\%\ w_{ss} = 1.25\ rad/s$

$$t_{br} = \int_{50}^{1.25} \frac{1}{-10 - 0.6w} dw = 2.19\ sec$$

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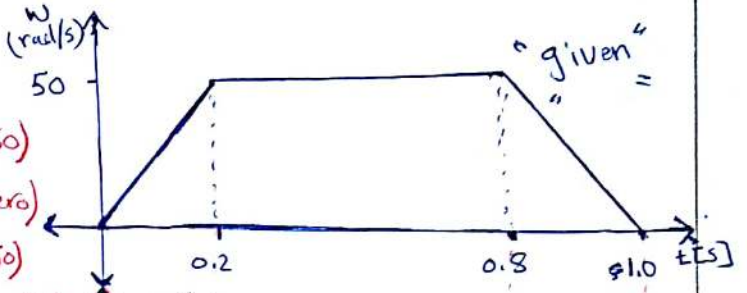
* sketching of loading characteristics :-

* هنا في هذا الموضوع كلنا سنرسم بناءً على معادلة الحركة (equ. of motion)

*draw speed, torque and power profile

example: if $y_\xi = 0.02\ kg.m^2$



① $\frac{dw}{dt} = \frac{\Delta w}{\Delta t}$ for linear

- $t=0 \to 0.2$ $\left(\frac{dw}{dt} = 250\right)$
- $t=0.2 \to 0.8$ $\left(\frac{dw}{dt} = zero\right)$
- $t=0.8 \to 1$ $\left(\frac{dw}{dt} = -250\right)$

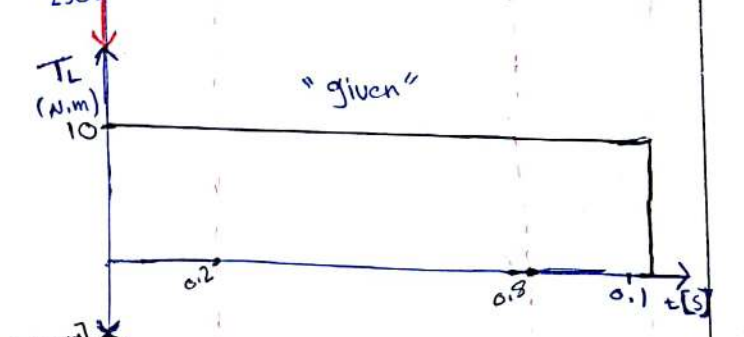
② Find the (T_m) if T_L is given

$$T_m = T_L + \frac{dw}{dt} \cdot y_\xi$$

$t(0 \to 0.2)$ $T_m = 10 + \frac{250}{5} = 60\ N.m$

$t(0.2 \to 0.8)$ $T_m = 10 + 0 = 10\ N.m$

$t(0.8 \to 1.0)$ $T_m = 10 - 250 * 0.2 = -40\ N.m$



③ sketching the power profile

$P_2 = T_m * w$

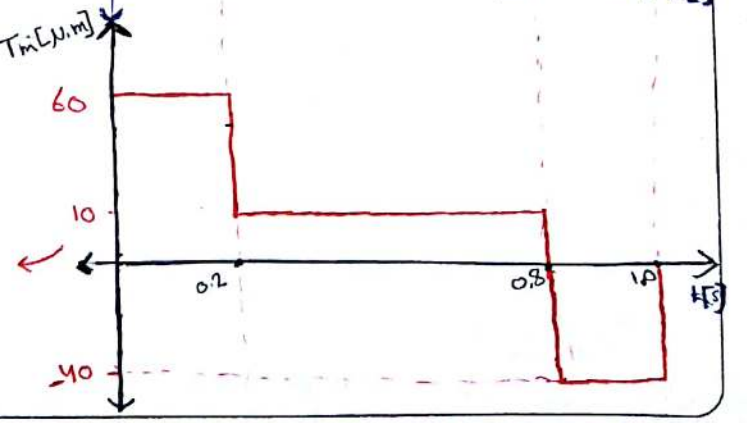
$(t=0 \to 0.2)$ at zero $\to P_2 =$ zero watt

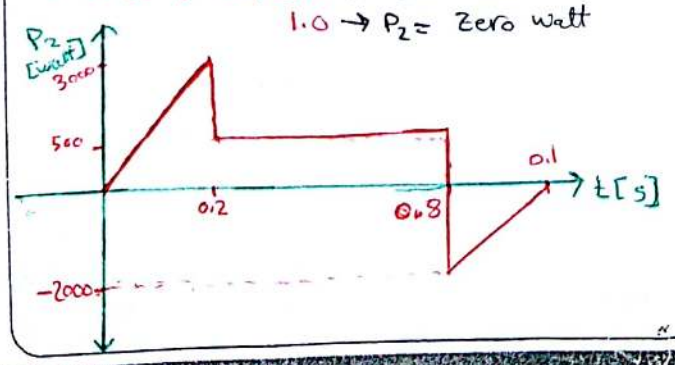
$0.2 \to P_2 = 3000$ watt

$t = (0.2 \to 0.8)$ $P_2 = 500$ watt

$t = (0.8 \to 1)$ at $0.8 \to P_2 = -2000$ watt

$1.0 \to P_2 =$ zero watt





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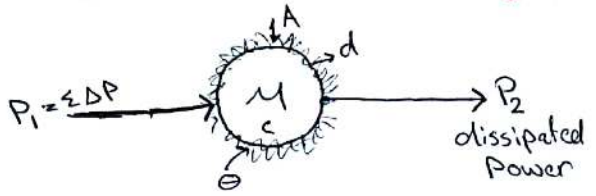
*motor selection:-

(A) Heating process → heating & cooling

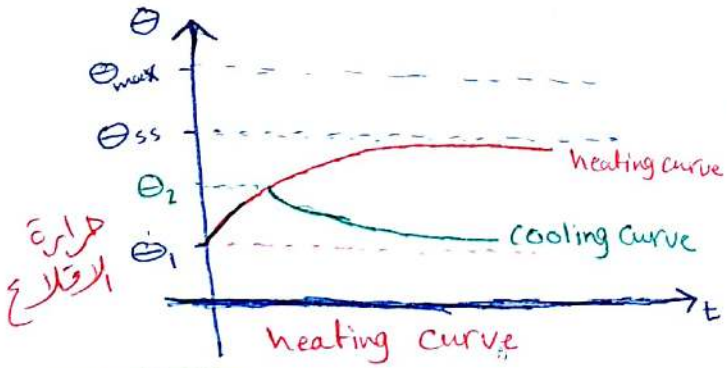


A:- heating surface
c:- specific heat.
Θ:- increase in Tempreture
d:- معامل انتقال الحرارة

من قانون حفظ الطاقة

$$P_1 dt - P_2 dt = mc\, d\theta \quad \text{....(1)}$$

m:- mass
C:- Thermal capacitance (السعة الحرارية)



For safe operation $\Theta_{ss} \leq \Theta_{max}$

$\Theta_2 < \Theta_{ss}$

For heating

$$\Theta = \Theta_{ss}(1 - e^{-t/T_h}) + \Theta_1 e^{-t/T_h}$$

→ where $T_h = \frac{m.c}{d.A}$ [heating time const.]

For cooling

$$\Theta = \Theta_{ss}(1 - e^{-t/T_c}) + \Theta_2 e^{-t/T_c}$$

→ T_c :- [cooling time constant]

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(B) Duty cycle

① continous duty cycle. (constant torque)

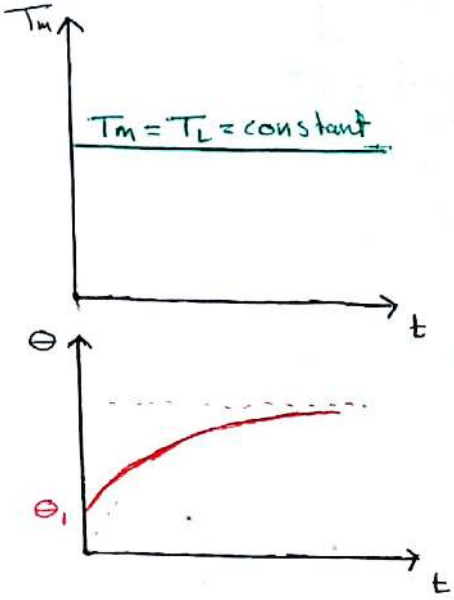
$T_m = T_L$ = constant

$w = w_{ss}$ = constant

$t_{on} \gg T_h \longrightarrow \theta_2 > \theta_{ss}$

but θ_{ss} must be less than θ_{max} to cause damge

example :- ① compresor ② bump

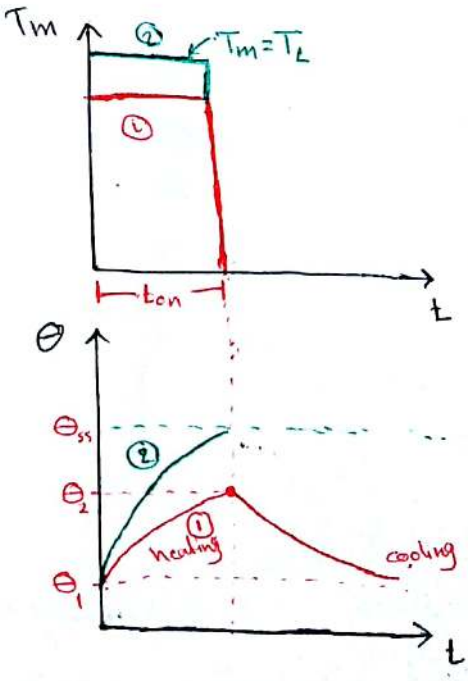


② short time duty cycle

$t_{on} \ll T_h \longrightarrow \theta_2 < \theta_{ss}$

$t_{off\ (rest)} \gg T_c \longrightarrow$ هذه الفترة كافية حتى يرجع الحرارة الى وضعه الطبيعي

example :- ① crane drive
② bulb drive
③ valve drive
④ drive for position system.



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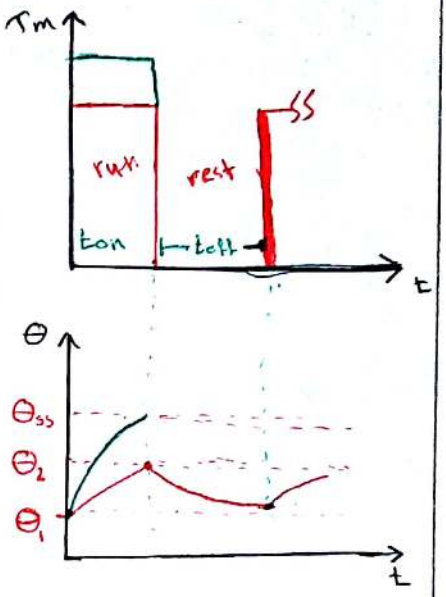
③ Intermittent periodic duty cycle.

$t_{on} < T_h \longrightarrow \theta_2 < \theta_{ss}$

$t_{off} < T_c$

* motor can be over loaded during running time.

example :- impressing & cutting & melling machine



④ Intermittent duty with starting

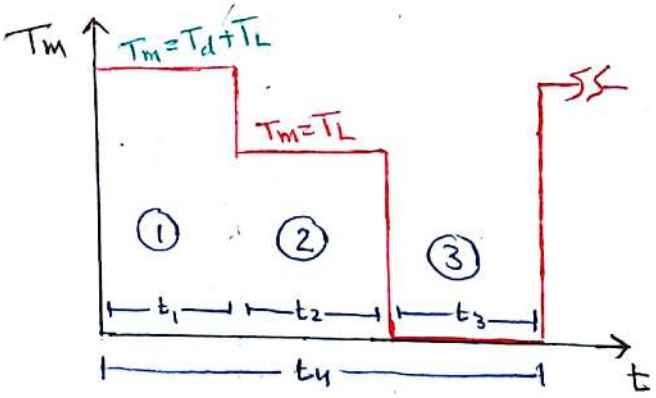
t_1 :- starting time (run) :- (t_{run})

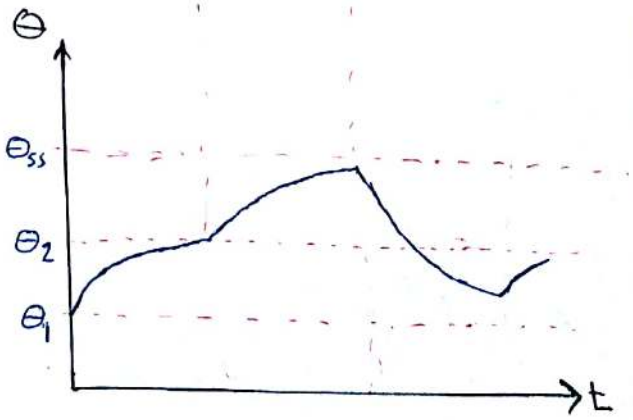
t_2 :- steady state time (t_{ss})

t_3 :- rest (off) time (t_{rest})

t_4 :- period time (cycle) (T_{cycle})

example :- اوفحان البوكية .





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⑤ Intermittent duty with starting & braking

① $T_m = T_d + T_L$

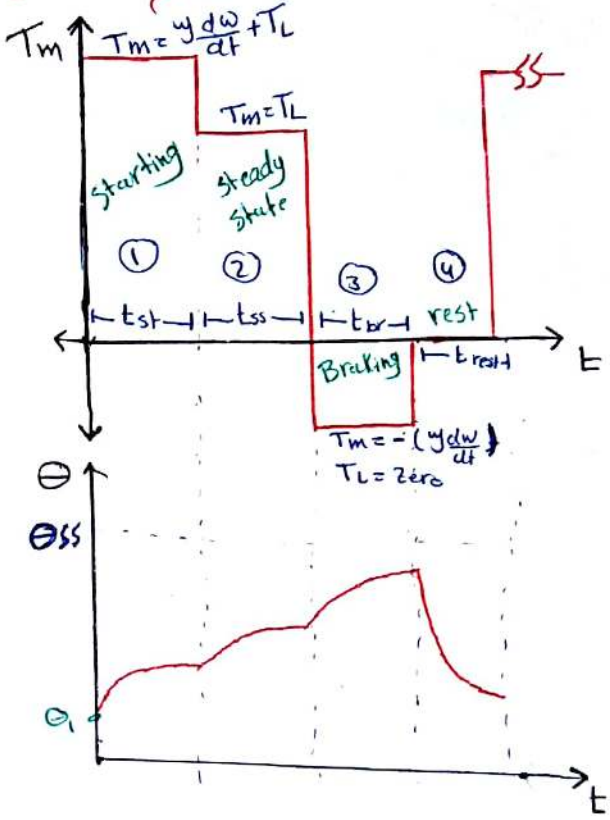
② steady state $T_m = T_L$

③ if the load is remove

$T_m = -T_d$

example :- ① Electric vichles

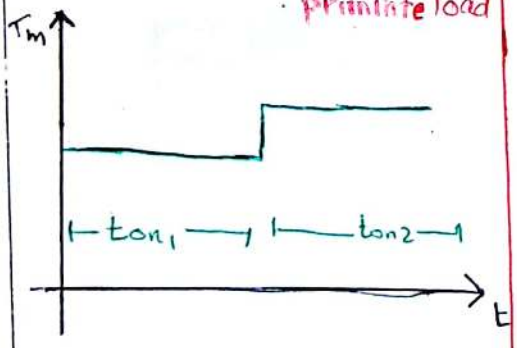
② Electric trains.



خلال فترة الإيقاف (braking) يكون العزم

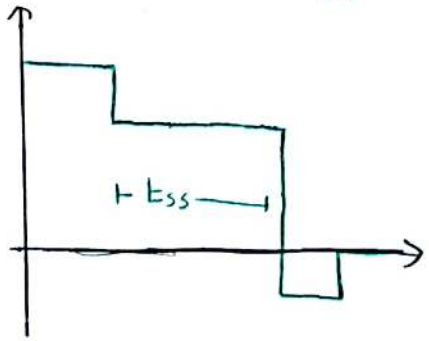
$T_m < 0$ اقل من صفر

⑥ continous with priminte load



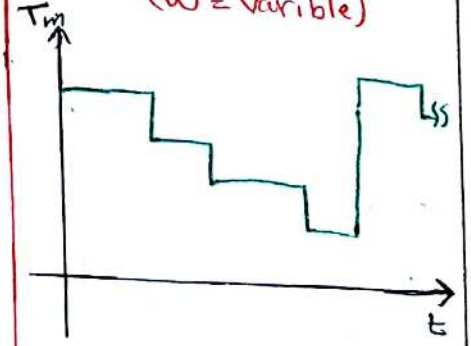
$t_{on_1} > T_h$

⑦ continous with steady state and braking



$t_{ss} >> T_h$

⑧ continous with (w = varible)



$t_{on} >> T_h$

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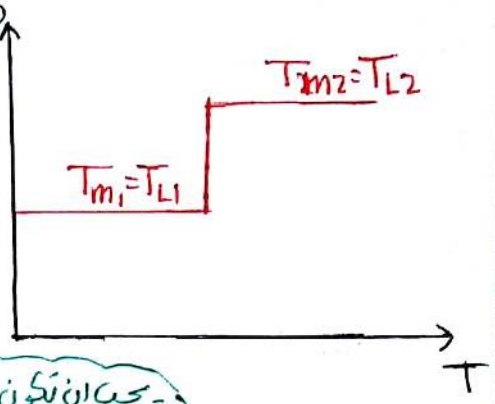
*Determination of motor ratings: (I_n, T_n, P_{2n})

① continous duty

* يتم من هذه الطريقة تحديد القدرة الميكانيكية الخارجة وذلك عن طريق أخذ أكبر قيمة للعزم
* حيث يعمل المحرك تحت حمل ثابت وسرعة ثابتة لوقت طويل



$$P_2 = T_m \cdot W_{ss}$$

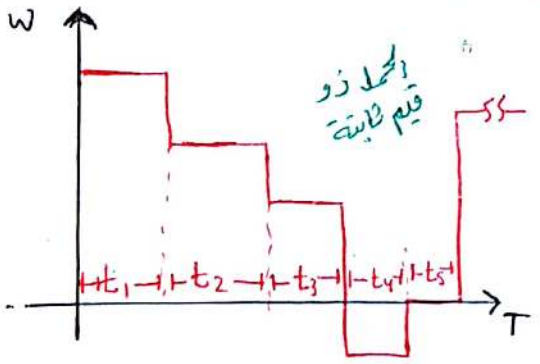
$$P_{2max} = T_{m2} \cdot W_{ss}$$

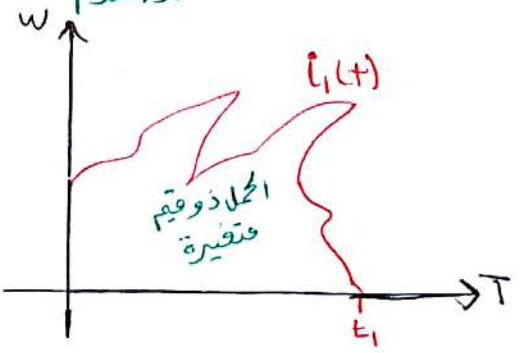
$$P_{2N} \geq P_{2max}$$

- يجب ان تكون قدرة المحرك الكهربي اكبر من اكبر قدرة خارجة للمحرك ((القدرة في الـ data sheet))

② Duty with fluctuating loads: (equivallent method)

* نريد تحويل التيار أو القدرة أو العزم من شيء متغير القيمة إلى قيمة ثابتة (مكافئة)





$$I_{eq} = I_{RMS} = \sqrt{\frac{I_1^2 t_1 + I_2^2 t_2 + \cdots}{T_{cycle}}} \quad \cdots (1)$$

$$I_{1_{eq}} = \sqrt{\frac{1}{t_1}\int_0^{t_1} i_1^2(t)\,dt} \quad \cdots (2)$$

(2) هذه المعادلة تحول القيمة المتغيرة إلى ثابتة (مكافئة)

* ممكن ان يكون السؤال خليط بين قيم متغيرة وثابتة فنجد مكافئة المتغيرة عبر القانون الثاني ونجد المكافئة الكلية عبر القانون الاول

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* over load cheak for the motor :-

$$\lambda_I = \frac{I_{max}}{I_{nominal}} \approx (3-3.5)$$

λ_I specified ← I_{max} from current profile

For DC motor

$$I_{nominal} \geq \frac{I_{max}}{\lambda_I}$$ if λ_I isn't specified (محدد)

For AC motor

$$\lambda_T = \frac{T_{max}}{T_{nominal}}$$ λ_T is specified

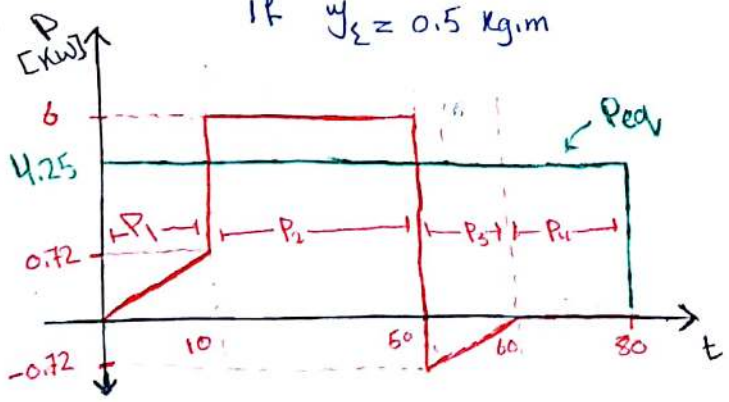
$$T_{nominal} \geq \frac{T_{max}}{\lambda_T}$$ λ_T isn't specified

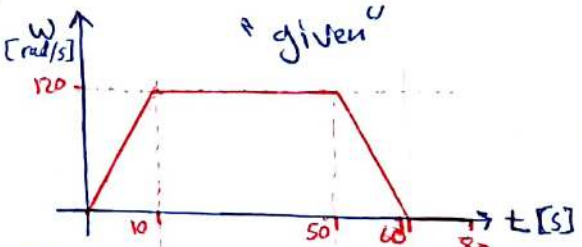
For syncrownous motor $\lambda_T = 2-2.25$

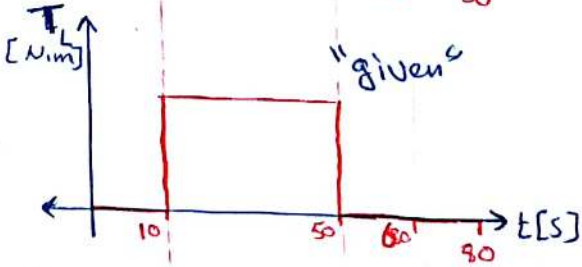
For induction motor $\lambda_T = 1.65-3$

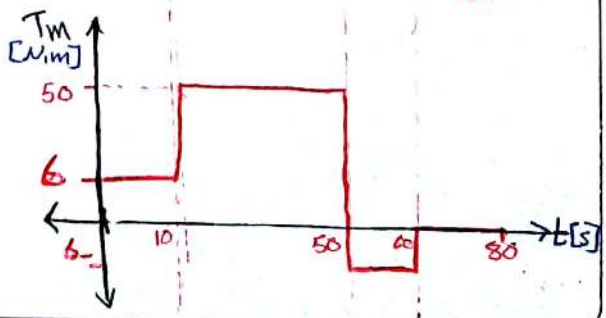
Example :- Find The equavillent power

If $y_2 = 0.5$ Kg.m









$P_1 = 0.072\,t$ KW

$P_2 = 6$ KW

$P_3 = -0.72 + 0.072\,t$ KW

$$P_{1e} = P_{2e} = \sqrt{\frac{1}{10}\int_0^{10}(0.072t)^2 dt} = 0.173 \text{ KW}$$

$$P_{eq} = \sqrt{\frac{P_{1e}^2 t_1 + P_2^2 t_2 + P_{3e}^2 t_3}{T_{cycle}}} = 4.25 \text{ KW}$$

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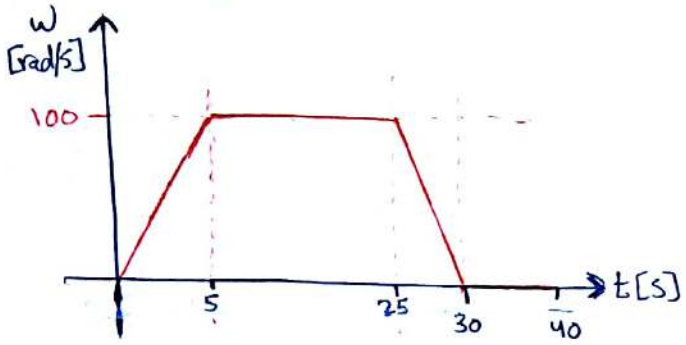
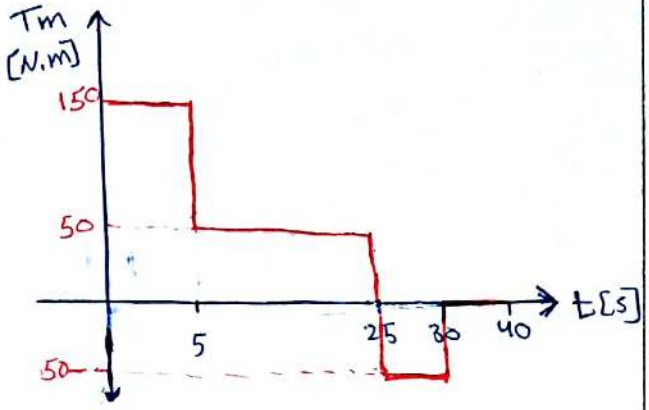
Example (2) :-





① name the type of duty cycle?

Intermittent duty with starting and braking

② Find the T_L at steady state?

$T_{Lss} = T_{mss} = 50$ N.m

③ Find T_{eq}?

$$T_{eq} = \sqrt{\frac{150^2*5 + 50^2*20 + (-50)^2*5}{40}} = 66.43 \text{ N.m}$$

④ Find P_{2max}?

$P_{2max} = T_{m_{max}} \cdot \omega_{ss} = 150 * 100 = 15$ KW

⑤ Find P_2 at steady state?

$P_{2ss} = T_{mss} \cdot \omega_{ss} = 5$ KW

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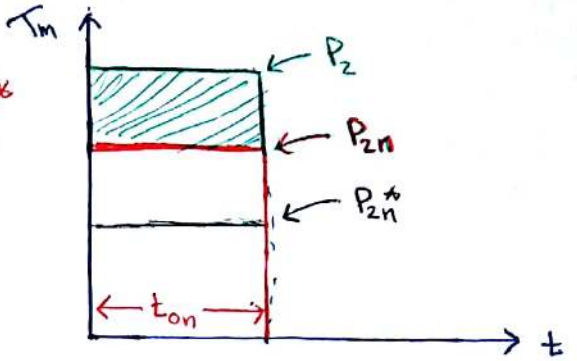
③ short time duty cycle.

* نريد هنا رفع θ_2 اى حالة الـ steady state وذلك عن طريق زيادة الحمل على المحرك ($P_2 \uparrow$) وذلك حسب المعادلة



$$P_2 = K\, P_{2n}$$

↳ over loading factor

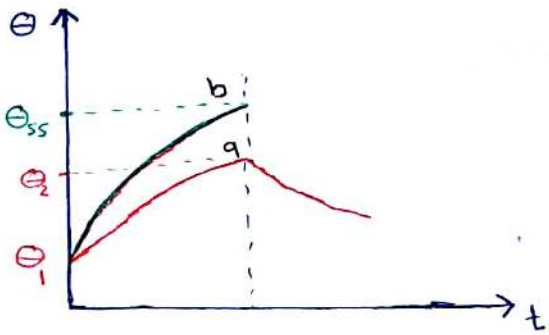
$$K = \frac{P_2}{P_{2n}} = \sqrt{\frac{1+\alpha}{1-e^{-t_{run}/T_h}} - \alpha}$$

t_{run} ← running time T_h ← heating time

$$\alpha = \frac{\Sigma \Delta P_c}{\Sigma \Delta P_{cu}} \simeq \text{zero}$$

ΔP_c ← constant power losses

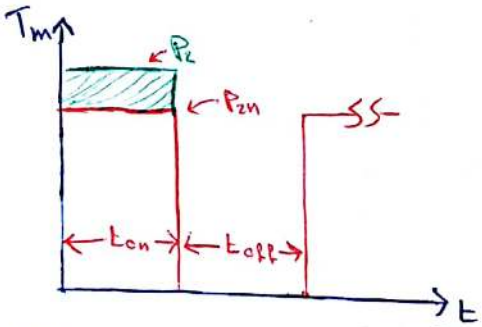
ΔP_{cu} ← copper losses



ملاحظة مهمة :- أنه لم نستطع زيادة الحمل على المحرك عندها نقوم بتغيير المحرك الموجود بمحرك ذا قدرة اقل ($\downarrow P_{2n}$) اي بمعنى اخر نقلل θ_{ss} حتى نصل الى θ_2

④ Intermittent periodic duty cycle

(K) * نفس كلام الذي بالاعلى لكنه يختلف قانون K

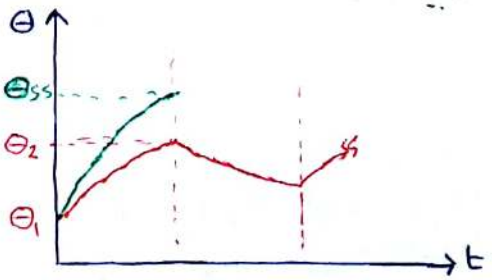


$$P_2 = K\, P_{2n}$$

$$K = \sqrt{(\alpha+1)\,\frac{1-e^{-(t_{on}/T_h + t_{off}/T_c)}}{1-e^{-(t_{on}/T_h)}}}$$

$$\alpha = \frac{\Sigma \Delta P_c}{\Sigma \Delta P_{cu}} \approx \text{zero}$$

اذا اوجدت $\theta_2 = \theta_{ss}$ فأن $P_2 = P_{2max}$



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Example (1) :- $P_{2n} = 20\,KW$; $\tau_h = \tau_c = 60\,min$

Find P_{2max} if

① duty cycle is short time duty & $t_{on} = 10\,min$.

$$\alpha = 0 \rightarrow K = \sqrt{\frac{1}{1 - e^{-(t_{on}/\tau_h)}}} = 2.55$$

$$P_{2max} = K P_{2n} = 51\,KW$$

② duty cycle is intermittent periodic & $t_{on} = t_{off} = 10\,min$

$$\alpha = 0 \rightarrow K = \sqrt{\frac{1 - e^{-(t_{on}/\tau_h + t_{off}/\tau_c)}}{1 - e^{-(t_{on}/\tau_h)}}} = 1.257$$

$$P_{2max} = K P_{2n} = 25.14\,KW$$

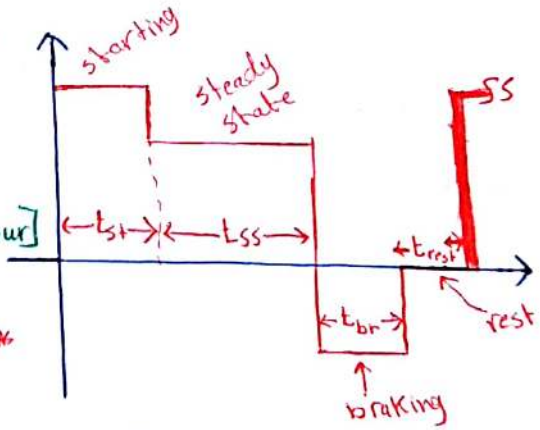
---

* Frequancy of operation For a motor subjective on intermittent perodic duty cycle

* هنا نريد حساب t_{rest} وذلك لحساب الـ maximum freq.

$$T_{cycle} = t_{st} + t_{ss} + t_{br} + t_{rest}$$

$$F_{max} = \frac{1}{T_{cycle}} = \frac{3600}{t_{st} + t_{ss} + t_{br} + t_{rest}} \; [cycle/hour]$$



* لحساب الـ t_{rest} يجب أن يكون هناك حفظ طاقة :-

$$\Delta E_{st} + \Delta P_n \cdot t_{ss} + \Delta E_{br} = \Delta P_N (\gamma t_{st} + t_{ss} + \gamma t_{br} + \beta t_{rest})$$

$\beta = (0.3 - 0.7)$ $\quad \gamma = \frac{1+\beta}{2}$

* ملاحظة مهمة :- إذا كان الـ F_{max} عدد كسري فأننا نختار العدد الصحيح الأقل منه (نحذف الكسر العشري)

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Example :- DC motor :- $I_{an} = 500\ A$ $R_a = 0.01\ \Omega$ $\Sigma \Delta P_c = 1\ KW$ ↳ constant power

$\beta = 0.5$

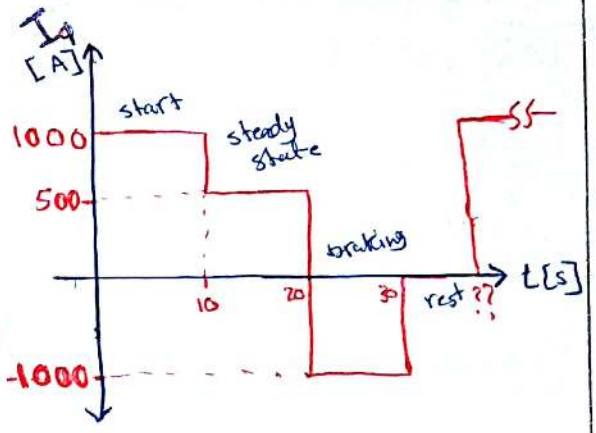
Sol:- $\beta = 0.5 \rightarrow \gamma = \frac{1+0.5}{2} = 0.75$

$\Delta E_{st} = (\underbrace{I_{st}^2}_{(10+1)*10} R_a + \Delta P_c) t_{st} = 110\ K\,joule$

$\Delta E_{ss} = (\underbrace{I_{ss}^2 R_a}_{\Delta P_{ss}} + \Delta P_c) t_{ss} = 35\ K\,joule$

$\Delta E_{br} = (I_{br}^2 R_a + \Delta P_c) t_{br} = 110\ K\,joule$

$\Delta P_n = I_{an}^2 R_a + \Delta P_c = 3.5\ KW$



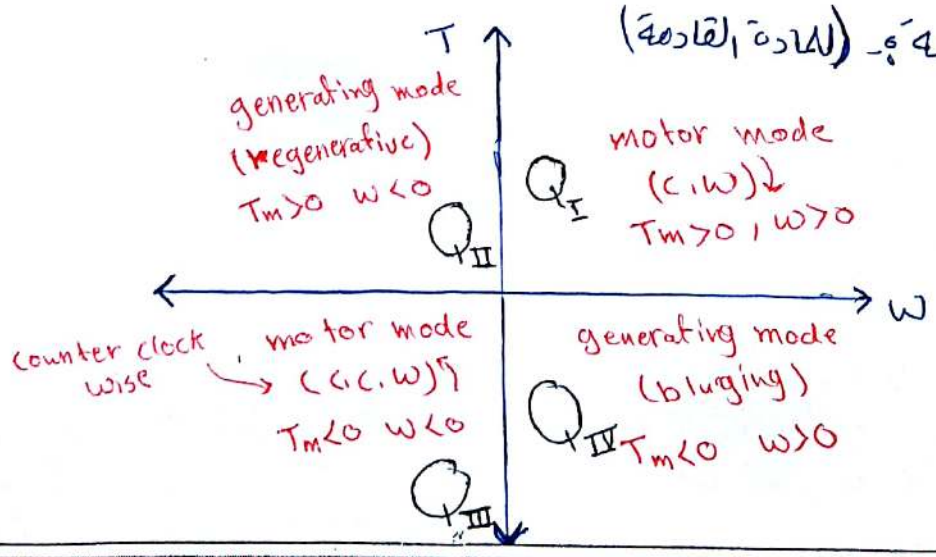
$\Delta E_{st} + \Delta P_{ss} \cdot t_{ss} + \Delta E_{br} = \Delta P_n (\gamma t_{st} + t_{ss} + \gamma t_{br} + \beta t_{rest})$

↳ $t_{rest} = 95.1\ s$

$f_{max} = \frac{3600}{10+10+10+95.7} = 28.64\ cycle/hour$

$f_{max\ selected} = 28\ cycle/hour$

وال طريقة دي :- (المدونة الفاضية)



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* static characteristic of DC motors-

note:- 1- $I_a = f(\omega)$ or $I_a = f(n)$ called electro mechanical char.

2- $T = f(\omega)$ or $T = f(n)$ called mechanical char. or (T/S) (Torque / speed)

note(2):- $E = K\phi\omega$ & $T = K\phi I_a$ اكثر قوانين استخداماً

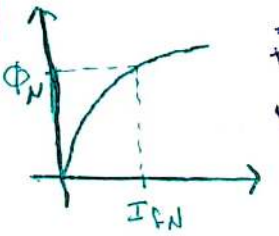
⇒ Type of DC machine depends on exictation:-

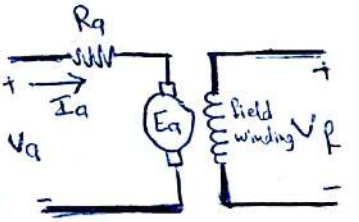
① separatly exicted DC motor

$I_f \neq f(I_a) \rightarrow \phi \neq f(I_a)$

$\rightarrow \phi$ = constant

* linear characteristic





② shunt dc motor

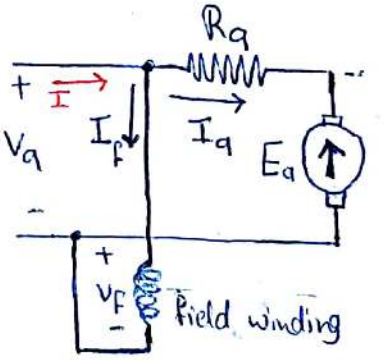
$I = I_a + I_f$

$R_{FW} \gg R_a \rightarrow I_f \ll I_a$, $I = I_a$

هذا النوع والذي قبله نفس الخصائص

* $I_{fN} \approx$ constant $\rightarrow \phi \approx$ constant

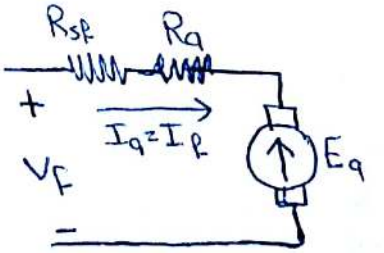
* linear characteristic



③ series DC motor

$I_{sf} = I_a \rightarrow \phi = f(I_a)$

* non linear characteristic



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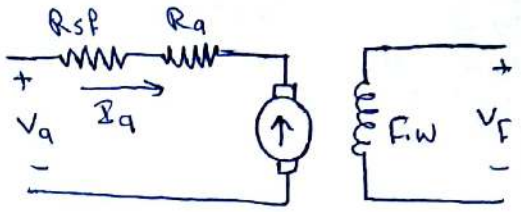
④ Compound Dc motor

$$\Phi_\varepsilon = \Phi_{s.f} + \Phi_{sep}$$

* هذا النوع يعتمد على المجال الاقوى فالمجال الاقوى يكسب المحرك خصائصه

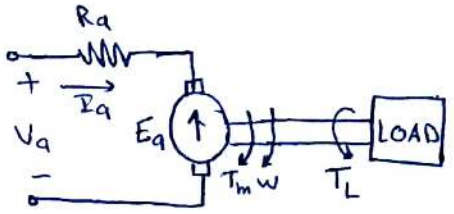


* charisteristic of Dc machine.

$$V_a = E_a + I_a R_a \quad \ldots \rightarrow E_a = K\phi w$$

$$V_a = K\phi w + I_a R_a$$

$$\Rightarrow w = \frac{V_a}{K\phi} - \frac{I_a R_a}{K\phi}$$ ... electromecanical Characteristic



$R_a = r_a + r_p + r_c$

armature resistance — addi. pole resistance — carbon resistance

but $T = K\phi I_a$

$$\Rightarrow w = \frac{V_a}{K\phi} - T\frac{R_a}{(K\phi)^2}$$ ... mechanical characteristic or (T/s) //

$$w = w_o - \Delta w$$

$$w_o = \frac{V_a}{K\phi}$$ ideal no. load speed $(w|_{I_a=0})$. السرعة لاسمية

$$\Delta w = \frac{I_a R_a}{K\phi} = \frac{T R_a}{(K\phi)^2}$$ .... Drop in speed due to Load الهبوط في السرعة ناتج عن الحمل.

ϕ :- flux

$$K = \frac{(P/2)\,N}{2\pi a}$$

P :- no. of poles

N :- no. of turns

a :- no. of parallel winding

$$w = \frac{2\pi n}{60}$$

angular speed للتحويل من speed الى

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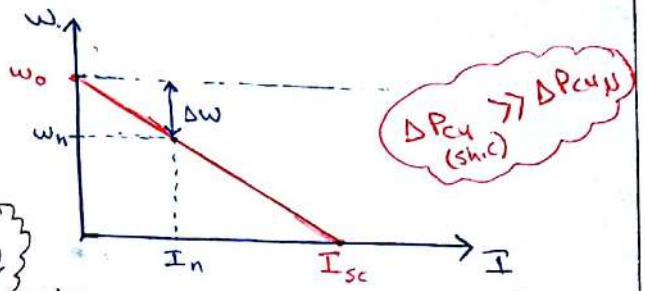
* static char. For separatly & shunt dc motor

$$\omega = \frac{V_a}{K\phi} - \frac{I_a R_a}{K\phi}$$

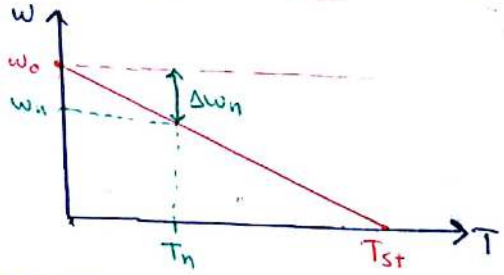
$$I_a(\omega=0) = I_{sc} = \frac{V_a}{R_a}$$



ملاحظة: $K\phi$ ثابتة

$$\omega = \frac{V_a}{K\phi} - \frac{T R_a}{(K\phi)^2}$$

$$T(\omega=0) = T_{starting} = K\phi I_{sc}$$



Type of char. ① Natural char.

if the char. work under nominal/Rating/name plate data

اذا كان يعمل بدون اي اضافة على دائرة المنتج

Van, Ran, Ian, Tn

② Regulated char.

if any Value is change

اذا حدث اي تغيير أو اضافة على دائرة/في القيم

* To constract natural c/c we need Two point

① no-load:-

$I_a = 0$, $T = 0$

$$\omega_o = \frac{V_{an}}{(K\phi)_n}$$

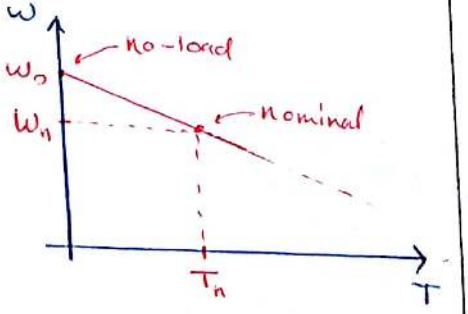
② nominal point

ω_{nom} (given) , I_{an} (given

$T_n = (K\phi)_n I_{an}$

$$(K\phi)_n = \frac{V_{an} - I_{an} R_a}{\omega_n}$$

طريقة حساب $(K\phi)_n$ الوحدة [Volt.sec]



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Example (1) :- 120 V (V_{an}) , 25 A (I_{an}) , 1250 rpm (n_n) , $R_a = 0.1\ \Omega$, (5 kW) (P_{2n}) عكن بطرح

Find :- ① $(K\Phi)_n = \frac{V_{an} - I_{an} R_a}{\omega_n}$ → but $\omega_n = \frac{2\pi n}{60} = 130.83$ rad/s

$= 0.89$ Volt.sec

② $T_n = (K\Phi)_n \cdot I_{an} = 0.89 * 25 = 22.25$ N.m

③ $T_{st} = (K\Phi)_n \frac{V_{an}}{R_a} = 0.89 * \frac{120}{0.1} = 1058$ N.m

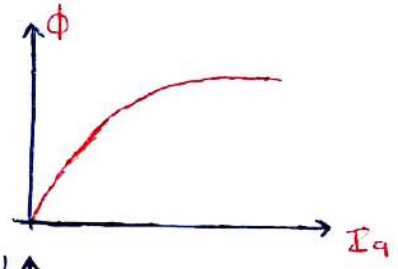
④ $\Delta\omega = \frac{I_{an} R_a}{K\Phi} = 2.809$ rad/s

---

* static char. of DC series motor

$\Phi = f(I_a) = K_1 I_a \longrightarrow I_a = \Phi / K_1$

$T = K\Phi I_a = K^* I_a^2 \longrightarrow I_a = \sqrt{T/K^*}$



* electromechanical char.

$\omega = \frac{V_a}{K^* I_a} - \frac{R_a + R_{sf}}{K^*}$

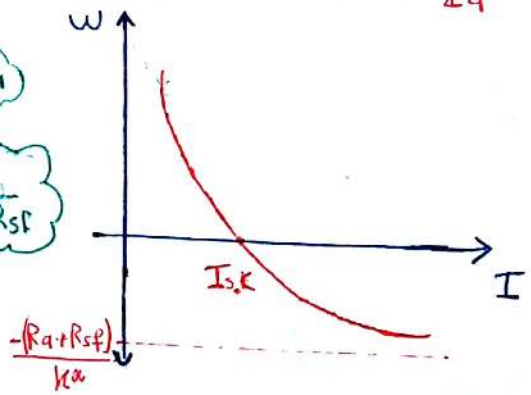
$\omega_o = \frac{V_a}{K^* I_a}$ $\Delta\omega$ - constant

$|\omega_o|_{max} \le 2\omega_n$

$I_{sc} = \frac{V_a}{R_a + R_{sf}}$

$I_a = 0 \longrightarrow \omega = \infty$ لذلك يمنع تشغيل المحرك بدون حمل

$I_a = \infty \longrightarrow \omega = -\Delta\omega$



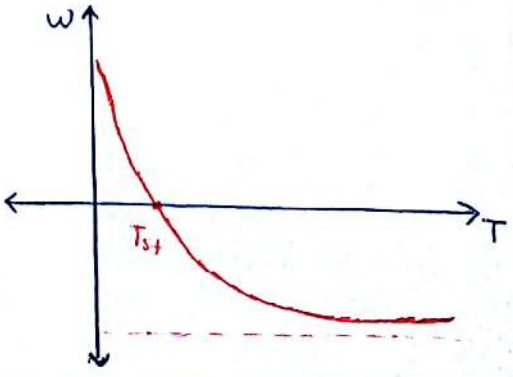
* mechanical char.

$I_a = \sqrt{\frac{T}{K^*}}$

$\omega = \frac{V_a}{\sqrt{K^*}\sqrt{T}} - \frac{R_a + R_{sf}}{K^*}$

$T_{st} = (K\Phi)_{st} I_{sc}$

$\phi \propto I_a \longrightarrow \frac{(K\Phi)_n}{(K\Phi)_{st}} = \frac{I_{an}}{I_{s.c}}$



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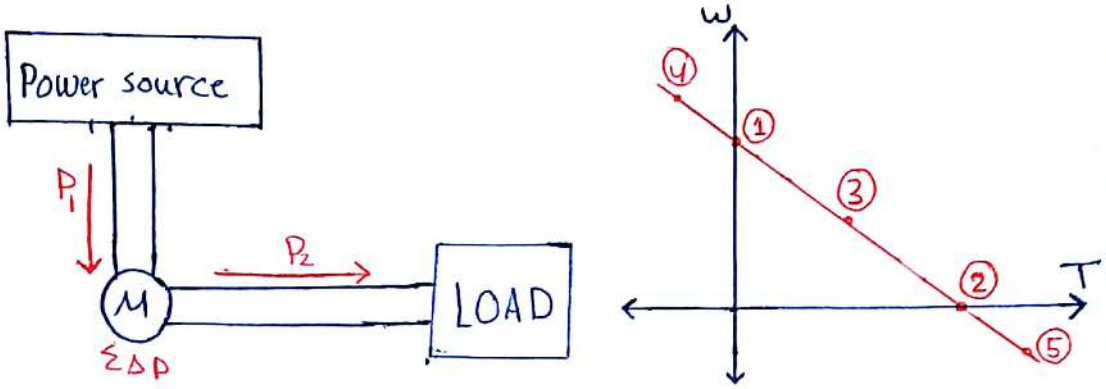
* operatings modes of DC motor (power flow) :-



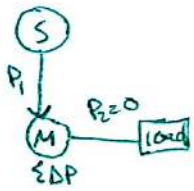
$P_1 = P_2 + \Sigma\Delta P$ ... motor operation
الحالة الطبيعية .

⇒ at point (1) (Ideal no Load)

$w = w_0$ $T = zero$ ⇒ $I_a = 0$ ⇒ $P_2 = T.w = zero$

* $P_1 = \Sigma\Delta P$ → $\eta = zero$

* تذهب كل الطاقة على شكل خسائر كهربائية (قيمتها قليلة نسبياً)

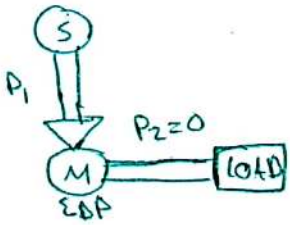


⇒ at point (2) (starting point)

$w = zero$ $T = T_{st}$ → $I = I_{sc}$ → $P_2 = T.w = zero$

⇒ $P_1 = \Sigma\Delta P$ → $\eta = zero$

$= I_{sc}^2 . R$ ← خسائر عالية جداً تذهب على شكل حرارة .



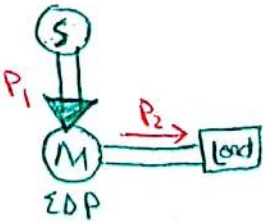
⇒ at point (3) (motoring mode)

$T > 0$, $w > 0$ الحالة الطبيعية للمحرك .

$P_1 = V_a I_a > 0$ $P_2 = T.w > 0$

$P_1 = \Sigma\Delta P + P_2$

(C . W) تتحول مع عقارب الساعة



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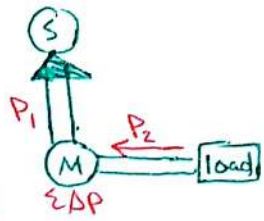
⇒ at point (4) (Regenerative braking) (generating mode)

$\omega > 0 , T_m < 0$

$\omega_1 > \omega_0 \rightarrow E_a > V_a \rightarrow I_a = \frac{V_a - E_a}{R_a} < 0$

$P_1 = V_a I_a < 0 \qquad P_2 = T \cdot \omega < 0$

$P_2 = P_1 + \Sigma \Delta P$ * هنا يعيد الحمل الطاقة إلى المصدر



* motor operate as generator in parallel to the source

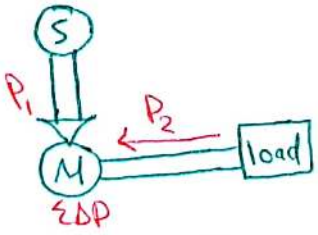
يعمل الحمل كمولد على التوازي مع المصدر.

⇒ at point (5) (plugging mode) (generating mode)

$\omega < 0 , T_m > 0 , I_a > 0$

$P_1 > 0 , P_2 < 0$

$\Sigma \Delta P = P_1 + P_2$ خسائر كهربائية عالية على شكل حرارة



* motor operate as generator inseries to the source

توالي

ملاحظات مهمة :-

① * لتفهم عمل المحرك (motor, regenerative, plugging) ارجع للملاحظة رقم ٨

② يأتي السؤال على هذا الموضوع بطريقة الشرح بحيث يذكر النقطة ويطلب اسم النقطة وشرح عنها كما تم ذكره

ملاحظة اخرى

سيتم شرح نفس النقاط والشرح في محركات ال AC

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Example (1): shunt dc motor, 200V, 2000rpm, $R_a = 0.5\Omega$, $R_F = 400\Omega$

$T_L = T_m = \text{constant}$.

⇒ Find the speed motor if the source voltage drop to 175?

$\phi \propto I_s \propto V_a$

بما ان المحرك (shunt DC) فأن ملفات التهييج متصلة مع الجهد الرئيسي مما يؤدي الى

$$\frac{\phi_n}{\phi^*} = \frac{V_{an}}{V_a^*} = \frac{200}{175}$$

$$\phi^* = 0.875\,\phi_n$$

$T_m = T_L = \text{constant}$

$(K\phi)^* I_a^* = (K\phi)_n I_{an}$

but $I_{au} = I_s - I_f = 10.5 - \frac{200}{400} = 10A$

$I_a^* = 11.4A$

$E_{an} = V_{an} - I_{an} R_a = 195V$

$E_a^* = V_a^* - I_a^* R_a = 169.3A$

$E_a = K\phi\omega$

$$\frac{E_{an}}{E_a^*} = \frac{K\phi_n \omega_n}{K\phi^* \omega^*} \longrightarrow n^* = 1984{,}5 \text{ rpm}$$

ملاحظة

① في الامتحان هناك أسئلة تعويض مباشر لكنه تعتمد على فهمك في المعلومات التي تم شرحها في بداية الموضوع

② الاسئلة التي يتم فيها تغيير معطى معين ونريد التغير الحاصل في جزء معين، تتطلب منك اتقان المعلومات التي شرحت بحيث تكون قادر معرفة القيم التي تغيرت والتي بقيت ثابتة

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Example (2) DC series motor, 230V, 25A, 750 rpm, $R_a = 0.5\Omega$, $R_{sf} = 1.5\Omega$

Find ① $I_{sc} = \frac{V_{an}}{R_a + R_{sf}} = 115\Omega$

$(K\phi)_n = \frac{V_{an} - I_{an}R_a}{\omega_n} = 2.293 V.s$

② $T_n = (K\phi)_n \cdot I_{an} = 57.325 N.m$

but $\omega_n = \frac{2\pi n}{60} = 78.5 rad/s$

③ $T_{st} = (K\phi)_{st} \cdot I_{sc} = 1213 N.m$

$\frac{(K\phi)_{st}}{(K\phi)_n} = \frac{I_{sc}}{I_n} \Rightarrow (K\phi)_{st} = 10.55 V.s$

Example (3):- DC separatly exicted motor has, 220V, 500A, 600 rpm

$R_a = 0.02\Omega$

$\omega_n = \frac{2\pi n}{60} = 62.8 rad/s$

(Part A) Find ① no.load speed (ω_o)

$\omega_o = \frac{V_{an}}{(K\phi)_n} = \frac{220}{3.344} = 65.79$

$(K\phi)_n = \frac{V_{an} - I_{an}R_a}{\omega_n} = 3.344 V.s$

② the starting torque (T_{st})

$T_{st} = (K\phi)_n I_{st} = 3.344 \times \frac{220}{0.02} = 36784 N.m$

③ The angular speed when the sys. drive a load with ($T_L = 0.5 T_n$)

$\omega = \frac{V_{an}}{(K\phi)_n} - \frac{I_a^* R_a}{K\phi} = 64.29 rad/s$

$T_L = 0.5 T_n$

$(K\phi)_n I_a^* = 0.5 (K\phi)_n I_{an}$

$I_a = 250 A$

④ The nominal effecincy of the motor

$\eta = \frac{P_{2n}}{P_{in}} = 95.5\%$

$P_{in} = V_{an} I_{an} = 110 KW$

$P_{2n} = T_n \omega_n = 105.002 KW$

$= (K\phi)_n I_n \omega_n$

(Part B) ⑤ explan the relationship between the droop speed and load torque in DC series motor

No Relationship $\rightarrow \Delta\omega = \frac{R_a + R_{sf}}{K^2}$

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DC Motors Characteristics- Problems

1. A 220V, 100A, 750 rpm separately excited DC motor has $R_a = 0.1\Omega$. The motor is used to drive a load, whose torque is expressed as $T_L = 500 - 0.3n, N.m$, where n is the motor speed in rpm.

2. Calculate the armature current and speed when the armature voltage is reduced to 110V. What will be the drop in speed at full load and 110V armature voltage?

3. A DC series motor has the following ratings: 200V, 15A, and 300 rpm. The armature and field resistances are 1.5Ω and 2.5Ω, respectively. If the magnetic circuit is linear, calculate the nominal torque and starting torque.

4. A DC series motor has the following ratings: 200V, 15A, and 900 rpm. The armature and field resistances are 1.5Ω and 2.5Ω, respectively. The motor is used to drive a load with constant torque $I_L = 0.5I_n$. If the magnetic circuit is linear, calculate the motor torque and speed for $V_a = 0.9V_n$.

5. A DC series motor has the following ratings: 230V, 25A, and 750 rpm. The armature and field resistances are 0.4Ω and 0.6Ω, respectively. The motor is used to drive a load with fan characteristic. If the magnetic circuit is linear, calculate the nominal torque and

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8. A 500V, 37.3kW, 1000 rpm DC shunt motor has nominal efficiency= 0.9. The armature circuit resistance is 0.24Ω. The field current is 1.8A. Determine the full-load drawn current.

9. A 250V shunt motor runs at 1000 rpm at no-load and takes 8A. The total armature and shunt field resistances are respectively 0.2Ω and 250Ω. Calculate the speed when loaded and taking 50A, assume the flux to be constant.

Prof., Dr. Hussein Sarhan

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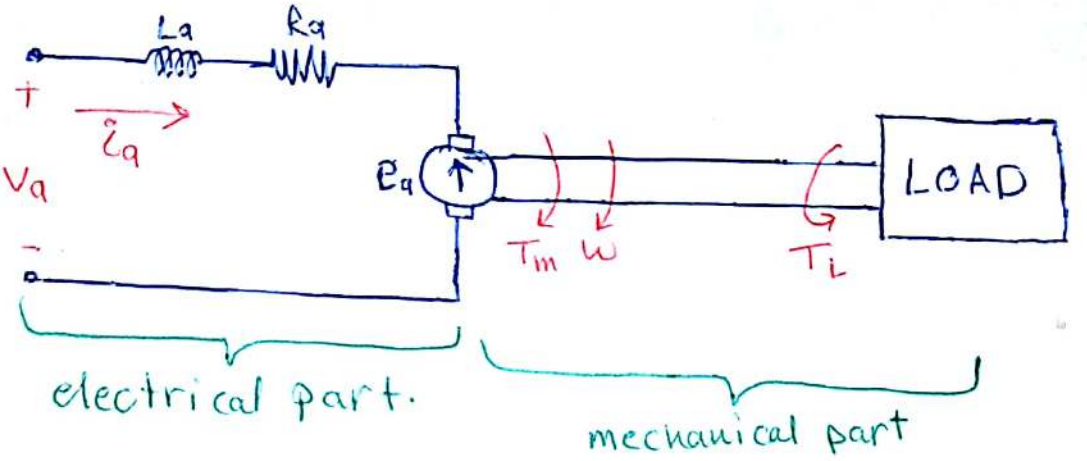
*Dynamic model of DC motor (Transient operation)



* في هذه الموضوع نريد تمثيل العلاقة بين الجهد (Va) والسرعة (w) عن طريق بلوكات (blocks) لحتى في مادة التحكم نقوم دراسة الاستجابة للسرعة بناءً على بعض من المعطيات سوف نراها لاحقاً

* لإشتقاق العلاقة ← هنالا جزئين ميكانيكي وكهربائي سوف نشتق الكهربائي ومن ثم الميكانيكي

* العلاقات التي تهمني في الإشتقاق (الإشتقاق حفظ)

⇐ قانون لابلاس ← $\mathcal{L}\{y''(t)\} = s^n Y(s)$

← في الجزء الكهربائي ← جهد الحاثة. $V_L = L \frac{di}{dt}$

قانون أوم $V = i \cdot R$

ثابت الزمن الكهربائي $\tau_e = \frac{La}{Ra}$

← في الجزء الميكانيكي ← معادلة الحركة $T_m - T_L = J_2 \frac{dw}{dt}$

قانون العزم $T = (K\phi)\, Ia$

قانون Eq $Ea = K\phi w$

ثابت الزمن الميكانيكي $\tau_m = \frac{J_2 \cdot Ra}{(K\phi)^2}$

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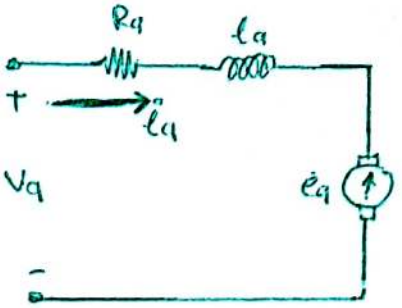
⇒ For electrial part.



$$V_a = L_a \frac{d i_a(t)}{dt} + i_a \cdot R_a + e_a$$

→ for lap iace transfer.

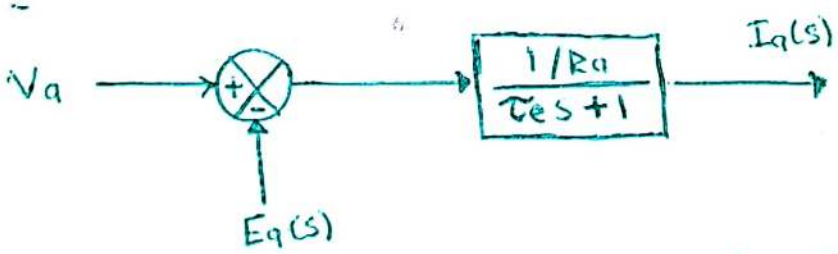
$$\rightarrow V_a(s) = L_a \cdot s \cdot I_a(s) + R_a I_a(s) + E_a(s)$$

$$\rightarrow V_a(s) - E_a(s) = I_a(s)(L_a s + R_a)$$

$$\rightarrow \frac{V_a(s) - E_a(s)}{I_a(s)} = L_a s + R_a$$

$$\rightarrow \frac{I_a(s)}{V_a(s) - E_a(s)} = \frac{1}{L_a s + R_a}$$ → but $\tau_e = \frac{L_a}{R_a}$

$$W_1(s) = \frac{I_a(s)}{V_a(s) - E_a(s)} = \frac{1/R_a}{\tau_e s + 1}$$ → Transfer function for electrical circuit.



"Block digram of DC motor Electrical part"

((first order time lag element))

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* For mechanical part

$$T_m - T_L = J_\epsilon \frac{dw}{dt}$$

$$T = K\phi I$$

$$K\phi (I_a - i_L) = J_\epsilon \frac{dw}{dt}$$

→ laplace transfer.

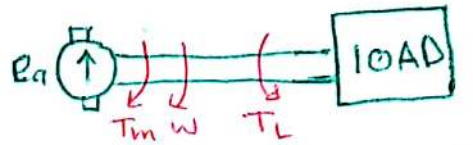
$$K\phi (I_a(s) - I_L(s)) = J_\epsilon \cdot s \Omega(s)$$

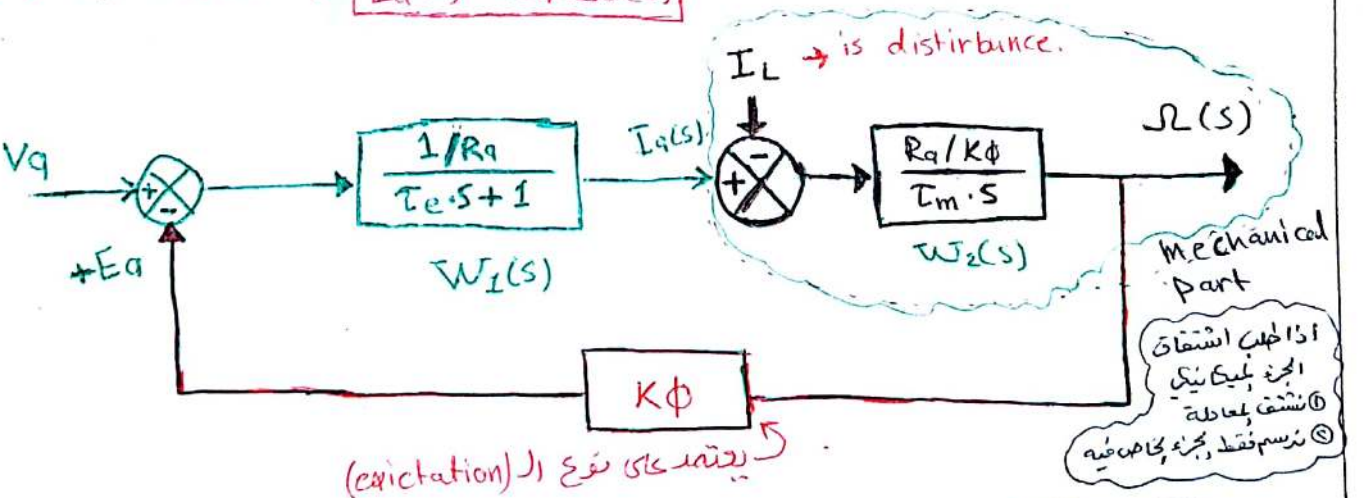
$$\frac{\Omega(s)}{I_a(s) - I_L(s)} = \frac{K\phi}{J_\epsilon \cdot s}$$

but $\tau_m = \frac{J_\epsilon R_a}{(K\phi)^2} \rightarrow J_\epsilon = \frac{\tau_m (K\phi)^2}{R_a}$

$$W_2 = \frac{\Omega(s)}{I_a(s) - I_L(s)} = \frac{R_a/(K\phi)}{\tau_m \cdot s}$$ → Transfer function of mechanical part.

$$\Rightarrow e_a = K\phi w \rightarrow E_a(s) = K\phi \Omega(s)$$





← ملاحظة مهمة:

* بأي السؤال عن هذا الموضوع بشكل اشتقاق اما اشتق الجزء الميكانيكي أو الكهربائي

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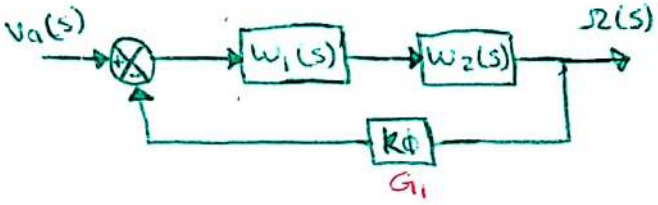
* For shunt or separatly exicted DC motor

→ apply superposition principle

• case ① $V_a(s) \neq 0 \quad I_L(s) = 0$

$$W_A(s) = \frac{\Omega(s)}{V_a(s)} = \frac{W_1 W_2}{1 + W_1 W_2 G_1}$$

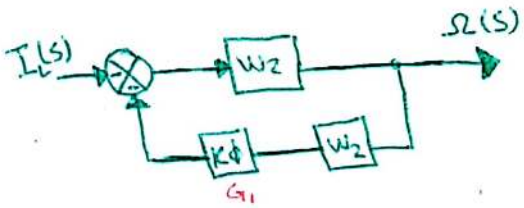
$$= \frac{1/K\Phi}{\tau_m \tau_e s^2 + \tau_m s + 1}$$



• case ② $V_a(s) = 0 \quad I_L(s) \neq 0$

$$W_B(s) = \frac{\Omega(s)}{-I_L(s)} = \frac{W_2}{1 + W_1 W_2 G_1}$$

$$= \frac{\frac{R_a}{K\Phi}(1 + \tau_e s)}{\tau_e \tau_m s^2 + \tau_m s + 1}$$



$\tau_e \tau_m s^2 + \tau_m s + 1 = 0$ → Chareristeristic equation of system

⇒ The Dynamic depend on the roots of char. equation. (s_1, s_2)

$$s_{1,2} = \frac{-\tau_m \mp \sqrt{\tau_m^2 - 4\tau_m\tau_e}}{2\tau_m\tau_e} \qquad D = \tau_m^2 - 4\tau_m\tau_e$$

① $D > 0 \longrightarrow \tau_m^2 - 4\tau_m\tau_e > 0 \longrightarrow \frac{\tau_m}{\tau_e} > 4$

→ s_1, s_2 :- real number, unequal Root

→ overdamped response.

② $D = 0 \longrightarrow \frac{\tau_m}{\tau_e} = 4$

→ s_1, s_2 :- Real, equal roots

→ critically damped response

③ $D < 0 \longrightarrow \frac{\tau_m}{\tau_e} < 4$

→ $s_{1,2} = -\sigma \pm j\Omega_0$

→ oscilletory response (الاستجابة التذبذبية)

→ $\sigma = 1/2\tau_e$ damping Factor

→ $\Omega_0 = \frac{\sqrt{\tau_m^2 - 4\tau_e\tau_m}}{2\tau_e\tau_m}$ Frequancy of free oscillation

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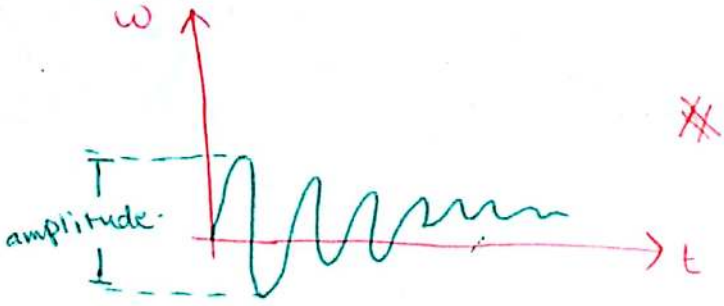
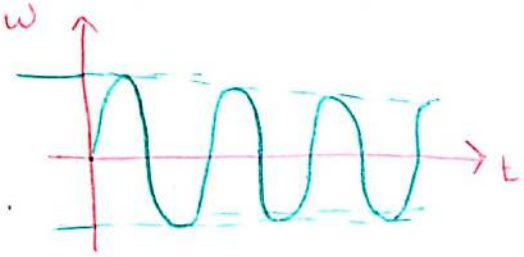
⇒ oscillatory response تخامد

① oscillations with damped (decaying) amplitude



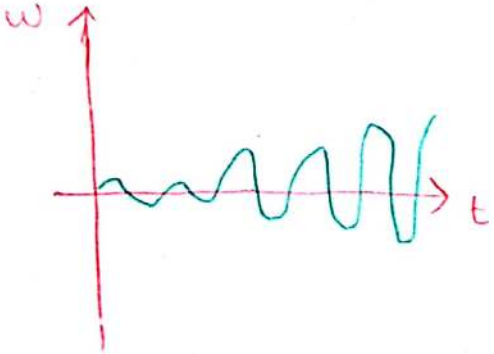
※ stable system.

② oscillations with constant amplitude



※ conditionary stable

③ un damped oscillation / with increase amplitude



※ unstable system.

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※Methods of transient operation of DC motor:-

نظرة عامة

① starting operation
- Direct start
- indirect start.

② ② Braking operation
- shunt / separatly exicted
  - Mechanical Braking
  - Electrical Braking
    - Dynamic
    - plugging
    - Regenerative
- series motor
  - mechanical Braking
  - Electrical Braking

③ speed control
- shunt / separatly exicted
  - mechanical method
  - Electrical method.
- series.

* في هذه الموضوع سوف ندرس كل حالة (آلية تطبيقها، تأثيرها على كل من
(static characteristic) الخاصية الطابعية ← الخاصية المتغيرة ←
(Dynamic ")
* موضوع (starting, Braking) مهم جدا آخر موضوع وهو (power and control circuit)

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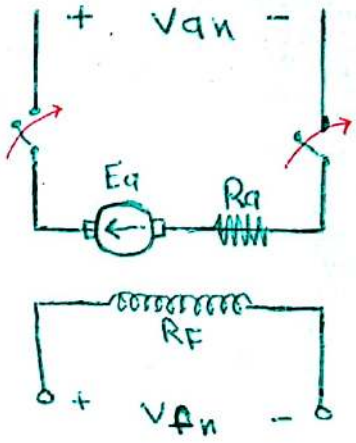
① starting Methods

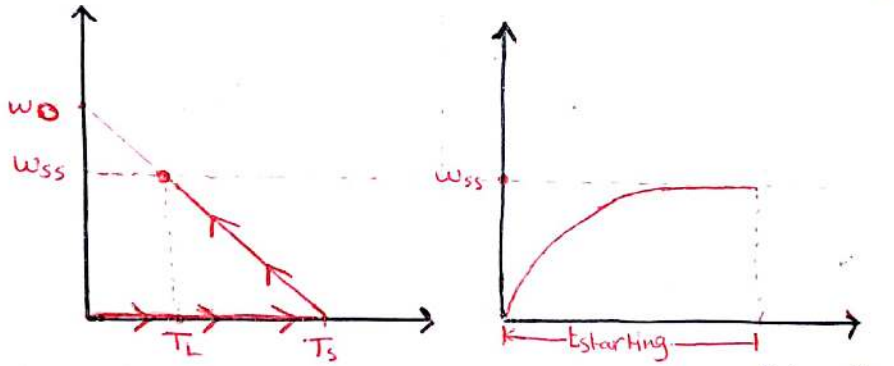
Ⓐ Direct start



⇐ تتم هذه الطريقة بدون أضافة اي وسائل في الدارة لتقليل تيار البدء

$I_{sc} = 20 I_n$ ⇐ تستخدم هذه الطريقة في المحركات الصغيرة بسبب

$\Delta P_{cu(st)} = I_{sc}^2 R_a \gg \Delta P_{cu}$ مما يعني وجود خسائر كبيرة والتي تكون على شكل حرارة heat losses.

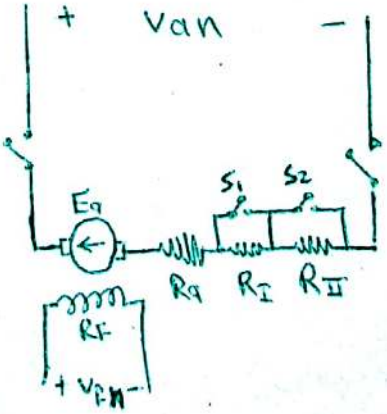


Static charisterstic Dynamic charisterstic

Ⓑ indirect start. → used to reduce ($I_{sc} \simeq 2-3 I_{an}$)

1) Insert additional Risistance (Radd) ⇒ (Reostatic start)

⇐ في هذه الطريقة يتم اضافة مقاومات على التوالي مع مقاومة الملف (R_a) خلال عملية البدء ثم فصل هذه المقاومات عندما يصل المحرك عند سرعة الاستقرار (w_{ss})



⇐ تكملة الشرح وطريقة حساب المقاومات في الصفحة التالية

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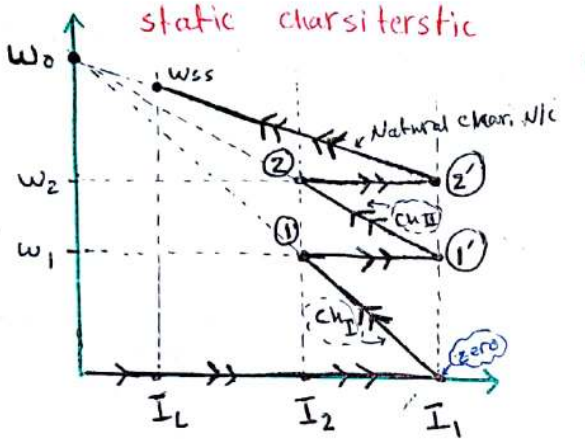

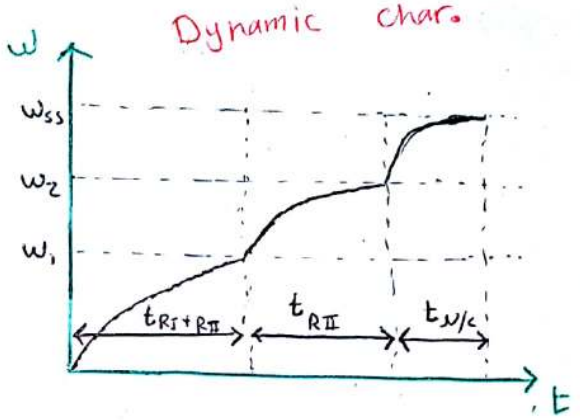

⇒ I_1 :- Maximum safe sh.cir current أعلى تيار آمن عند التشغيل أو sh.cir

⇒ I_2 :- switching current (Remove Radd) التيار الذي يتم فصل المقاومة إزالة عنده

⇒ $Char._I$, $Char._{II}$:- Regulated characterstic (R/C)

⇐ آلية العمل :- ① يتم تشغيل المحرك بوجود R_I ، R_{II} اي (S_1 and S_2 O.C) فعند وصول التيار I_1 تبدأ السرعة بالتزايد على ($Char._I$) وقيمة I_1 هي

$$I_1 = \frac{V_{an}}{R_a + R_I + R_{II}}$$

② عند وصول التيار لـ I_2 يتم اغلاق المفتاح S_1 اي حذف المقاومة R_I فتنتقل السرعة الى الخاصية $Char_{II}$ وتبدأ بالتزايد

③ عند وصول التيار مرة اخرى لـ I_2 يتم اغلاق المفتاح S_2 اي حذف المقاومة R_{II} فتنتقل السرعة الى الخاصية الطبيعية (N/c) حتى تصل الى (ω_{ss})

* طريقة حساب $R_{II} + R_I$

① عند نقطة التشغيل zero

$$I_1 = \frac{V_{an}}{R_a + R_I + R_{II}} \longrightarrow R_I + R_{II} = \frac{V_{an}}{I_1} - R_a$$

② مساواة السرعة في النقطة ① و ①'

$$\omega_1 = \omega_{1'}$$

(أثر الاستنتاج)

$$\frac{V_{an}}{K\phi} - \frac{I_2(R_a + R_I + R_{II})}{K\phi} = \frac{V_{an}}{K\phi} - \frac{I_1(R_a + R_{II})}{K\phi}$$

$$\hookrightarrow I_2(R_a + R_I + R_{II}) = I_1(R_a + R_{II}) \longrightarrow I_2 R_I + (I_2 - I_1) R_{II} = (I_1 - I_2) R_a$$

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ملاحظات

(two-step reostatic starting) ① ماتم شرحه هو بأضافة مقاومتين وهذا يدعى
و القواعين التي في الصفحة السابقة فقط لهذه الحالة

② فأذا كانت مقاومة واحدة فنحتاج فقط معادلة واحدة وهي عند التشغيل.

③ واذا كانت ٣ مقاومات نحتاج ٣ معادلات (غالباً ما ينتج بالامتحان) بين للعلم.

---

ex DC separatly exicted motor, 220V, 172A, 1000rpm, $R_a = 0.062$

one-step reostatic starting, $I_1 = 350A$, $I_2 = 250A$

① Find R_{add}.

$$I_1 = \frac{V_{an}}{R_a + R_{add}} \Rightarrow R_{add} = \frac{V_{an}}{I_1} - R_a = \frac{220}{350} - 0.062 = 0.567\,\Omega$$

② Find w_1 (switching speed)

$$w_1 = \frac{V_{an} - I_2(R_a + R_{add})}{(K\Phi)_n} = \frac{220 - 250(0.062 + 0.567)}{1.99} = 31.48 \text{ rad/s}$$

$$w_n = \frac{2\pi n}{60} = 104.72 \text{ rad/s}$$

$$(K\Phi)_n = \frac{V_{an} - I_{an} R_a}{w_n} = 1.99 \text{ V.s}$$

---

ex2 DC-separatly exicted motor, 100V, 20A, 1000rpm, $R_a = 0.05\,\Omega$

2-step reostatic start $I_1 = 3I_n$, $I_2 = 2I_n$

① Find R_I, R_{II} $I_1 = 60A$ $I_2 = 40A$

بالامتحان حلها عنه طريق الاشتقاق

$$R_I + R_{II} = \frac{V_{an}}{I_1} - R_a \Rightarrow R_I + R_{II} = \frac{97}{60}$$

$$I_2 R_I + (I_2 - I_1) R_{II} = (I_1 - I_2) R_a \Rightarrow 40R_I - 10R_{II} = 1$$

$$R_I = 0.555\,\Omega$$
$$R_{II} = 1.061\,\Omega$$

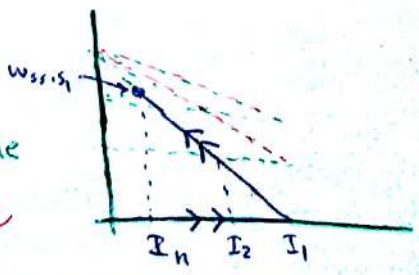
② if S_1 (Switch1) fault to close?

w↑ up to $w_{ss.s1}$ if S_2 fault

* if S_2 close

w will move at new char. depend on the value of R_I

يتشغل w على خاصية جديدة تعتمد على قيمة R_I



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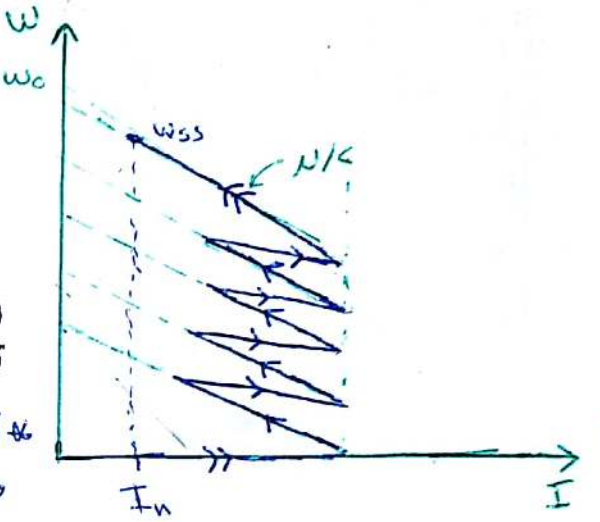
2) Reduce Terminal voltage (V_a)

$$\downarrow I_{sc} = \frac{V_a \downarrow}{R_a}$$

$$\omega = \frac{V_a}{(K\phi)_n} - \frac{I_a R_a}{(K\phi)_n} = \omega_o - \Delta\omega$$

$\Delta\omega \rightarrow$ constant $\rightarrow$ لذلك كل المنحنيات سوف تكون متوازية مع N/L

$\omega_o \rightarrow$ virable



* تستخدم لطريقة smooth starting . وتستخدم في حالة الحمل أو اذا كان الحمل أقل ما يمكن light load. unloaded

* Braking methods

① ⇒ of DC separaty exicted motor

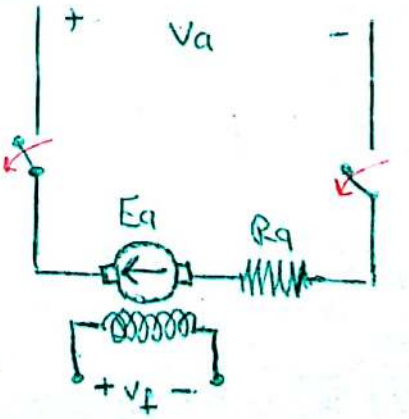
Braking بشكل عام يتم عكس T_d وهذا يكون اذا كانت $T_d < 0$
① $T_m = 0 \rightarrow$ mech. Braking
② $T_m < 0$] electrical Braking
③ $T_m < T_L$]

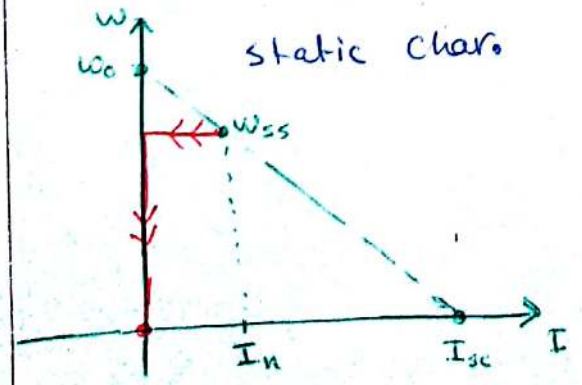
① mechanical Braking ($T_m = 0$)

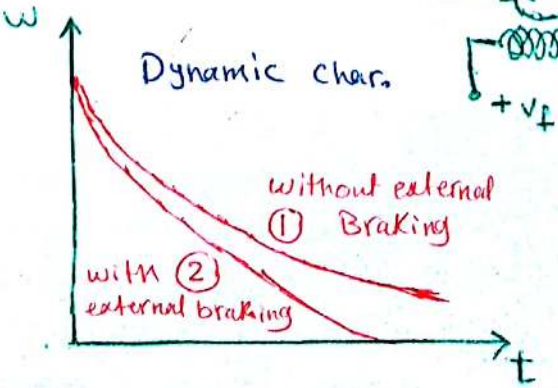
← في هذه الحالة يتم فصل الحمل مباشرة دون أضافة اي شيء

$T_m = 0, I_a = 0 \rightarrow T_d = -T_L \rightarrow \omega\downarrow$

← يحتاج الحمل فترة طويلة للتوقف للتبديد الطاقة الحركية المخزونة لذلك نحتاج لتقليل الوقت ويتم عن طريق إضافة حمل كالي الـ(motor shaft) (حمل ميكانيكي) عندما نقترب السرعة منه لتصفر







يستخدم في small motor

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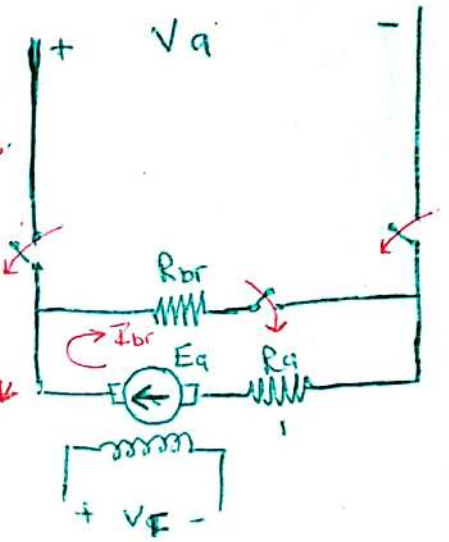
2. Electrical Braking.

(A) Dynamic Braking

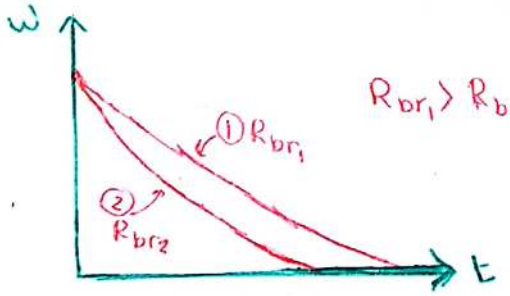
← يتم فصل التغذية عن دارة الجنتج وإضافة مقاومة R_{br} على التوازي.

$$I_{abr} = \frac{\overset{zero}{V_a} - E_a}{R_a + R_{br}} = \frac{-E_{abr}}{R_a + R_{br}} < 0 \qquad V_a = 0$$

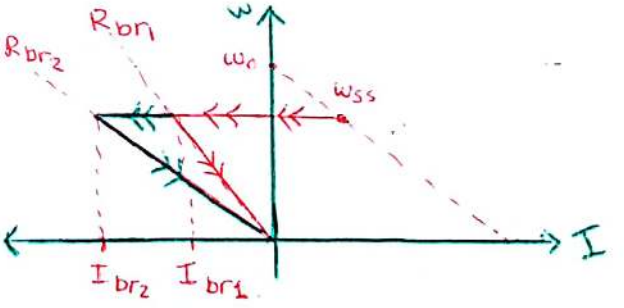
$$* \; T_{m_{br}} = (K\phi)_n \, I_{abr} < 0$$



← سبب إضافة R_{br} هو تقليل تيار الفصل (I_{abr}) وكلما زدنا R_{br} يقل تيار (I_{br}) لكنه يزداد زمن التوقف



$R_{br_1} > R_{br_2}$



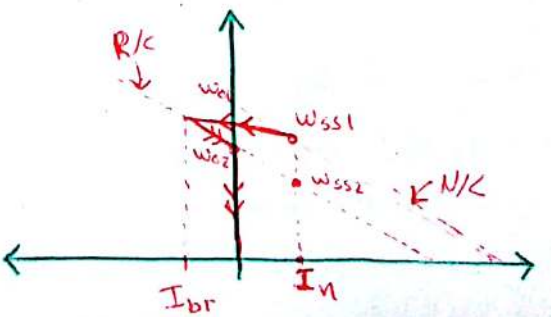
(B) Regenerative Braking.

$T_m < 0 \rightarrow T_d < 0 \rightarrow w \downarrow$

$$I_a = \frac{V_a - E_{abr}}{R_a} < 0 \quad \text{because } E_a > V_a$$

← الطريق الانتقال من ω_{ss1} إلى ω_{ss2}

لذلك تنتقل السرعة من N/C إلى R/C ثم تبدأ بالتباطؤ حتى تصل إلى ω_{ss2} لكنه عندما تصبح $\omega \simeq \omega_{ss2}$ نقوم بفصل المحرك (عندما يكون $I_a = 0$)



← تستخدم هذه الطريقة عندما نطبق على المحرك سرعة أقل من السرعة التي يعمل عليها

← تمتاز هذه الطريقة أنها ذات كفاءة عالية وتبدل الطاقة الحركية إلى طاقة كهربائية ونستفيد منها بدل ما تتحول لخسائر.

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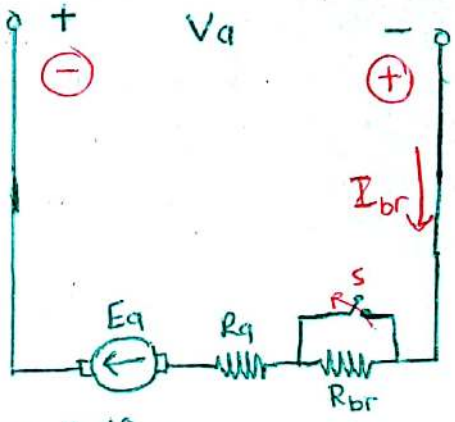
ⓒ plugging braKing



← في هذه الطريقة يتم عكس قطبية المحرك

$I_{a\,br} = \frac{-V_a - E_{a.br}}{R_a + R_{br}} < 0$, $T_m < 0$, $w \downarrow$

← نقوم بإضافة R_{br} لتظليل تيار I_{br} لأنه عالي جداً

← آلية الإغلاق

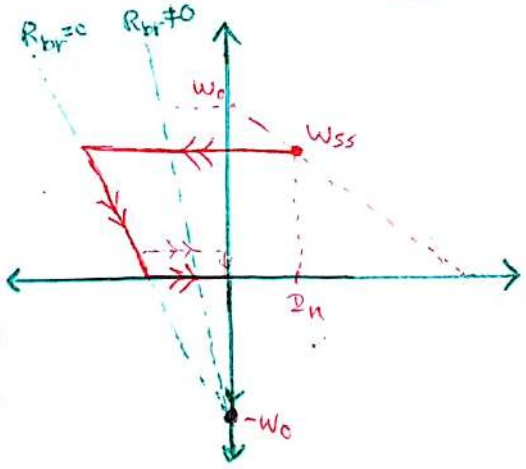
← عند عكس القطبية تنتقل السرعة إلى ناحية امتدادها من $(-w_o)$ ثم تبدأ السرعة بالتناقص وعندما تصبح $(w \simeq 0)$ نقوم بفصل المحرك

← تقترب عند الصفر

وذلك لمنع المحرك من الدوران بالاتجاه المعاكس

← مميزاتها ← زمن الفصل قليل

← سلبياتها ← خسائر كبيرة أثناء الإغلاق (braking)



ملاحظات مهمة

① حاول قرب المفاهيم مع المحتوى لأنه رح نمر عليها في AC motor ، series

- معناها بفصل إلى فاتح مباشرة من التيار والسرعة تقل مباشرة
- تنتقل السرعة ← إلى ناحية مشابهة لها امتدادها من الصفر

⇒ mechanical Braking :-

⇒ Dynamic Braking :-

- تنتقل السرعة إلى ناحية مشابهة لها امتدادها من $(-w_o)$
- تتميز بهذا الحالة الانتقال إلى ناحية ← مقاومة لها بسرعة أقل

⇒ blugging Braking :-

⇒ Regenerative Braking :-

② علاقة الـ I_{br} مطلوب حفظها لأنها حرفياً بتيجي بسؤال (فرع) ☺

③ $E_{q_{br}}$ ليس بالضرورة أن تكون مساوية E_q وذلك يعتمد على السرعة (w) عندما يكون الـ braking

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ex/ DC separatly exicted motor, 220V, 172A, 1000 rpm, $R_a = 0.062\Omega$

→ For dynamic Braking, $\omega_{br} = 0.5\omega_n$, $I_{br} = -I_n$

Find R_{br}.

$$I_{br} = \frac{-E_{a,br}}{R_a + R_{br}}$$

$$-172 = \frac{104.2}{R_{br} + 0.062}$$

$$\Rightarrow R_{br} = 0.544\ \Omega$$

but

$$E_{a_{br}} = (K\phi)_{br}\,\omega_{br}$$

$$(K\phi)_{br} = (K\phi)_n = \frac{V_{an} - I_{an}R_a}{\omega_n} = \frac{220 - 172 \times 0.062}{\frac{2\pi \times 1000}{60}} = 1.99\ V.s$$

$$\omega_{br} = 0.5\omega_n = 52.36\ rad/s$$

$$\Rightarrow E_{abr} = 104.2\ V$$

---

ex/ DC separatly exicted motor, 220V, 970 rpm, 100A, $R_a = 0.0525\ \Omega$.

Plugging Braking, $n_{br} = 1000$ rpm.

① Find R_{br}

$$I_{br} = \frac{-(V_a + E_{br})}{R_a + R_{br}}$$

but → $\frac{E_{br}}{E_{an}} = \frac{(K\phi)_n\,\omega_{br}}{(K\phi)_n\,\omega_n}$ ($n = \frac{60\omega}{2\pi}$) → but → $E_{an} = V_{an} - I_{an}R_a$

$$= 220 - 100 \times 0.0525 = 214.75\ V$$

$$-100 = \frac{-220 - 221.4}{0.0525 + R_{br}}$$

$$E_{br} = \frac{n_{br}}{n_n} E_{an} = 221.4\ V$$

$$\Rightarrow R_{br} = 2.154\ \Omega$$

② Find $T_{br} = (K\phi)_n I_{br} = \frac{E_{an}}{\omega_n} \cdot I_{br} = \frac{214.75}{\frac{2\pi \times 970}{60}} \times 200 = -422.83\ N.m$

③ Find The torque at zero speed.

$$T_{(\omega=0)} = (K\phi)_n I_{br(\omega=0)}$$

but → $I_{(\omega=0)} = \frac{-V_a - E_{abr}}{R_a + R_{br}}$ (zero → $\omega = 0$)

$$= -99.7\ A$$

$$T_{(\omega=0)} = \frac{E_{an}}{\omega_n} \cdot I_{(\omega=0)} = -210.78\ N.m$$

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* Braking methods

② ⇒ of DC series motor

1. Mechanical Braking

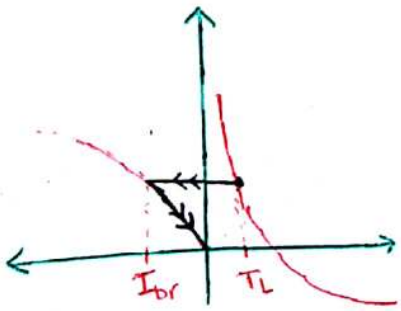
DC separatly exicted ⇐ * نفس شرح الـ

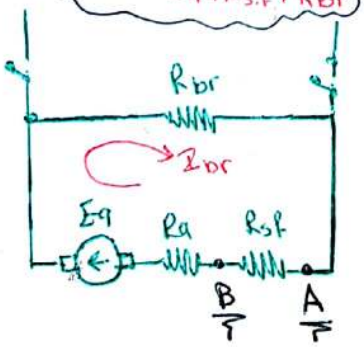
2. Electrical Braking

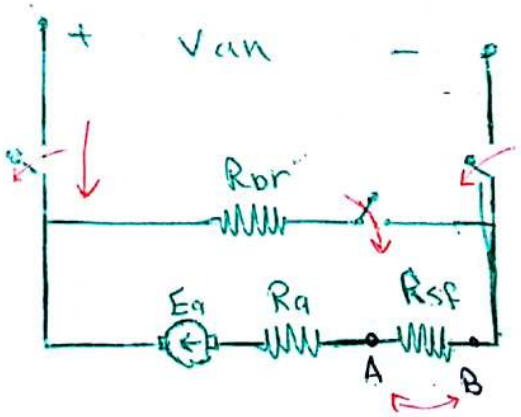
Ⓐ Dynamic Braking ① self-exictation

* في هذه الحالة يتم فصل المصدر وعكس اتجاه ملف المجال ليبقى التيار بنفس الاتجاه

$$I_{br} = \frac{E_{q.br}}{R_q + R_{s.f} + R_{br}}$$



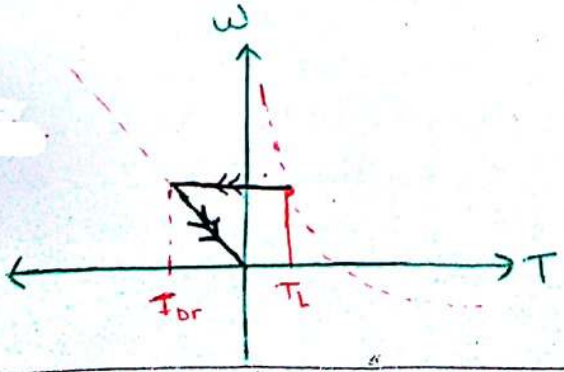


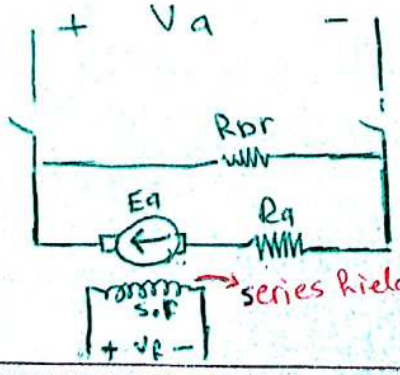


② separatly-exictation

⇐ في هذه الحالة يتم فصل المصدر مع دارة المنتج (Armture) مع بقاء ملفات المجال موصولة

* يعني زي الـ separatly أو الـ shunt





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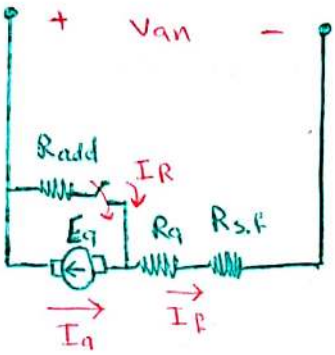
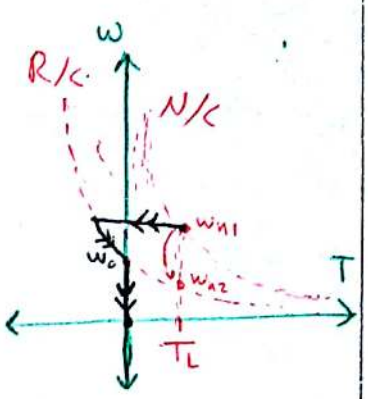
(B) Regenerative braking

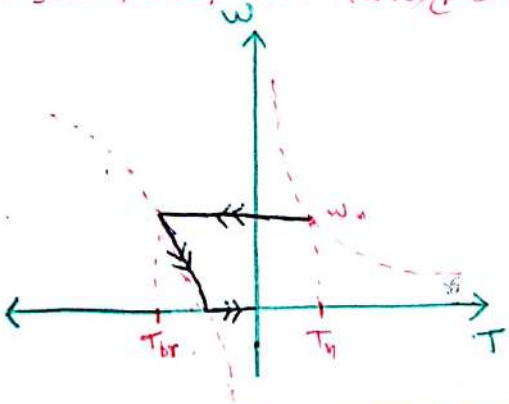
* عند إضافة مقاومة على التوازي مع المنتج فأن $(I_f = I_R + I_a)$ وهذا يعني ان (I_f) سوف يزداد وبالتالي يزداد الفيض المغناطيسي وبالتالي سوف تقل السرعة. وعندما تصل $(w = w_{o2})$ تقوم بفصل التغذية

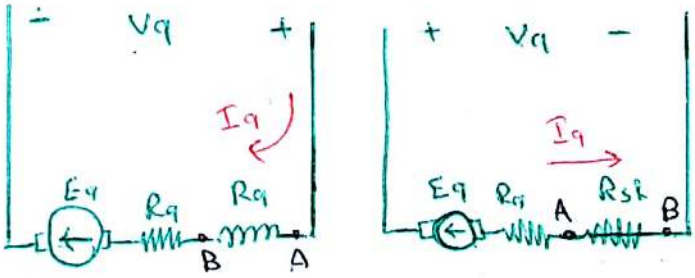




(C) Plugging-braking

* في هذه الحالة تقوم بعكس قطبية المصدر وفي الوقت نفسه عكس قطبية ملف المجال (field) ليبقى التيار في الاتجاه نفسه وعندما تصبح $(w = 0)$ تقوم بفصل التغذية.





ex DC series motor, 220V, 1000 rpm, 100A, T_L-constant, $R_a = R_{sf} = 0.05\Omega$

* For Dynamic braking, $T_{br} = 2T_n$, $n_{br} = 800$ rpm, Find R_{br}.

$$I_{br} = \frac{E_{a,br}}{R_a + R_{sf} + R_b}$$

$$141.4 = \frac{237.5}{0.1 + R_{br}}$$

$$R_{br} = 1.58\Omega$$

ثابت التناسب

$$\frac{T_{br}}{T_n} = \frac{K^* I_{br}^2}{K^* I_n^2} = 2 \rightarrow I_{br} = \sqrt{2 I_n^2} = 141.4A$$

$$\frac{E_{abr}}{E_{an}} = \frac{K^* I_{br} w_{or}}{K^* I_n \cdot w_n} \rightarrow E_{abr} = \frac{n_{br} I_{br}}{n_n I_n} E_{an} = 237.5$$

$$E_{an} = V_{an} - I_{an}(R_a + R_{sf}) = 210V$$

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✱ speed control of DC motor

⇒ in shunt or separatly exicted DC motor

ⓐ mechanical method (using gear)

ⓑ Electrical method.

$\omega = \frac{V_a - I_a R_a}{K\phi}$ Then ω effect by V_a & R_a & ϕ

1- add R_{add} in series with the armture :-

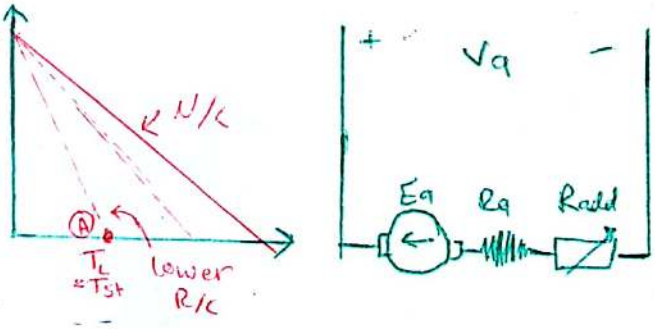
$R_{add} \uparrow$, $(V_a$ & $\phi)$ - constant.

$\omega_o = \frac{V_a}{K\phi}$ = constant , $I_{s.c} = \frac{V_a}{R_a + R_{add}\uparrow} \downarrow$, $\Delta\omega \uparrow = I_a \frac{R_a + R_{add}\uparrow}{K\phi}$

For point Ⓐ holding point

$T_L = T_{st} = K\phi I_a$

$= K\phi \left(\frac{V_a}{R_a + R_{add}}\right)$



Properities :- $0 < \omega < \omega_n$, SS↓ , β↓ , Δω↑ , η↓ , step control

⇒ $R_{add} = \left(\frac{K\phi \cdot V_a}{T_L}\right) - R_a$

قانون آخر ← $R_{add} = \frac{V_a}{I_a} - R_a$ ← مشتق من قانون s.h.c

وهنا في حال T_L = const.

ملاحظة اذا كانت $T_L \neq$ const نجد قيمتها عند $\omega = 0$ ونعوض في القانون الاول

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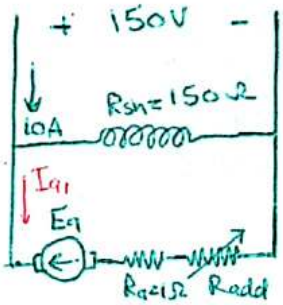
ex| shunt dc motor, 150 V, 1200 rpm, $R_a = 1\Omega$, $R_{sh} = 150\Omega$, $I = 10A$

$T_L = T_n$ = constant



① Find Radd for $n^* = 600$ rpm

عند تغيير السرعة E_a تتغير فقط اذا كانت T = const. الـ

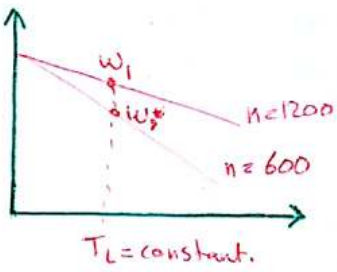
قبل تغيير السرعة:

$$I_{a1} = I - I_f = 10 - \frac{150}{150} = 9A$$

$$E_{a1} = V_a - I_{a1} R_a = 150 - 9*1 = 141\ V$$

سوف تتأثر بتغير التيار ⇒ $\frac{T_{L1}}{T_L^*} = \frac{(K\Phi)_n I_{a1}}{(K\Phi)_n I_a^*}$ but $T_{L1} = T_L^* =$ constant

$$I_a^* = I_{a1} = 9A$$



$$\frac{E_{a1}}{E_a^*} = \frac{(K\Phi)_n \omega_1}{(K\Phi)_n \omega^*} = \frac{n_1}{n^*}$$

$$E_a^* = \frac{n^*}{n_1} E_{a1} = \frac{600}{1200} * 141 = 70.5\ V$$

$$\Rightarrow E_a^* = V_a - I_a^*(R_a + R_{add}) \rightarrow R_{add} = \frac{V_a - E_a^*}{I_a^*} - R_a = 7.833\Omega$$

② Find the efficiency For each, if $\Delta P_{constant} = 100W$

$P_1 = V_a \cdot I = 150 * 10 = 1500W$

$$\eta_1 = \frac{P_1 - \Sigma\Delta P_1}{P_1} * 100\% = 77.93\ \%$$

$\Sigma\Delta P_1 = \Delta P_F + \Delta P_C + \Delta P_a = I_F^2 \cdot R_F + \Delta P_C + I_a^2 R_a$

$= (1)^2 * 150 + 100 + 9^2 * 1 = 331W$

$$\eta_2 = \frac{P_1 - \Sigma\Delta P_2}{P_1} * 100\% = 35.63\ \%$$

$\Sigma\Delta P_2 = \Delta P_F + \Delta P_C + I^2(R_a + R_{add}) = 950W$

③ Find Radd at holding point.

$$R_{add} = \frac{V_1}{I_a} - R_a = 15.667\Omega.$$

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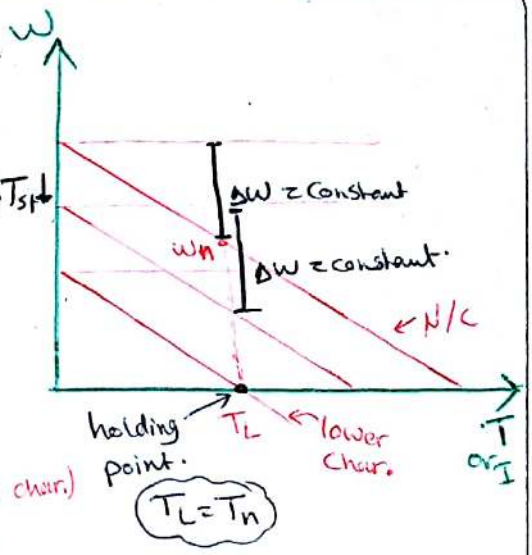
2. Variation of terminal voltage.

$V_a \leq V_{an}$, ϕ – constant , $R_{add} = 0$

$\downarrow \omega_o = \frac{V_a \downarrow}{K\phi}$, $\Delta\omega = \frac{I_a R_a}{K\phi}$ = constant , $\downarrow I_{sc} = \frac{V_a \downarrow}{R_a}$, $T_{st} \downarrow$

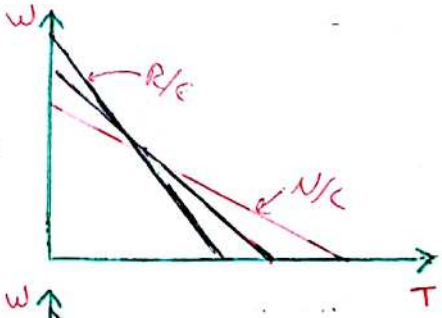


Properities ① $\Delta\omega$-const. → β constant
s.s operation don't effect
② $0 \leq \omega \leq \omega_n$ (wide range)
③ efficient method ($\eta\uparrow$)
④ smooth control (unlimited no. of char.)
⑤ below Natural char. (N/C)

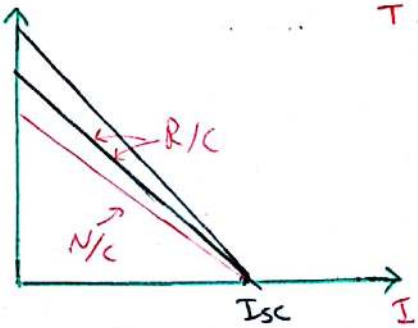
3. Variation of flux. (ϕ)

$\phi \leq \phi_n$, V_a-constant , $R_{add} = 0$.

← يتم تغيير الضوء الخافت ليس وذلك عن طريق تقليل (V_a) عن القيمة الاسمية
← لا يتم زيادة (V_a) لانها تؤدي الى منطقة الاشباع ومن ثم تلف بالغاز



$\uparrow \omega_o = \frac{V_a}{K\phi \downarrow}$, $\uparrow \Delta\omega = \frac{I_a R_a}{K\phi \downarrow}$, $\uparrow T_{st} = K\phi \downarrow I_{sc}$, $I_{sc} = \frac{V_a}{R_a}$ – constant



Properities :- ① $\Delta\omega\uparrow$, $\beta\uparrow$, effect on s.s operation
② above (N/C) $\omega \geq \omega_n$
③ efficient method ($\eta\uparrow$)
④ smooth control
⑤ $\omega_n \leq \omega \leq \omega_{max}$

لا يتأثر بتغير ϕ

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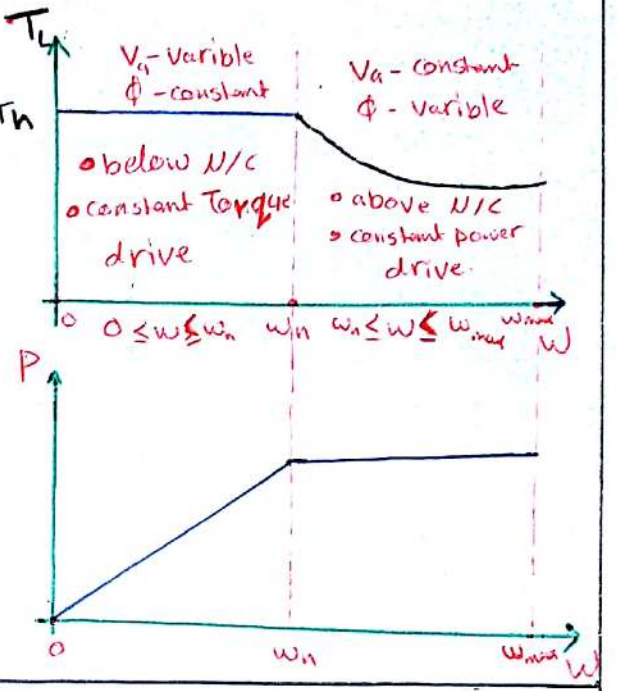
4. combined method.

- في هذه الطريقة نقوم بتغيير (ϕ) أو (V_a) وتثبيت القيمة الاخرى
- في الرسمة شرح الطريقة كاملة
- يعني أنه اذا اردنا خاصية أدنى من الخاصية الطبيعية نقوم بتغيير V_a وتثبيت ϕ
- و اذا اردنا خاصية اعلى نقوم بتغير ϕ وتثبيت V_a



ex DC separatly exicted motor, 220 V, 500 A, 600 rpm, $R_a = 0.02\ \Omega$, $R_f = 10\ \Omega$

$T_L = 2000 - 2n$, ∴ used combiend speed control

Find ① V_{a_1} and I_{a_1} when $n_1 = 450$ rpm.

$n_1 < n_n \rightarrow V$ is varble $\quad \phi_1 = \phi_n$

$T_{L1} = (K\phi) I_{a1} \rightarrow I_{a1} = \frac{T_{L1}}{(K\phi)_n}$

but $T_{L1}(450) = 2000 - 450 \times 2 = 1100\ N.m$

$\rightarrow (K\phi)_n = \frac{V_a - I_a R_a}{w_n} = 3.34\ v.s$

$I_{a_1} = 329\ A$

$\frac{E_{a1}}{E_{an}} = \frac{(K\phi) w_1}{(K\phi) w_n} = \frac{n_1}{n_n} = \frac{V_{a_1} - I_{a_1} R_a}{V_{an} - I_{an} R_a}$ ← المجهول فقط V_{a1}

$\rightarrow V_{a1} = 164\ V$

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② Find V_{R2} and I_{a2}? when $n_2 = 750$ rpm

$n_2 > n_n \longrightarrow \phi$ - Varible $\quad V = V_{an}$.

$$I_{a2} = \frac{T_{L1}}{(K\phi)_2} = \frac{500}{(K\phi)_2} \longrightarrow T_{L1} = 2000 - 2 \times 750 = 500\ N.m.$$

$$V_{a2} = V_{an} = 220 = E_a + R_a I_{a2} = (K\phi)_2 \omega_2 + R_a I_{a2}$$

$$220 = (K\phi)_2 \cdot \frac{2\pi n_2}{60} + 0.02 I_{a2} \longrightarrow \text{but} \quad (K\phi)_2 = \frac{500}{I_{a2}}$$

$$\Rightarrow 0.02 I_{a2}^2 - 220 I_{a2} + 39270 = 0$$

✓ $I_{a2_1} = 181.5\ A$ $\quad I_{a2_2} = 10818.5\ A \longrightarrow \gg I_{an} \longrightarrow \times$

اذا كان التيارين اعلى من I_{an} نحسب الكفاءة للتيارين ونختار صاحب الكفاءة الاعلى.

$$(K\phi)_2 = \frac{500}{I_{a2}} = 2.755\ V.S$$

$V_R \propto \phi$

$$\frac{V_{Rn}}{V_{R2}} = \frac{(K\phi)_n}{(K\phi)_2} \longrightarrow (K\phi)_2 = 3.34\ V.S$$

من الفرع السابق.

$$V_{R2} = \frac{(K\phi)_2}{(K\phi)_n} V_{Rn}$$

$V_{Rn} = V_{an}$

$V_{R2} = 181.3\ V$

* نصيحة
- حل الاسئلة كل DC motor يحتاج قليل من التفكير (مع ان قوانينه قليلة لكنه يحتاج لطريقة كيفية لاستخدامهم)
- لذلك لحل السؤال بطريقة سريعة وسليمة ← ضع القانون النهائي الذي سوف يتم حل السؤال عليه على زاوية الصفحة ثم ضع المجاهيل التي تريد حسابها وارسم مسار للعثور على هذه المجاهيل

لا تحفظ حل السؤال

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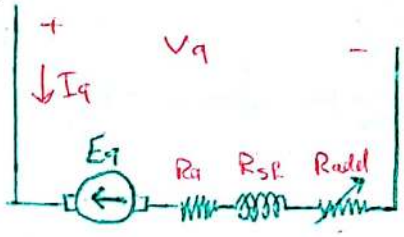
⇒ speed control in Series DC motor

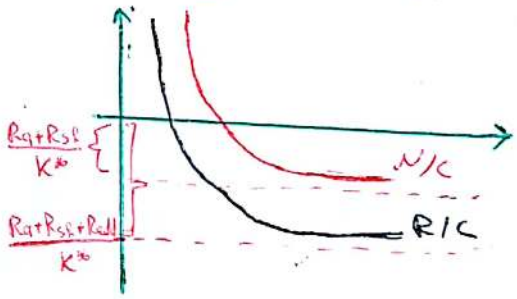
$$\omega = \frac{V_a}{K^* I_a} - \frac{R_a + R_{sf}}{K^*}$$ Parameters :- R, V_a

① adding Radd in series with armture



$$\omega = \frac{V_a}{K^* I_a} - \frac{R_a + R_{sf} + R_{add}}{K^*}$$

$\Delta\omega \uparrow = - \frac{R_a + R_{sf} + R_{add}\uparrow}{K^*}$

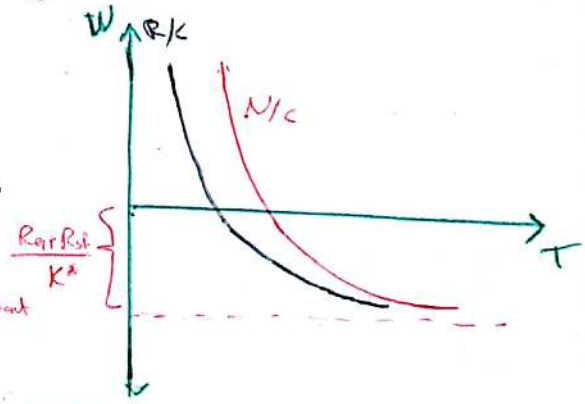


Properities ① $\beta\downarrow$ effect on S.S
② below N/C
③ $\eta\downarrow$ (losses ↑)
④ step-control

② Variation of armture voltage ($V_a\downarrow$)

$$\omega = \frac{V_a}{K^* I_a} - \frac{R_a + R_{sh}}{K^*}$$

$\omega_o\downarrow$; $\Delta\omega$ ~ constant.



Properities :- ① $\Delta\omega$ - constant, β - constant
② below N/C
③ efficient method ($\eta\uparrow$)
④ $0 \le \omega \le \omega_n$
⑤ smooth control

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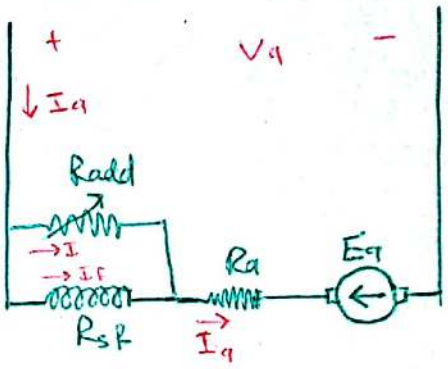
③ adding Radd in parallel with series field (Radd // R_{sf})

$$\omega = \frac{V_a}{K\phi I_a} - \frac{R_{aE}}{K\phi}$$

(ω = constant; R_{aE} ↓ → $\Delta\omega$ ↓)

$$R_{aE} = R_a + (R_{add} // R_{sf}) = R_a + \left(\frac{R_{sf} \cdot R_{add}}{R_{add} + R_{sf}}\right)$$



Properities → above N/C

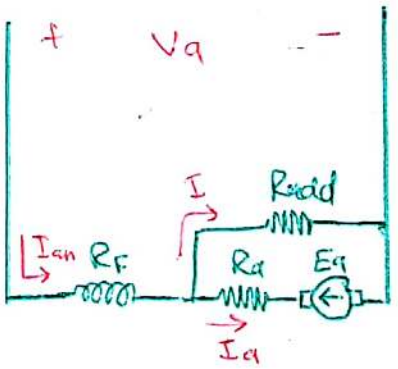
④ adding Radd in Paralled with armture (Radd // armture)

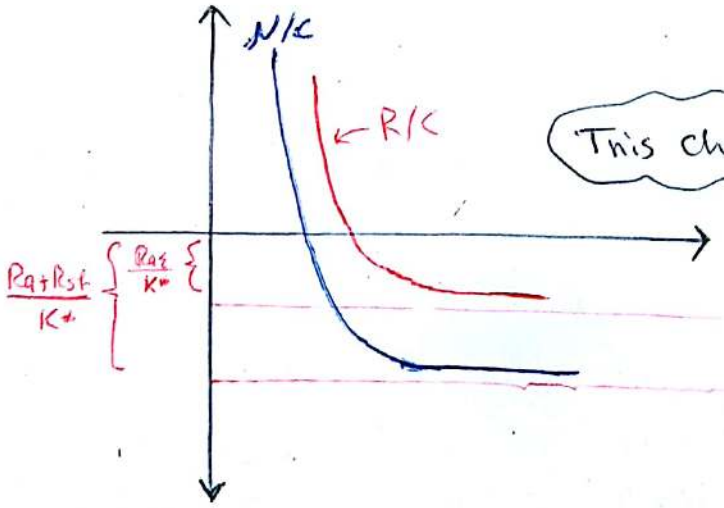
$$\omega = \frac{V_a}{K\phi I_a} - \frac{R_{aE}}{K\phi}$$

(ω ↑ → I_a ↓ → $\Delta\omega$ ↓)

$$R_{aE} = R_{sf} + (R_a // R_{add}) = R_{sf} + \left(\frac{R_a \cdot R_{add}}{R_a + R_{add}}\right)$$

Properities → above N/C





This char. for each ③ & ④

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Ex/ DC series motor, 320V, 60 N.m, 600 rpm, $R_a = 2\Omega$, $R_{sf} = 3\Omega$

Find ① I_{sf}? $I_{sf} = I_a$

$T_n = K^* I_{sf}^2 \rightarrow 60 = K^* I_{sf}^2 \rightarrow \boxed{K^* = \frac{60}{I_{sf}^2}}$

$V_a = I_a(R_a + R_{sf}) + E_{a_n}$

$320 = I_{sf}(5) + \frac{3770}{I_{sf}}$

$5 I_{sf}^2 - 320 I_{sf} + 3770 = 0$

$E_{a_n} = (K\phi)\omega = K^* I_a \omega_n = \frac{60}{I_{sf}} \cdot \omega_n$

$\omega_n = \frac{2\pi n_n}{60} = 62.832$ rad/s

$I_{sf1} = 15.567$ A $\quad I_{sf2} = 48.43$ A ←

$K_1^* = 0.0256 \quad K_2^* = 0.248$

هنا يجب ايجاد P_1 و P_2 واختيار التيار القريب من P_2

$P_2 = T.\omega = 60 * 62.832 = 3768$ W

$P_1 = V_a I_a$

$P_{11} = 320 * 15.567 = 4976$ W ← هذه الاقرب

$P_{12} = 320 * 48.43 = 15500.8$ W

$I_{sf} = 15.567$ A

① Find V_a^*, if $n^* = 400$

$E_a^* = K^* I_a \cdot \omega^*$
$= 194.28$ V

$V_a = I_a(R_a + R_{sf}) + E_a^*$
$= 9 * 5 + 194.25$

$V_a = 239.28$ V

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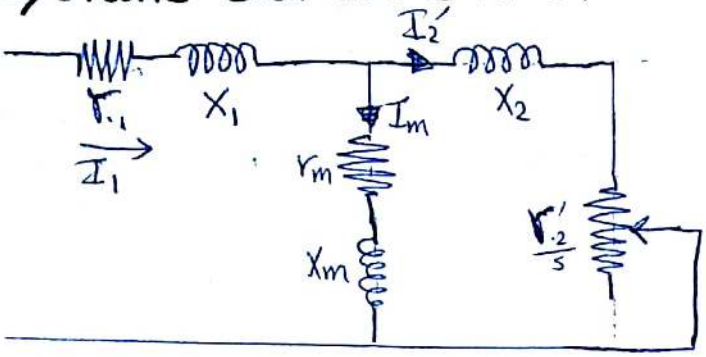
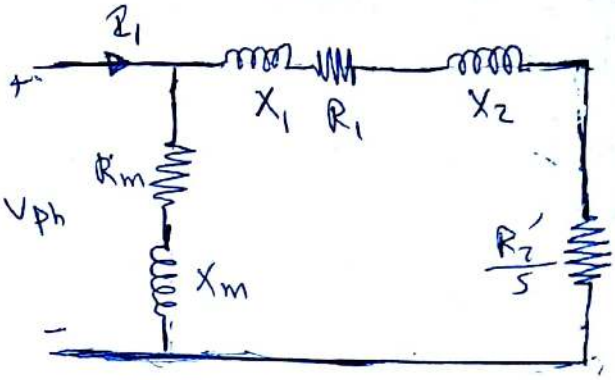
*AC motor

⇒static characteristic of AC drives-



T- Type (real prameter)



TT-Type (For calculation)

* العناصر التي بجانبها (1) تكون stator / * العناصر التي بجانبها (2) عناصر الروتر.

* For convert from T-Type to TT type

$R_1 = C_1 r_1$ $\quad R_2' = C_1^2 r_2'$ $\quad X_1 = C_1 X_1$ $\quad X_2 = C_1^2 X_2$

$R_m = r_m + r_1$ $\quad X_m = X_1 + X_m$

$$C_1 = 1 + \frac{X_1}{X_m}$$

$$S = \frac{n_0 - n}{n_0} = \frac{w_0 - w}{w_0}$$

$$n_0 = \frac{60 f_1}{(P/2)}$$

$$w_0 = \frac{2\pi f_1}{(P/2)}$$

$$w = w_0 (1 - S)$$

$$I_1 = I_2' = \frac{V_{1ph}}{\sqrt{\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_{sc})^2}}$$

$$X_{sc} = X_1 + X_2'$$

$$T = \frac{3 V_{ph}^2 R_2'/s}{w_0 \left[\left(R_1 + \frac{R_2'}{s}\right)^2 + X_{sc}^2\right]}$$

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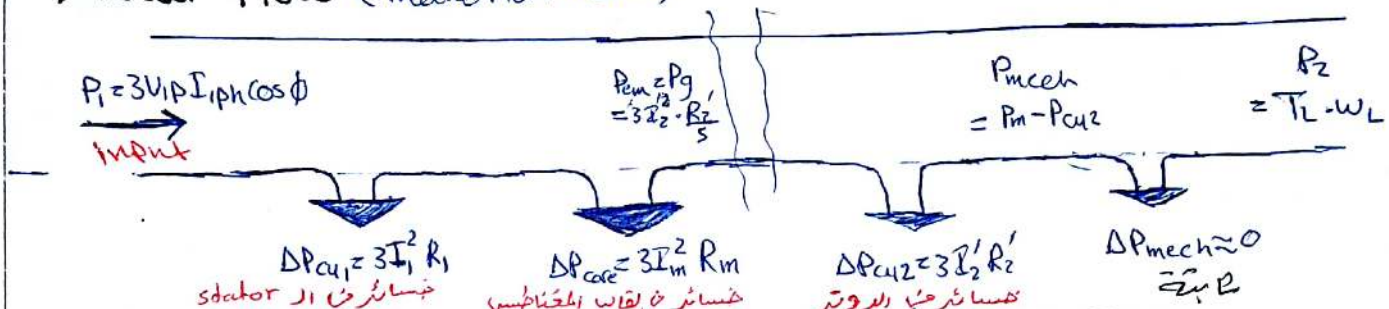
⇒ Power flow. (induction motor)



$$T = \frac{P_{rm}}{\omega_0} = \frac{V_{ph}^2 R_2'/s}{\omega_0 \left[(R_1 + \frac{R_2'}{s})^2 + X_{sc}^2\right]}$$

هذا الموضوع اخذناه بالتفصيل في ماشين 2

* electromechanical characteristic

at point ① [ideal no load]

$I = 0 \quad s = 0 \quad \omega = \omega_0$

at point ② [short circuit]

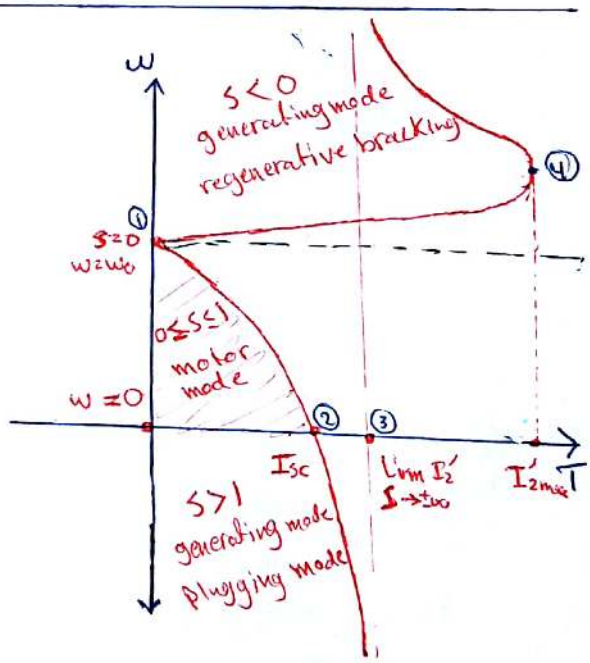
$I = I_{sc} \quad s = 1 \quad \omega = zero$

at line ③

$s = \pm\infty \quad I_2' = \lim_{s \to \pm\infty} I_2'$

at point ④ maximum current.

$s_{I_{max}} = \frac{-R_2}{R_1} \quad I_{2max} = \frac{V_1}{X_{sc}}$



Plugging mode

The motor operate as generator in series to the source regenerative braking

The motor operate as gen. in parallel to the source

ملاحظات مهمة

١- يأتي السؤال كما هذا الموضوع بأعطاء قيمة s وعليك رسم الرسمة وبيان جزء معين مثل (رسم العلاقة الكهروميكانيكية للـ AC عندما s = +1 وبين الـ motor mode)

٢- الجواب يكون كما قد السؤال يعني طلبا s = 0 يرسم الجزء الوسط والاسفل فقط

٣- الـ s تتغير بشكل عكسي يعني كلما ارتفعنا بالرسمة بتقل (والعكس)

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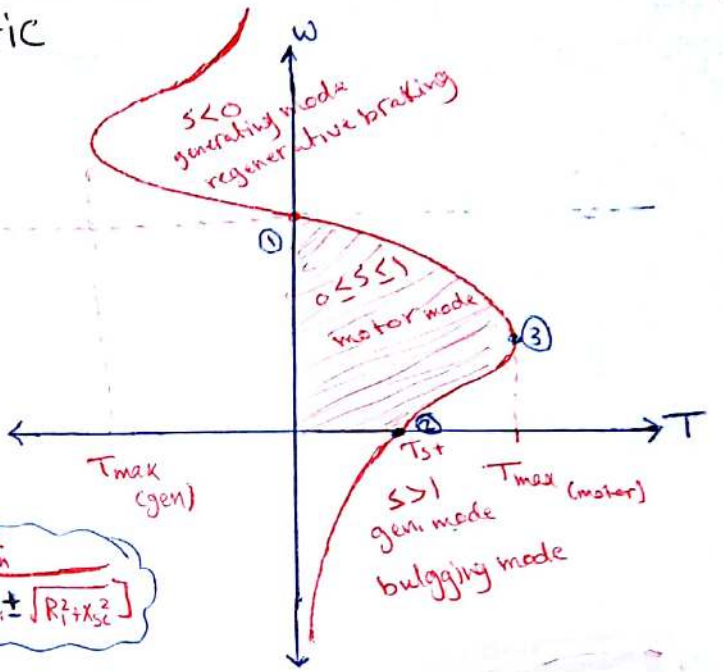
* mechanical characteristic



at point ① [no-load]

$w = w_0 \quad s = 0 \quad T = 0$

at point ② [starting point]

$w = 0 \quad s = 1 \quad T = T_{st}$

at point ③

$T = T_{max}$ when $\frac{dT}{dw} = 0 \quad s = s_{cr}$

$$s_{cr} = \pm \frac{R_2'}{\sqrt{R_1^2 + X_{sc}^2}}$$

$$T_{max} = \frac{3V_{ph}^2}{2\omega_0 \left[R_1 \pm \sqrt{R_1^2 + X_{sc}^2}\right]}$$

→ generating + motoring

spical case

$$T = \frac{2\,T_{max}\,(1 + a\,s_{cr})}{\frac{s}{s_{cr}} + \frac{s_{cr}}{s} + 2\,a\,s_{cr}} \qquad a = \frac{R_1}{R_2'}$$

When $a\,s_{cr} \simeq 0$

$$T = \frac{2\,T_{max}}{\frac{s}{s_{cr}} + \frac{s_{cr}}{s}} \longrightarrow s_{cr} = \frac{R_2'}{X_{sc}}$$

$s \gg s_{cr}$ → non-linear relationship $T \propto \frac{1}{s}$

$s_{cr} \gg s$ → linear relationship $T \propto s$

$$\lambda = \frac{T_{max}}{T_n}$$

$$s_{cr} = s_n\left(\lambda \pm \sqrt{\lambda^2 - 1}\right)$$

فائدة

for induction motor $T \propto V_1^2$

For synchronous motor $T \propto V_1$

فائدة

* stable steady state operation

$w_0(1+s_{cr}) \geq w \geq w_0(1-s_{cr})$

maximum holding torque

$T_{L\,hold} \leq T_{max}$

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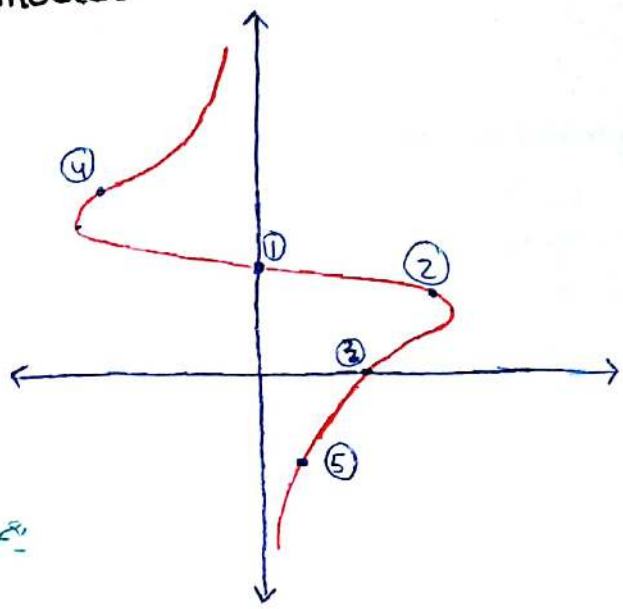
*Induction motor operating mode:-



① No load mode

$S=0 \quad \omega=\omega_0 \quad T_L=0 \quad I_2'=0$

$P_2 = T \cdot \omega = \text{zero}$

$P_1 = \Sigma\Delta P = \Delta P_{cu1} + \Delta P_{core}$

$\eta = \text{zero}$

② motoring mode

$1 \geq S \geq 0 \quad , \quad \omega_0 > \omega > 0 \quad ; \quad 0 < T < T_{st}$

يعمل المحرك في الوضع الطبيعي عند القيم المقدرة (rated)

③ short circuit mode

$S=1 \quad , \quad \omega=0 \quad , \quad T = T_{s.c} = T_{st}$

$I_1 = I_2' = I_{s.c} \quad , \quad I_m = 0 \quad , \quad P_2 = T \cdot \omega = 0$

$P_1 = \Sigma\Delta P = \Delta P_{cu1} + \Delta P_{cu2}$ قيمة عالية جداً

④ regenerative braking (generating mode)

$\omega > \omega_0 \quad S < 0 \quad P_2 < 0 \quad P_1 < 0$

$P_2 = P_1 + \Sigma P_2$ المحرك يزود الشبكة بقدرة P_1 وتغطي الضياعات

* The motor operate as generator in parallel To The source

⑤ blugging braking (generating mode)

$S > 1 \quad , \quad \omega < 0 \quad , \quad T_L < T_{st} \quad , \quad P_2 < 0 \quad , \quad P_1 > 0$

$\Sigma\Delta P = P_1 + P_2$ قيمة عالية جداً

* The motor operate as generator in series to the source.

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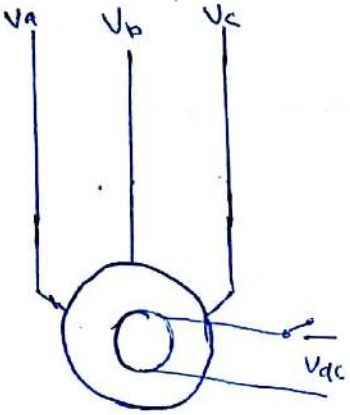
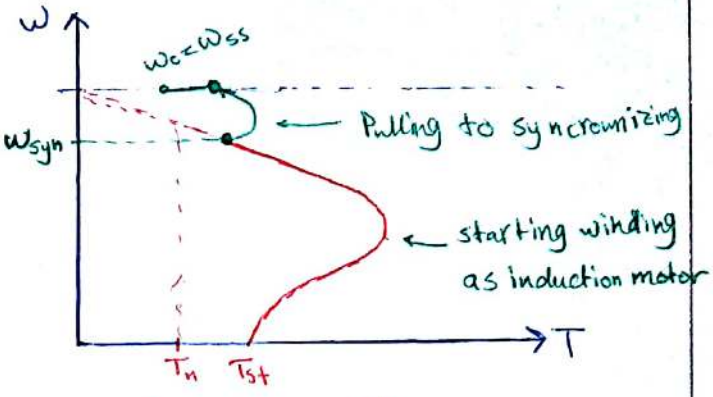
*Charecteristic of syncrounous 3ph motor

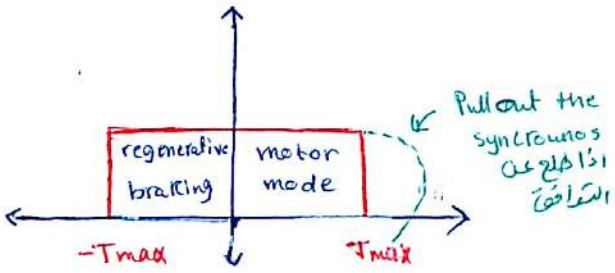




starting operation

شروط ← $w_{syn} = (90-95)\% \, w_0$

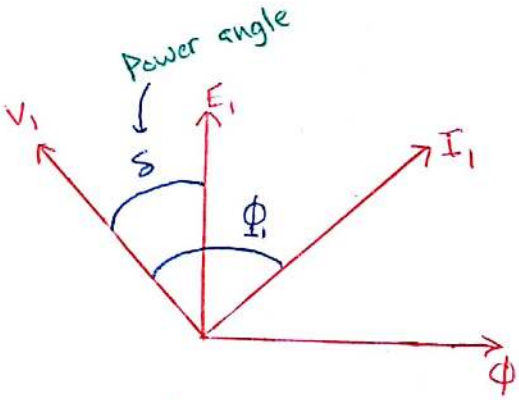
$T_{syn} > T_{Load}$



steady state

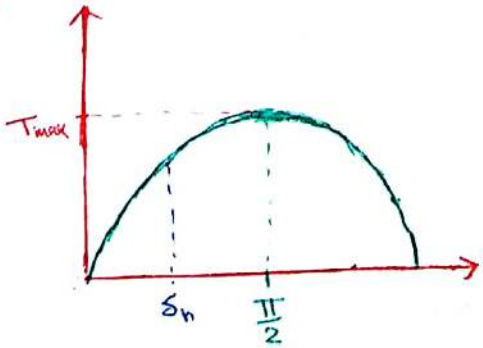
$T_L < T_{max}$

$$T = \frac{P}{w_0} = \frac{3 V_1 E}{X_d} \sin\delta$$



Phasor digram of syn. motor

$$P = \frac{V_1 E_1}{X_d} \sin\delta$$



$\delta_n = (20-30)^\circ$

$$\lambda = \frac{T_{max}}{T_n} \simeq (2-3)$$

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Example (1) :- 3-ph induction motor, Δ-connection, 25KW, 400V, 50Hz

$P=6 \quad \lambda=2.5 \quad s_{cr}=0.2$

Find ① nominal angular speed (ω_n)

$P=6 \rightarrow n_o = \frac{60f}{(P/2)} = 1000 \text{ rpm}$

$s_{cr} = s_n(\lambda \pm \sqrt{\lambda^2-1}) \rightarrow s_n = 0.042$

$n_n = (1-s_n)n_o = 958 \text{ rpm} \rightarrow \omega_n = \frac{2\pi n}{60} = 100.29 \text{ rad/s}$

② nominal torque (T_n)

$T_n = \frac{P_{2n}}{\omega_n} = \frac{25 \text{ KW}}{100.29 \text{ rad/s}} = 249.32 \text{ N.m}$

③ maximum torque (T_{max})

$T_{max} = T_n\left(\frac{s_n}{s_{cr}} + \frac{s_{cr}}{s_n}\right) = 623.3 \text{ N.m}$

ملاحظة :-
الحل على هذا القانون يكون فقط بحالة اذا لم يكن هناك معلومات كافية للحل على الطرق الاخرى

④ starting torque (T_{st})

$s=1$

$T_{st} = \frac{2T_{max}}{\left(\frac{s_{cr}}{1} + \frac{1}{s_{cr}}\right)} = 239.73$

⑤ For V_{1ph} decrease by (5%) Find (T_{st})

$V_{1ph}^* = 0.95 V_{1ph} \quad T \propto V^2$

$\frac{T_{st}^*}{T_{st}} = \left(\frac{V_{1ph}^*}{V_{1ph}}\right)^2 \rightarrow$

$T_{st}^* = (0.95)^2 T_{st} = 216.36 \text{ N.m}$

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Example (2) :- 3ph-induction motor, 1450 rpm, 50 Hz, 380 V, λ-conn.

$R_1 = 0.5\ \Omega$ $R_2' = 1.5\ \Omega$ $X_{sc} = 5\ \Omega$

Find ① over loading capacity factor (λ)

$$\delta_n = \frac{1500-1450}{1450} = 0.033 \qquad \delta_{cr} = \pm\frac{R_2'}{\sqrt{R_1^2+X_{sc}^2}} = \pm 0.3$$

$V_{ph} = \frac{380}{\sqrt{3}}$ ↘

$$T_n = \frac{3V_{ph}^2 R_2'/s_n}{\omega_0\left[(R_1+\frac{R_2'}{s_n})^2+X_{sc}^2\right]} = 19.6\ N.m \qquad \omega_0 = \frac{2\pi\times n}{60} = 157\ rad/s$$

$$\omega_n = \frac{2\pi n_n}{60} = 151.8\ rad/s$$

$$T_{max} = \frac{3V_{ph}^2}{2\omega_0\left[R_1 \pm \sqrt{R_1^2+X_{sc}^2}\right]} = 83.3\ N.m \text{ motor}$$

or $-101.7\ N.m$ generator

$$\lambda = \frac{T_{max}}{T_n} = 4.25$$

② nominal power (P_n)

$$P_n = T_n \cdot \omega_n = 2.975\ KW$$

③ maximum Rotor current (I_{2max}') and $\delta(I_{2max}')$

$$I_{2max}' = \frac{V_{1ph}}{X_{sc}} = 44A \qquad \delta(I_{2max}') = \frac{-R_2'}{R_1} = -3$$

④ The angular speed at I_{2max}'

$$\omega(I_{2max}') = \omega_0(1-s_{max}) = 628\ rad/s$$

⑤ range of stable steady state operation

$$\omega_0(1+\delta_{cr}) \geq \omega \geq \omega_0(1-\delta_{cr})$$

$$204.1 \geq \omega \geq 109.9$$

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Question 3 A **(14 points)**

Part (a): A three-phase, star-connected induction motor has the following ratings: 400V, 50Hz, 1470 rpm, $R_1 = 0.8\Omega$, $R'_2 = 0.3\Omega$, $X_1 = 0.55\Omega$, and $X'_2 = 0.45\Omega$

Find:

1. The over-loading capacity λ for this motor.
2. The maximum current of the motor and angular speed at which the maximum current occurs.
3. What will happen to the starting torque if the supply voltage has been reduced by 10%?

Part (b): Plot the electro-mechanical characteristic of three-phase induction motor for positive slip c and show on it the operating mode(s) of the motor.

Answer:

$V_{ph} = \frac{400}{\sqrt{3}} = 231V$; $n_0 = 1500$ rpm; $\omega_0 = 157$ rad/s

$S_n = \frac{1500-1470}{1500} = 0.02$

① $T_n = \frac{3V_{ph}^2 \cdot R'_2/S_n}{\omega_0\left[(R_1 + \frac{R'_2}{S_n})^2 + X_{s.c}^2\right]} = \frac{(400)^2 \times 0.3/0.02}{157\left[(0.8 + \frac{0.3}{0.02})^2 + 1^2\right]} = 61\ N\cdot m$

$T_{max} = \frac{3V_{ph}^2}{2\omega_0\left[R_1 + \sqrt{R_1^2 + X_{s.c}^2}\right]} = \frac{(400)^2}{2\times 157\left[0.8 + \sqrt{0.8^2+1^2}\right]} = 245$

$\lambda = \frac{T_{max}}{T_n} = \frac{245}{61} = 4.02$

② $I'_{2\,max} = \frac{V_{ph}}{X_{s.c}} = \frac{231}{1} = 231A$

$S(I'_{max}) = -\frac{R'_2}{R_1} = -\frac{0.3}{0.8} = -0.375$

$\omega(I'_{max}) = \omega_0(1 - S(I'_{max})) = 157 \times 1.375 = 215.9$ rad/s

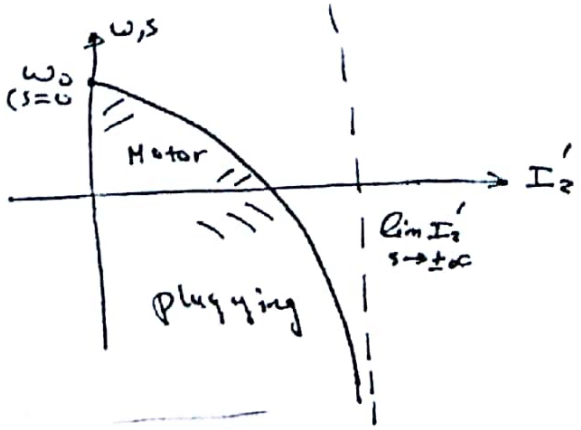
③ $T \propto V_{ph}^2 \Rightarrow T_{st}^* = T_{st\,(N)} \cdot (0.9)^2 = 0.81 T_{st\,(N)}$

$\Rightarrow T_{st}$ ↓ by 19%.

OR: $T_{st\,(n)} = \frac{(400)^2 \times 0.3}{157\left[(0.8+0.3)^2+1^2\right]} = 138.34\ N\cdot m$

$T_{st}^* = 0.81 \times 138.34 = 112.05\ N\cdot m$

(b)



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A three-phase, delta- connected induction motor has the following ratings:

440V, 970rpm, 50Hz, $R_1 = 1.5\Omega$, $R'_2 = 1\Omega$, $X_{s.c} = 4\Omega$

Find:

1. The starting torque and maximum torque of the motor.
2. The over-loading capacity λ.
3. The range of speed for stable steady-state operation.
4. The drop in speed for natural mechanical characteristic
5. The maximum holding torque under re-generative braking operation.
6. The maximum current
7. The speed at which the maximum current occurs
8. The power rating of the motor
9. What will happen to the over-loading capacity if the supply voltage has been reduced by 15%?

Prof. Dr. Hussein Sarhan

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*Transient operation of AC motor

starting operation

- Direct start
- indircet start
  - by reduce V_{iph}
    - Δ-Y method
    - auto transformer.
  - by R_2 add
  - by X_1 add
  - by R_1 add

← الهدف من indirect start هو تقليل (I_{sc}, T_{st})

Braking operation

- Dynamic braking.
- Electrical Braking
  - Dynamic braking
    - AC Dyn. Braking
    - DC Dynamic braking
  - plygging Braking
  - regenerative braking

speed control

- by R_1 add → only for squral cage
- by X_1 add
- reduce V_{iph}
- variation of frequancy
- variation no. of poles
- V/f control

(by X_1 add, reduce V_{iph}, variation of frequancy, variation no. of poles, V/f control) → for squral cage or wound rotor

- By R_2 add → only for wound rotor

← ح يكون أغلب شرحنا عن الـ induction motor لأن الـ syncrounos يعمل كـ محرك فقط في حالة البدء والإطفاء.

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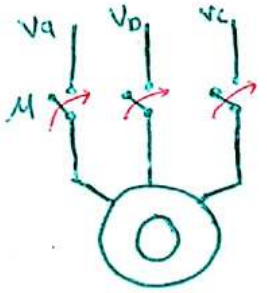
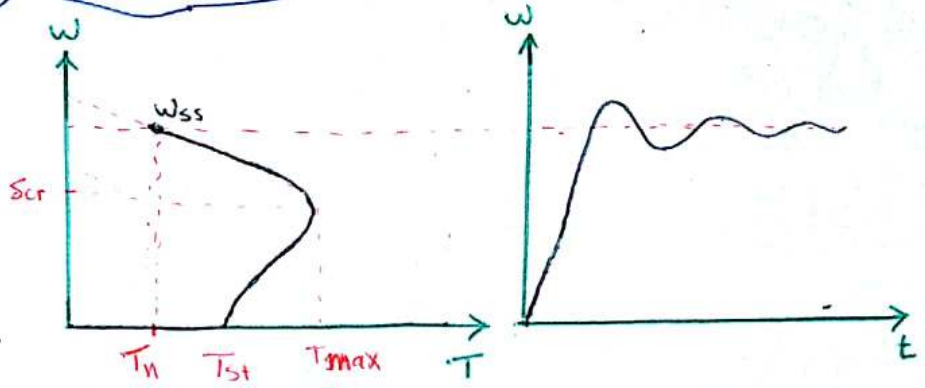
* starting method in induction method

① Direct start





← في هذه الحالة يتم فيها تغذية مباشرة دون أضافة اي عنصر للدارة.

② indirect start.

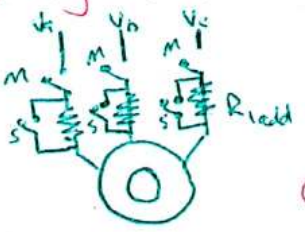
← يتم فيها أضافة عنصر للدارة لتقليل تيار البدء

← لمعرفة ما سيحصل في الخاصية يجب معرفة ما سيحصل في كل من ($T_{st}, I_{sc}, s_{cr}, T_{max}$)

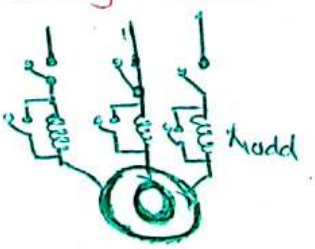
$$I_{sc} = \frac{V_{1ph}}{\sqrt{(R_1+R_2')^2+(X_{sc})^2}} \quad , \quad T_{st} = \frac{3V_{1ph}^2 \cdot R_2'}{\omega_0[(R_1+R_2')^2+X_{sc}^2]} \quad , \quad s_{cr} = \frac{R_2'}{\sqrt{R_1^2+X_{sc}^2}} \quad , \quad T_{max} = \frac{3V_{1ph}^2}{\omega_0[R_1+\sqrt{R_1^2+X_{sc}^2}]}$$

Ⓐ insert R_{1add} or X_{1add} in series with the stator

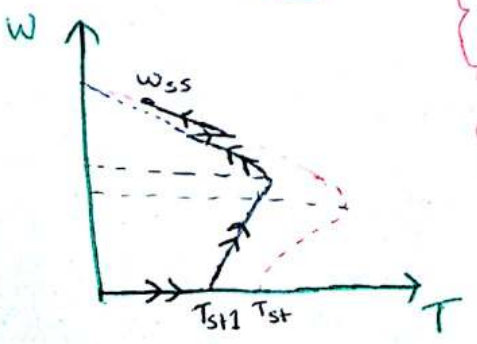
* adding R_{1add}.

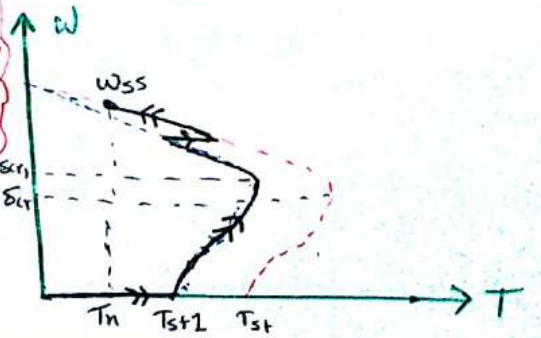


* adding X_{1add}.



في كلا الحالتين سوف يقل
$I_{sc}\downarrow, T_{st}\downarrow$
$s_{cr}\downarrow, T_{max}\downarrow$
لكن X_{1add} أفضل لأن
$\Delta P_{cu}\downarrow$





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* how to calculate X_{1add} & R_{1add}.

$$I_{s.c(N)} = \frac{V_{1ph}}{Z_{s.c(N)}} \qquad Z_{s.c(N)} = \sqrt{X_{sc}^2 + (R_1 + R_2')^2}$$

(N/c)

$$I_{sc(starting)} = \frac{V_{1ph}}{Z_{s.c(s)}}$$

add X_{1add} $Z_{s.c(s)} = \sqrt{(X_{s.c} + X_{1add})^2 + (R_1 + R_2')^2}$

add R_{1add} $Z_{s.c(s)} = \sqrt{X_{sc}^2 + (R_1 + R_{1add} + R_2')^2}$

$$\text{reduction factor}(\alpha) = \frac{I_{s.c(s)}}{I_{sc(N)}} = \frac{Z_{s.c(N)}}{Z_{sc(s)}} < 1$$

$$\Rightarrow Z_{s.c(s)} = \frac{Z_{s.c(N)}}{\alpha}$$

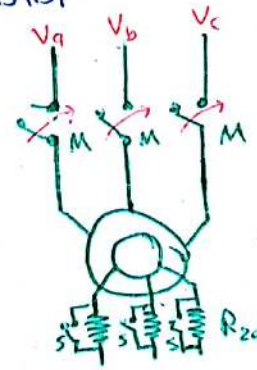
For R_{1add} $\longrightarrow$ $R_{1add} = \sqrt{\left(\frac{Z_{s.c(N)}}{\alpha}\right)^2 - X_{s.c}^2} - R_1 - R_2'$

For X_{1add} $\longrightarrow$ $X_{1add} = \sqrt{\left(\frac{Z_{s.c(N)}}{\alpha}\right)^2 - (R_1 + R_2')^2} - X_{s.c}$

تطبيق مباشر

(B) insert R_{2add}' in series with the rotor

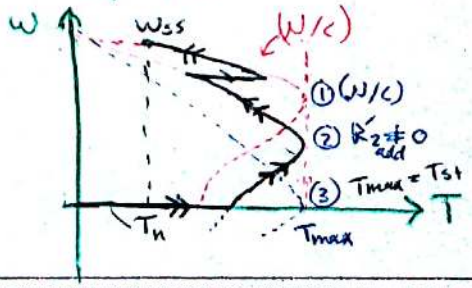
$R_{2add}' \uparrow$, $I_{sc} \downarrow$, $T_{st} \uparrow$, T_{max} is constant, $S_{cr} \uparrow$



⇒ disadvanges of this method.
increase of rotor's losses

Find R_{2add}' when $T_{st} = T_{max}$.

(chap 3) in the figure.



$$\Rightarrow S_{cr} = 1 = \frac{R_2' + R_{2add}'}{\sqrt{R_1^2 + X_{sc}^2}}$$

$$\Rightarrow R_{2add}' = -R_2' + \sqrt{R_1^2 + X_{sc}^2}$$

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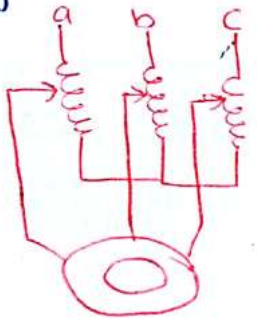
ⓒ reduce V_{iph}

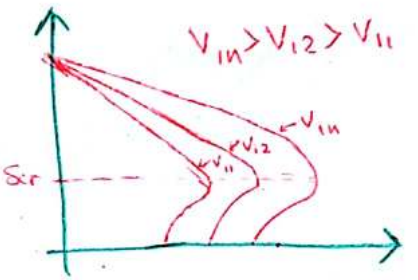
$I_{sc} \propto V_{iph}$, $T_{st} \propto V_{iph}^2$, $T_{max} \propto V_{iph}^2$, Scr-constant

← عند استخدام هذه الطريقة يجب مراعاة ما يلي ① t_{st} سوف يزداد .

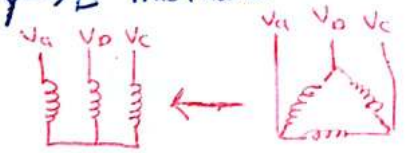
ⓒ ان تبقى $T_{st} > T_{Load}$

1. Using Auto transformer



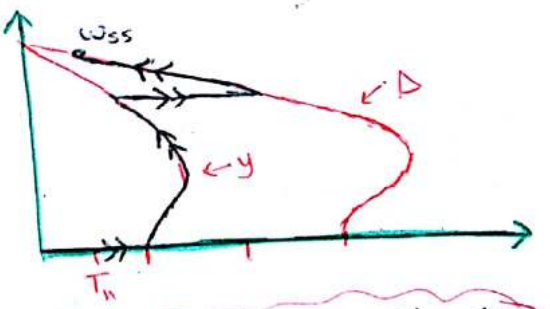


2. Y → Δ method:



$$V_{iph_Y} = \frac{V_{iph_\Delta}}{\sqrt{3}} \quad , \quad I_{sc_{(Y)}} = \frac{I_{sc(\Delta)}}{\sqrt{3}}$$

$$T_{st_Y} = \frac{T_{st(\Delta)}}{3} \quad , \quad T_{max_{(Y)}} = \frac{T_{max(\Delta)}}{3}$$



ملاحظة: من الرسمة يجب ان تبين ان T_{st} قلت بمقدار $\frac{1}{3}$

ex induction motor , $R_1 = 3\Omega$, $R_2' = 2\Omega$, $X_{sc} = 10\Omega$, find?

① Find The reduce of T_{st} if V_{iph} reduce by 20%?

$T_{st} \propto (V_{iph})^2 \rightarrow \frac{T_{st(st)}}{T_{st(N)}} = \frac{(0.8 V_{iph})^2}{(V_{iph})^2} \rightarrow T_{st(st)} = 0.64\, T_{st(N)}$ (T_{st} ↓ by 36%)

② Find R_2' add if, $T_{max} = T_{st}$

$R_{2add}' = \sqrt{R_1^2 + X_1^2} - R_2' = 7.54\Omega$

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ex|2 3-phase induction motor, 400V, 50Hz, 1420 rpm, 100A, Δ-connection

For direct start → $I_{sc} = 8I_n$, $T_{st} = 2.2T_n$

Find (T_{st} & I_{sc}) For Y→Δ start.

$$I_{sc(Y)} = \frac{1}{\sqrt{3}} I_{sc(\Delta)} = 4.62\, I_n$$

$$T_{st(Y)} = \frac{1}{3} T_{st(\Delta)} = 0.73\, T_n$$

ex|3 3-ϕ induction motor, 400V, Y-connection, 960 rpm, 50 Hz

$R_1 = 0.08$, $R_2' = 0.1\,\Omega$; $X_1 = X_2' = 0.3\,\Omega$

Find R'_{2add} that make $I_{sc} = 2I_n$

$$I_n = \frac{V_{1ph}}{\sqrt{\left(R_1 + \frac{R_2'}{s_n}\right) + X_{sc}^2}} \qquad \text{but } s_n = \frac{1000-960}{1000} = 0.04$$

$$\Rightarrow I_n = 87.18\,A$$

$$I_{sc} = 2I_n = \frac{V_{1ph}}{\sqrt{(R_1 + R_2' + R'_{2add})^2 + X_{sc}^2}} \quad \rightarrow \quad R'_{2add} = 1\,\Omega$$

---

* starting for syncronous motor.

يلزم التشغيل بعمل المحرك التزامني كمحرك حثي ←

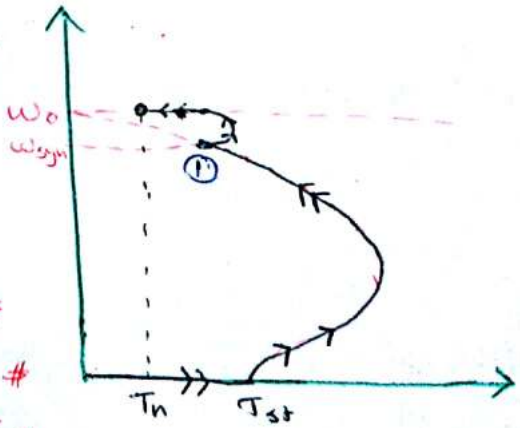
⇒ the motor has two winding.

① AC → for starting

② DC → for produce magnatic field

① عندما تصل السرعة (ω_{syn}) عند النقطة ← يتم فصل الملفات (AC) (عند بتأكد ان السرعة وصلت الـ ω_{syn})

يتم وصل تغذية الـ DC



$$\omega_{syn} = (0.9 - 0.95)\,\omega_o$$

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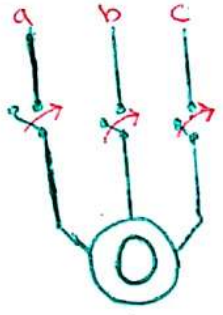
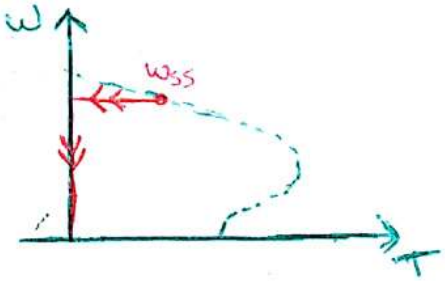
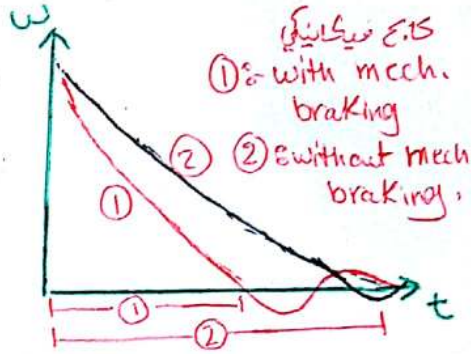
* Braking method of induction motor

1. Mechanical Braking



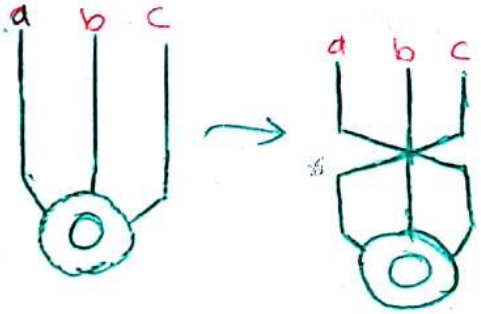


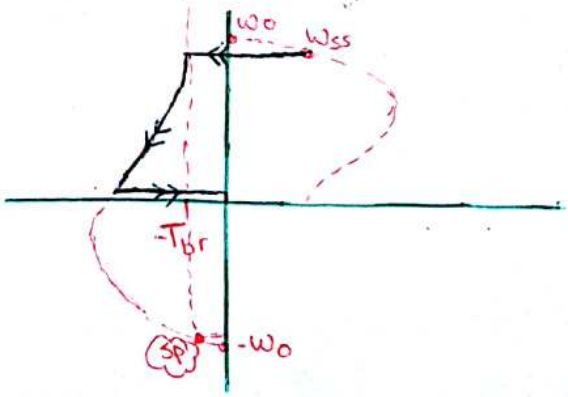


2. Electrical Braking

(A) plugging Braking

⇐ يتم في هذه الطريقة تغيير التوصيل بين فازين





⇐ يتم فصل التغذية عندما تصبح السرعة $w \simeq 0$

⇐ اذا كان المحرك من نوع Wound rotor يمكننا أضافة Rheostat عندها $(-T_{max} = const \;\&\; S_{cr})$

⇐ اذا لم يتم الفصل يعمل المحرك في الاتجاه المعاكس

$$\Rightarrow S = \frac{w_o - w_{ss}}{w_o} \rightarrow w_{ss} = w_o(1-S)$$

& plugging $\Rightarrow sp = \frac{-w_o - w_{ss}}{-w_o} = 2 - S_n$

⇒ losses become very high because sp is very high.

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ex| 400V, 50Hz, 6 poles, $R_1 = R_2' = 1\Omega$, $X_1 = X_2' = 2\Omega$, Y-connection.

$N_{ss} = 950$ rpm, plugging Braking

① Find I'_{2br} & T_{br}

$S_p = 2 - S_n = 1.95$ $\longrightarrow S_n = \frac{n_o - n_n}{n_o} = \frac{1000-950}{1000} = 0.05$

$$I'_{2br} = \frac{V_{1ph}}{\sqrt{(R_1 + \frac{R_2'}{S_p})^2 + X_{sc}}} = \frac{400/\sqrt{3}}{\sqrt{(1 + \frac{1}{1.95})^2 + 4^2}} = 54\,A$$

$$T_{br} = \frac{3 I'^2_{2br} \cdot R_2'/S_p}{\omega_o} = \frac{3 * 54^2 * \frac{1}{1.95}}{104.7} = 42.82\ N.m$$

① if $I_{br} = 1.5\, I_n$, Find R'_{2add} & T_{br}

$$I_n = \frac{V_{1ph}}{\sqrt{(R_1 + \frac{R_2'}{S})^2 + X_{sc}}} = 10.8\,A$$

$$I_{br} = 1.5\, I_n = 16.2 = \frac{V_{1ph}}{\sqrt{(R_1 + (\frac{R_2' + R'_{2add}}{S_p})^2 + X_{sc}^2}} \quad (R_2' + R'_{2add})$$

$\Rightarrow R'_{2add} = 23.73\Omega \rightarrow T_{br} = \frac{3 I'^2_{2br} \cdot R_2'/S_p}{\omega_o} = 95.35\ N.m$

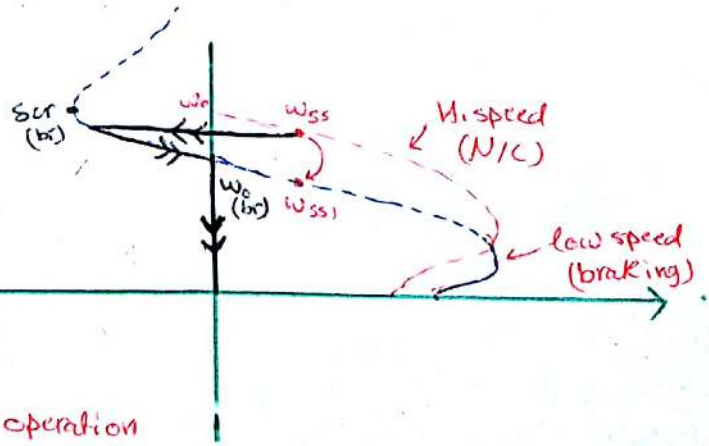
③ regenerative Braking.

$\Rightarrow \omega > \omega_{o(br)}$

$\Rightarrow S < 0 \Rightarrow T_m < 0$

$\Rightarrow P_2 = P_1 + \Delta P_{losses}$.

$\Rightarrow$ maximum holding Torque $= -T_{max}$



$\Rightarrow$ range for speed for safe operation

$$\omega_{o(br)} \le \omega \le (1 + S_{cr(br)})\, \omega_{o(br)}$$

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ex 3-ϕ induction motor, Y-connection, 6-poles, 50Hz, 400V
$R_1 = R_2' = 1\Omega$, $X_1 = X_2' = 2\Omega$, For regenerative braking in N/C

Find ① maximum holding torque

$n_o = 1000 rpm \rightarrow \omega_{o(br)} = 104.7 rad/s$

$\frac{R_2'}{\sqrt{R_1^2 + X_{sc}^2}} = s_{cr} = -0.242 \rightarrow I_{2br}' = \frac{V_{1ph}}{\sqrt{(R_1 + \frac{R_2'}{s_{cr}})^2 + X_{sc}^2}} = 45.5A$ ($s = s_{cr}$)

$T_{max} = \frac{3 I_{2br}' \cdot R_2'/s_{cr}}{\omega_o} = -244.5 N.M$

② range of speed for s.s point.

$n_{o_{br}} \leq n \leq (1 + s_{cr})_{br} n_{o_{br}}$

$1000 \leq n \leq 1242 rpm$

③ ω^* when $T_L = 100 N.m$.

$-100 = T_L = \frac{3 V_{ph} \cdot R_2'/s^*}{\omega_{o_{br}}[(R_1 + \frac{R_2'}{s^*})^2 + X_{sc}^2]} \Rightarrow s_1^* = -0.959 \quad \because |s_{cr}| < |s_1^*|$

$s_2^* = -0.063$ ✓

$\Rightarrow n^* = n_{o(br)}(1 + s_2^*) = 1063 rpm$

$\omega^* = 111.32 rad/s$

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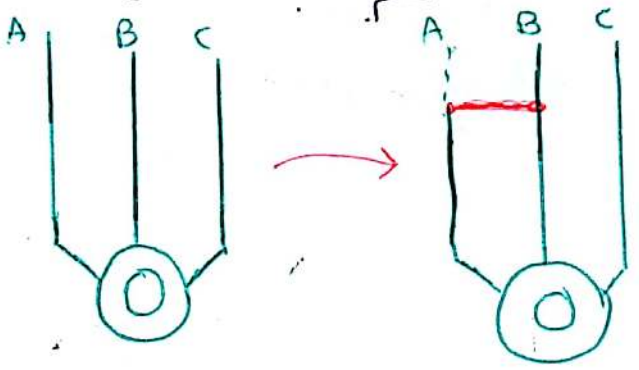
ⓒ Dynamic Braking

1 AC-Dynamic Braking

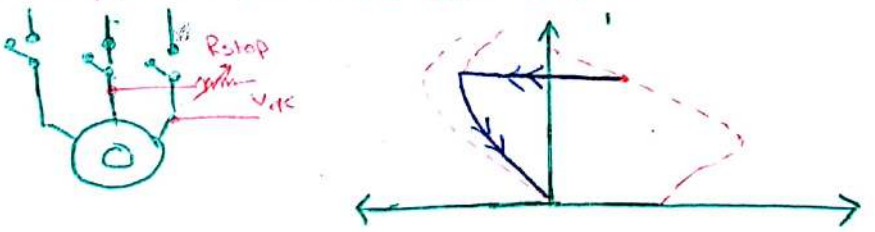


* في هذه الحالة يتم فصل فاز وقصره مع إلفاز الآخر
← اذا وصلناه مع إلفاز الآخر سمي التوصيلة 3-lead connection
← اذا فصلناه تسمي 2- lead connection

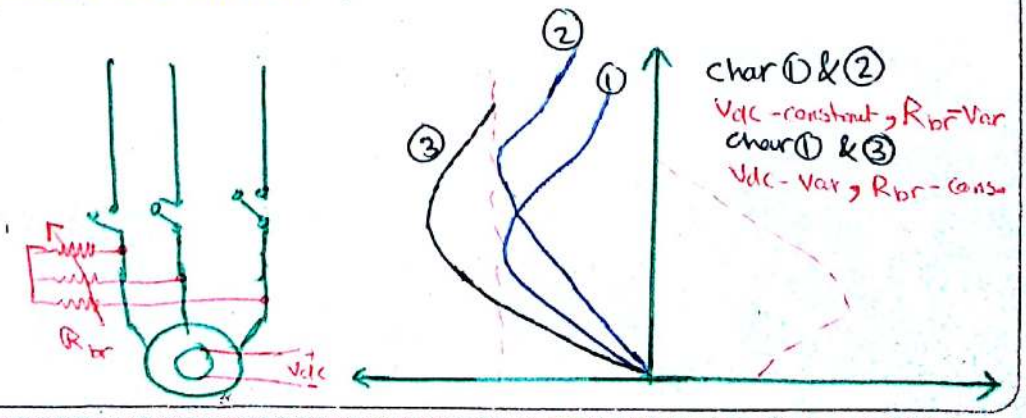
2 DC- dynamic torque

1. For induction motor.

← يتم في هذه الحالة فصل تغذية الـ AC ووصل مصدر تيار مباشر مع لوريد عن طريق مقاومة للتحكم بالتيار
← عندما تقترب السرعة من الصفر يتم فصل التغذية (DC) والمقاومات



2. For syncrounous motor



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* speed control

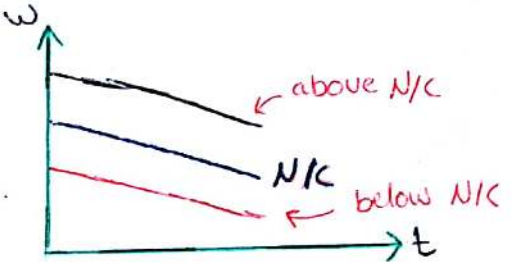
→ Transient form char.1 to another char. for same load ← هو التغير المتعمد في خواص المحرك للحصول على خواص تنظيمية

- Factors effect the selection of speed control method
العوامل المؤثرة في اختيار طريقة التحكم بالسرعة ((للقهوة))

① Direction of speed control

$\omega > \omega_n$:- above N/C

$\omega < \omega_n$:- below N/C



② Range of speed control

$\omega_{min} \leq \omega \leq \omega_{max}$

$0.5 \leq \omega \leq \omega_n$ → narrow range

$0 \leq \omega \leq \omega_n$ → wide range

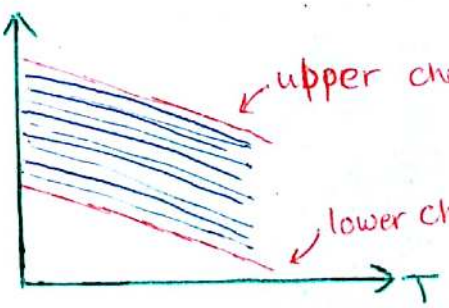
$\omega_{max} \leq 2\omega_n$ → condition for mech. consindration

③ smoothness of speed control

← اذا كانت عدد الخواص التنظيمية (Regulated char.) محدد بين اعلى واقل خاصية فتتحكم عالية التحكم

← كانت يكون في الخطوات (step - control)

← واذا كان غير محدد (تدريجي) (smoth control)



④ margin for S.S operation.

depend on hardness (β) → β↓ → steady state↓ → the stablity decrease

⑤ effecincy (η)

← دائماً نأخذ الطريقة الأكثر كفاءة

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* speed control of 3-ϕ AC machine

$$T = \frac{3\, V_{ph}^2\, R_2'/s}{\omega_o\left[\left(R_1 + \frac{R_2'}{s}\right)^2 + X_{sc}^2\right]}$$

Parameters :- R_1 (only for squirrel cage), X_1, V_{1ph}, $[\omega_o, F, P]$ (for squirrel cage and wound rotor), R_2' (only for wound rotor)

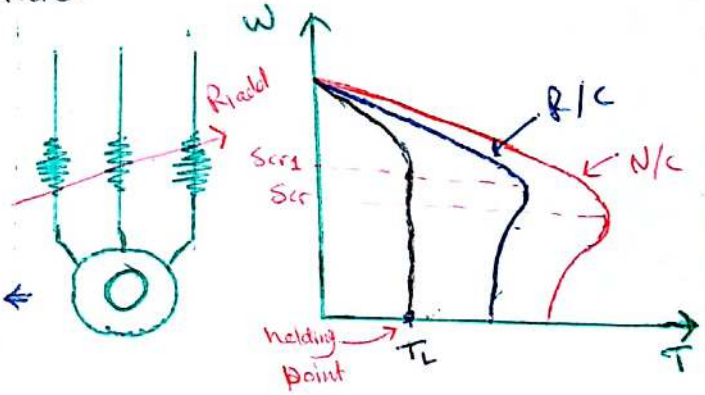
① insert R_{1add} to the stator

$R_{1add}\uparrow, T\downarrow, s_{cr}\downarrow, \Delta\omega\uparrow$

⇒ at holding point $T_{st} \geq T_L$

⇒ $\eta\downarrow$ because losses ↑

← لايجوز ان تكون T_{st} اقل من T_L لان المحرك لن يقلع.



② insert X_{1add} to the stator

← نفس الطريقة السابقة لكن الكفاءة اقل لان الخسائر الكهربائية اكبر قليلا.

③ reducing V_{1ph} $(V_{1ph} \leq V_{ph(n)})$

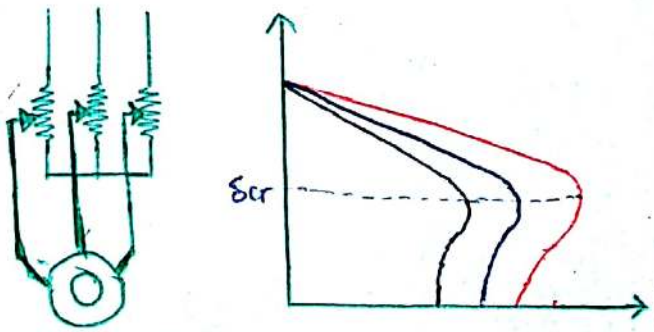
$V_{ph}\downarrow, T\downarrow, s_{cr}\downarrow$ ←

← كيف هذه الطريقة ان العزم (T) يتأثر بشكل كبير من (V) $(T \propto V^2)$

← يتم استخدام (autotrans) في هذه الطريقة

← يتم استخدامها في الاحمال الخفيفة

← مسارها ← كفاءتها جيدة ← smooth control



(auto transformer)

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* For method ①, ②, ③

For constant λ

$$\lambda = \frac{T_{max(n)}}{T_{Load(n)}} = \frac{T_{max(R)}}{T_{Load(R)}} = \text{constant}$$

$T_{L(R)} = T_{L(n)} \frac{T_{max(R)}}{T_{max(n)}}$ $\quad$ $T_{LR} \lesssim T_{L(n)}$

ex| 3-ϕ induction motor, 480V, 6-poles, 50Hz, $R_1 = R_2' = 0.5\Omega$

$T_L = 120$ N.m = constant.

Find ① n_n, For $V_{1ph(n)}$

$$T_n = \frac{3V_{1ph}^2 R_2'/s_n}{\omega_0 \left[\left(R_1 + \frac{R_2'}{s}\right)^2 + X_{sc}^2\right]}$$

$s_n \rightarrow$ small quantity $\quad s_n \cong 0$

$$T_n = \frac{3V_{1ph}^3 s_n}{\omega_0 \cdot R_2'} \rightarrow \omega_0 = \frac{2\pi f}{(P/2)}$$

$s_n = 0.0272$

$n_n = n_0(1 - s_n) = 973$ rpm

② Find n_{new}^* for $V_{ph(new)} = 0.8\, V_{1ph(n)}$

$$\frac{T_{new}}{T_n} = \frac{V_{1ph(new)}^2}{V_{1ph(n)}^2} \cdot \frac{s_{new}}{s_n}$$

$T_{new} = T_n = $ constant

$n_{new} = (1 - s_{new})\, n_0$

$= 957.5$ rpm

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[4] changing of no-load speed W_o.

$$W_o = \frac{2\pi \cdot f_1}{(P/2)}$$ Parameters :- P & f_1

(A) changing number of pole (P-var.)

← هنا يتم اعادة توصيل ملفات ال Stator وهذا توصيف من سرعة إلى

$W_o = 2W_{on}$ — H.S connection

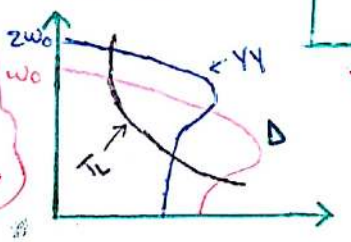
$W_o = \frac{1}{2} W_{on}$ — L.S connection

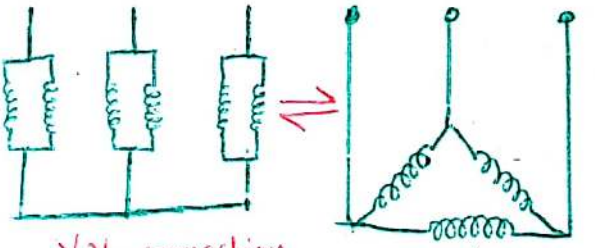
- 1o connection For constant power. [$P_2 \cong$ const.]

$T_{L(YY)} = \frac{1}{3} T_{L(\Delta)}$

$T_L \propto \frac{1}{W}$ → (P-constant)

هذا شرط هذه التوصيلة حتى يبقى الـ Power ثابت مثل cutting machine





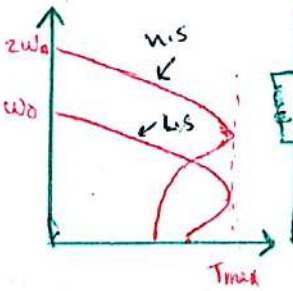
YY-connection
P = 6
High speed

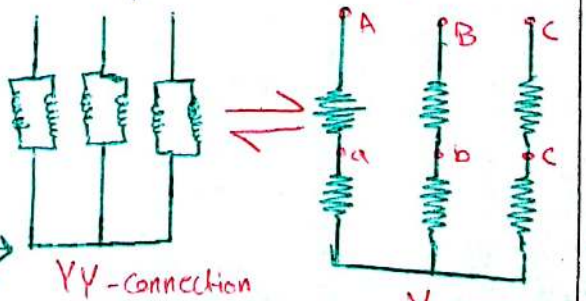
Δ-connection
P = 12
low speed

- 2o connection For constant torque (T = constant)

$T_L \simeq$ constant

$P_{2Y} = \frac{1}{2} P_{2YY}$





YY-connection
P = 6
high-speed

Y-conn.
P = 12
low-speed

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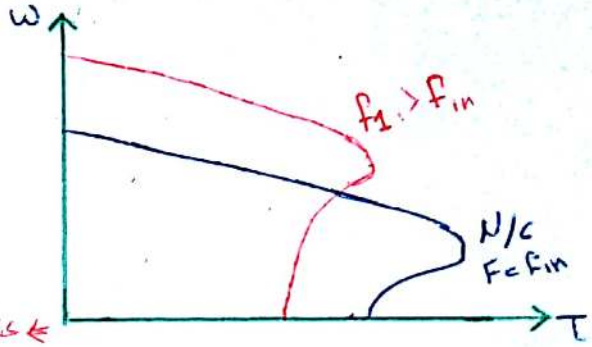
(B) variation of frequency (F_1 = Var)

$f_{1min} \leq f_1 \leq f_{max}$

$F_{1max} \leq 2 f_{in}$

$\rightarrow F \propto X_{sc} \propto \frac{1}{I_{sc}}$

$\rightarrow F \propto \omega_o \propto \frac{1}{T}$



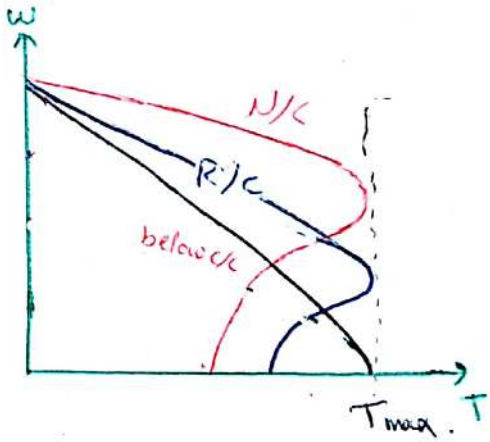
← عشان التردد اللي، عزم الحمل سوف يصبح قليل جداً
مما يؤدي اى عدم القدرة على العمل
← يتم استخدام هذه الطريقة لا

narrow range s.c ($0.5\omega_n \leq \omega \leq \omega_n$)

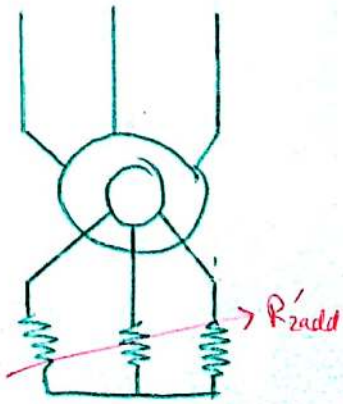
(5) Inserting R'_{2add} in the rotor

Wound rotor ← هذه الطريقة مستخدمة فقط لا

- R'_{2add} ↑ , S_{cr} ↑ , T_{max} - constant
- I'_2 ↓ , T_m ↓ , ω ↓
- at below char. $T_{ST} = T_{max}$



Properities :- ① narrow range
② below N/C
③ η ↓



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ex| 3-ϕ Induction motor, Y-connection, 30 hp, 480 V, 6 poles, 50 Hz
$R_1 = 0.5\,\Omega$ $R_2' = 0.4\,\Omega$, T = constant

① Find n_n?

(1 hp = 746 watt)

$$P_{2n} = T_n \omega_n \rightarrow T_n = \frac{P_{2n}}{\omega_n} = \frac{30 \times 746}{\omega_0(1-s_n)}$$

$$T_n = \frac{3 V_{iph}^2 R_2'/s_n}{\omega_0'\left[(R_1 + \frac{R_2'}{s})^2 + X_{sc}^2\right]} \rightarrow s_n \text{ is very small}$$

$$T_n = \frac{3 V_{iph}^2 s_n}{\omega_0 R_2'} = \frac{P_{2n}}{\omega_0[1-s_n]}$$

$$\omega_0 = \frac{2\pi f}{(P/2)} = 104.72 \text{ rad/s}$$

$$s_n^2 - s_n + \frac{P_{2n} \cdot R_2'}{3 V_{iph}^2} = 0$$

0.03885

$s_{n_1} = 0.96$ ✗ $s_{n_2} = 0.0405$ ✓

$$n_n = n(1-s_n) = 959.5 \text{ rpm}$$

② Find R_2' add when n reduce by 20%?

$n_{new} = 0.8\, n_n = 767.6$ rpm.

$$s_{new} = \frac{n_0 - n_{new}}{n_0} = 0.2324$$

$$T_{n_1} = T_{new}$$

$$\frac{3 V_{iph}^2 s_n}{R_2'} = \frac{3 V_{iph}^2 s_{new}}{R_2' + R_2' \text{add}}$$

R_2' add = 1.895 Ω

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(6) V/F Control

⇒ change in (V_{1ph}) and supply frequency (F), certian principle depend on the type of load and assumption.

$V_1 = E_a + I \cdot R \rightarrow R \approx \text{zero} \rightsquigarrow E_a \approx V_1$

$V_1 = 4.44\, \phi_{max}\, f_1 \cdot N$

$V_1 \propto \phi \cdot f \rightarrow \phi \propto \frac{V}{F}$

Case 1 V_{1ph} = Varible $f_1 = f_{in}$ = constant

$\phi \propto V \rightarrow V\downarrow, \phi\downarrow, T\downarrow$

$V\uparrow, \phi = \phi_n = \text{constant} \rightarrow$ over heating.

Case 2 V = constant f_1 = Varible

$\phi \propto 1/f \rightarrow f\uparrow, \phi\downarrow, T\downarrow$

$f\downarrow, \phi\uparrow, \phi_{max} = \phi_n \rightarrow$ over heating.

Case 3 $V_1 \leq V_{in}$, f = varible:

$\lambda = \frac{T_{max}}{T_{Load}} = \text{constant}$ ↑↑

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هذا المطلوب → $\lambda_n = \lambda_{(R/C)} = \text{constant}$

$$\frac{T_{max(N/C)}}{T_{Load(N/C)}} = \frac{T_{max(R/C)}}{T_{Load(R/C)}}$$

$$\frac{3V_{1ph(n)}^2}{2\omega_0\left[R_1+\sqrt{R_1^2+X_{sc}^2}\right]}\cdot\frac{1}{T_{Ln}} = \frac{3V_{1ph(R/C)}^2}{2\omega_0\left[R_1+\sqrt{X_{sc}^2+R_1^2}\right]}\cdot\frac{1}{T_{L(R/C)}}$$

$X_{sc} \gg R_1$
$R_1 \approx$ zero
$X_{sc} = 2\pi f L_{sc}$
$\omega_0 = \frac{2\pi f}{(P/2)}$

بعد التبسيط ↓

$$\frac{V_{1ph(n)}^2}{f_{1(n)}^2}\cdot\frac{1}{T_{Ln}} = \frac{V_{1ph(R/C)}^2}{f_{1(R/C)}^2}\cdot\frac{1}{T_L}$$

$$\rightarrow \frac{V_{1ph(R/C)}}{f_{1(R/C)}} = \frac{V_{1ph(n)}}{f_{1n}}\sqrt{\frac{T_L}{T_{Ln}}}$$

K = Constant.

general priciple of V/f control

$$T_L = T_0 + (T_{Ln} - T_0)\left(\frac{\omega}{\omega_n}\right)^X$$

$$K = \frac{V_{1ph(n)}}{f_{1n}}$$

If $T_0 = 0$ → $T_L = T_{Ln}\left(\frac{\omega}{\omega_n}\right)^X$

→ (case 1) $X = 0$ "constant torque drive"

$T_L = T_n = \text{constant} \rightarrow V_{1ph} = K \cdot f_1$

→ (case 2) $X = 1$ "constant power drive"

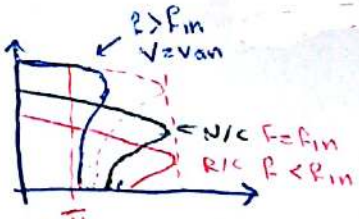
$T_L \propto \frac{1}{\omega} \rightarrow \frac{T_L}{T_{Ln}} = \frac{\omega_n}{\omega} = \frac{f_{1n}}{f_1}$

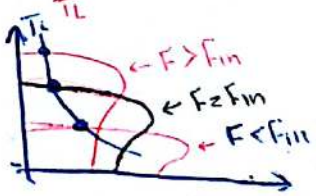
$\rightarrow V_{1ph} = K\sqrt{T_1} \longrightarrow K = \frac{V_{1ph(n)}}{\sqrt{f_{1n}}}$

→ case (3) $X = 2$

$T \propto \omega^2 \rightarrow \frac{T_L}{T_{Ln}} = \frac{\omega^2}{\omega_n^2} = \frac{f_1^2}{f_{1n}^2}$

$V_{1ph} = K f_1^2 \longrightarrow K = \frac{V_{1ph(n)}}{f_{1n}^2}$





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ex/ 480 V, 2 poles, 60 Hz, Y-connection, $X_{sc} = 4\Omega$, $R_1 = 0.2\Omega$, $R_2' = 0.3\Omega$

T_L = constant = 60 N.m at 3500 rpm.

⊛ at $\frac{V}{f}$ - constant, find the angular speed when $f_1 = 50$ Hz then find I_{2sh}'?

$$\frac{V_{1ph}}{f_1} = \frac{V_{1ph(n)}}{f_{in}} = \frac{480/\sqrt{3}}{60} = 4.62 \text{ V/Hz}$$

For $f_1 = 50$ Hz $\rightarrow$ $V_{1ph} = 231.2$ V

$n_{o1} = 3000$ rpm.

عندما يكون s صغيرة يمكن تجاهل هذا المقاومة لان قيمتها كبيرة

For s = small $\rightarrow$ $T \cong \frac{3V_{ph}^2 s_1}{R_2' \omega_o}$ $\rightarrow$ $\boxed{s_1 = 0.0353}$

$n_1 = n_{o1}(1 - s_1) = 2894$ rpm $\rightarrow$ $\boxed{\omega_1 = 303 \text{ rad/s}}$

$$I_{sc}[60\text{Hz}] = \frac{V_{1ph}}{\sqrt{(R_1 + R_2')^2 + X_{sc}^2}} = 68.75\text{A}$$

$$I_{sc}[50\text{Hz}] = \frac{V_{1ph}}{\sqrt{(R_1 + R_2')^2 + \left(\frac{5}{6}X_{sc}\right)^2}} = 68.50\text{A}$$

ملاحظة

لا تنسى أبداً ان X_{sc} تتأثر بتغير الـ frequency لذلك عند حساب اي شيء عندما تقول الشارد تقوم بإيجاد X_{sc} أولاً $X_{sc\,new}$

$$X_{sc\,new} = X_{sc\,old} \cdot \frac{f_{1(new)}}{f_{1\,old}}$$

$$\omega_{o\,new} = \omega_{o\,old} \cdot \frac{f_{1(new)}}{f_{1(old)}}$$

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ex/ 440V, 50Hz, 6-poles, Y-connection, $R_1 = 0.6\,\Omega$, $X_1 = X_2' = 1\,\Omega$, $s_n = 0.05$, V/f control?

① For F = 10Hz → Find T_{max} ??

$$\frac{V_{1ph}}{f_1} = \frac{V_{1ph(n)}}{f_{1n}} = \frac{(440/\sqrt{3})}{50} \longrightarrow V_{1ph} = 50.87\ V$$

$$n_{o(n)} = \frac{60 f_n}{(P/2)} = 1000 rpm \longrightarrow \omega_{o_n} = \frac{2\pi n_{o(n)}}{60} = 104.7\ rad/s$$

$$\omega_o(10Hz) = 20.93\ rad$$

$$T_{max(n)} = 344\ N.m \longrightarrow T_{max}(10Hz) = 140.4\ N.m$$

Find K For $T_{max(n)} = T_{max}(10Hz)$

$$\frac{V_{1ph}}{f_1} = K^* \longrightarrow V_{1ph} = 10\,K^*$$

$$X_1 = X_2' = 2\pi f L \longrightarrow L[50Hz] = \frac{1}{2\pi 50}$$

$$X_1 = X_2'[10Hz] = 2\pi \cdot 10 K^* \frac{1}{2\pi 50} = \frac{1}{5}\,\Omega$$

$$T_{max}[10Hz] = 344 = \frac{3(10K^*)^2}{2\omega 20.93\left[0.6 + \sqrt{0.6^2 + 2^2}\right]}$$

$$\Rightarrow K^* = 7.96$$

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Power and control circuit for transient operation

⇒ using Electromagnatic relay and contactor.

⇒ ① speed principle
② Voltage principle
③ displacement
④ current principle
⑤ time principle ← هذا هو النوع المستخدم

* في هذا الموضوع نعطي نوع كاتة العابرة (start, break, ...) والمطلوب هو
① رسمة الـ power cir. ② control cir ③ مخططات الوايرز (Chart)

* Electromagnatic device. (contactor , relay)
Power circuit control circuit

contactor : activate when voltage applied to the coil.

① normally open

② normally closed

مبدأ العمل:- هي عبارة عن ملف يتم وصل أطرافه مع مصدر تغذية وعليه ملامسات عادية مفتوحة.
تتحرك عن وضعين، حيث عند وصل التغذية يتغير وضع الملامسات لعكسية
فإذا كانت مغلقة تفتح وإذا كانت مفتوحة تغلق.

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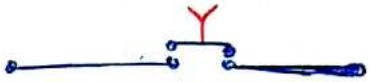
Time relay

(A) Time delay on

1.
N.O
(closed after time delay)

2. 
N.C
(open after time delay.

مبدأ العمل:- يتغير وضع ملامسات المرحل الزمني بعد فترة من وصول التغذية لملف المرحل وتعود الى الوضع الطبيعي (مباشرة) بعد فصل التغذية.

(B) Time delay off.

1.
N.O

2.
N.C

مبدأ العمل:- يتغير وضع الملامسات (مباشرة) عند وصول التغذية لملف المرحل وتعود الى الوضع الطبيعي (بعد فترة) من فصل التغذية.

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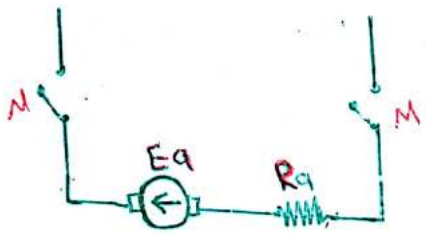
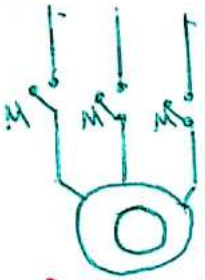
⇒ constract the power and control circuit For :-
(according to time response)

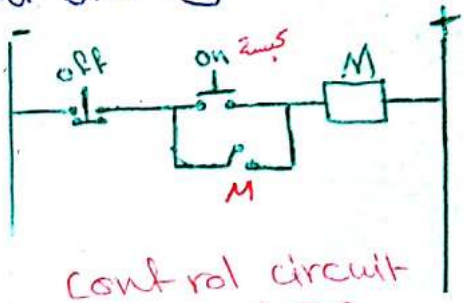
case 1 Direct start and mechanical braking.



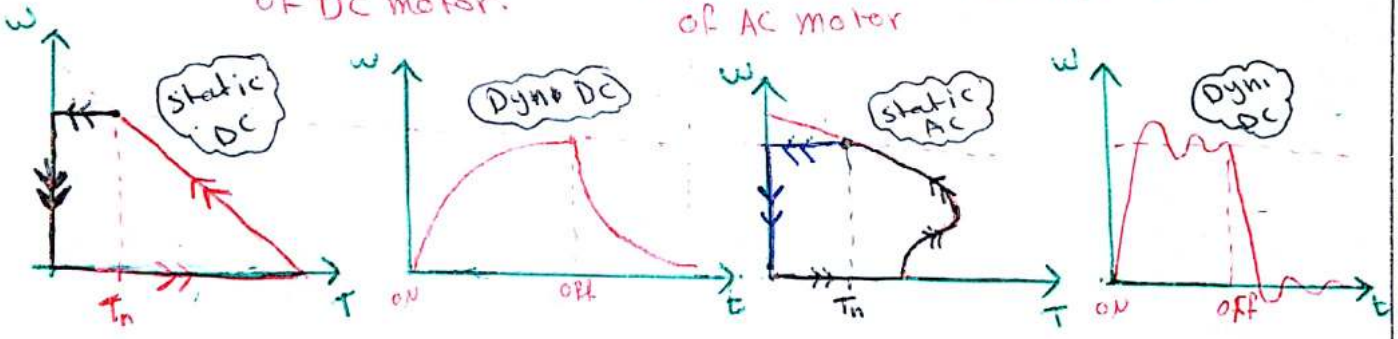
Power circuit of DC motor.



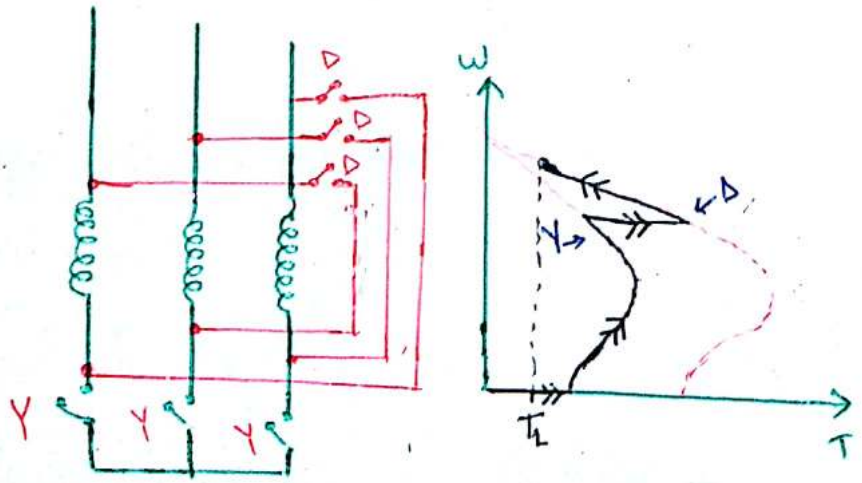
Power circuit of AC motor



Control circuit For AC & DC

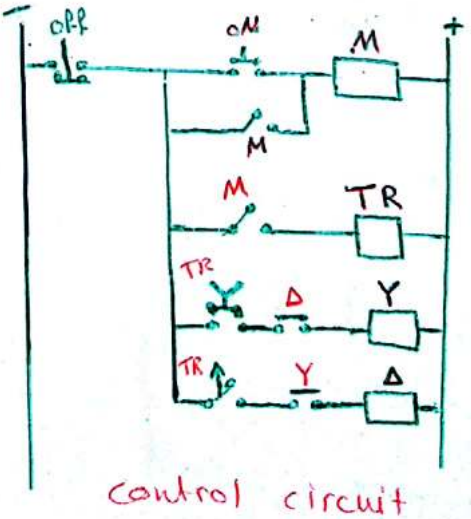


case 2 Y - Δ starting



Power circuit

طريقة توصيل Δ (3→1 ، 2→3 ، 1→2)



control circuit

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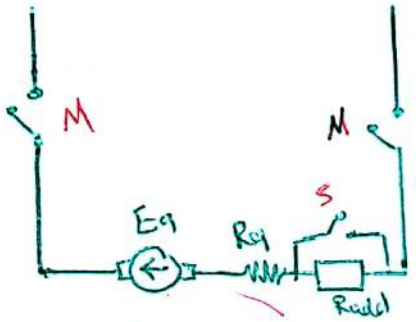
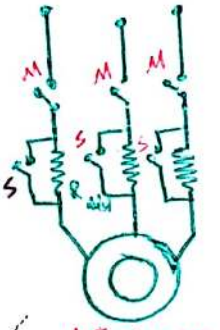
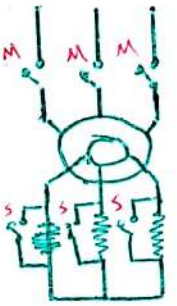
Case 3 one-step reostatic ~~start~~ and mechanical Braking



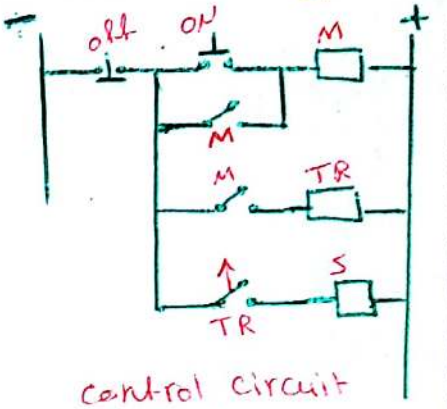
DC - power circuit



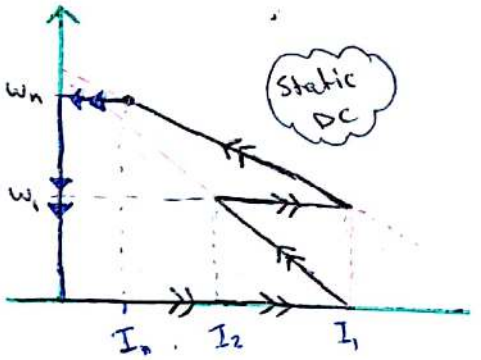
AC-power circuit squirral cage

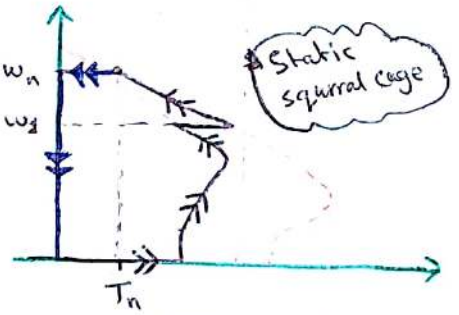


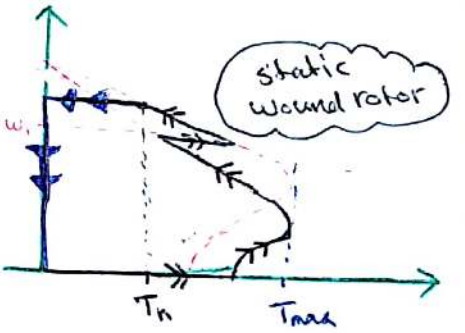
AC-power circuit wound rotor

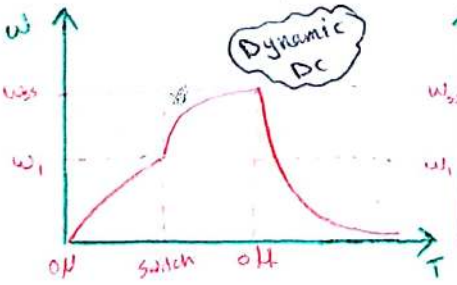


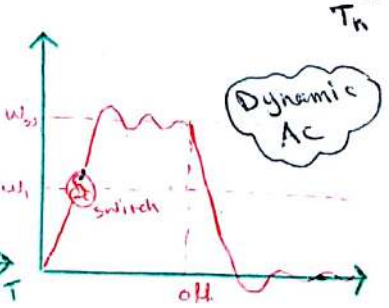
control circuit For AC & DC



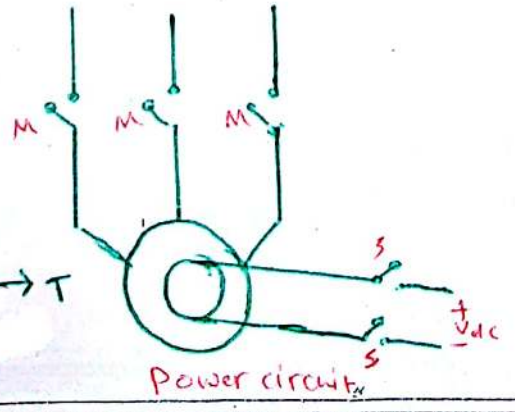




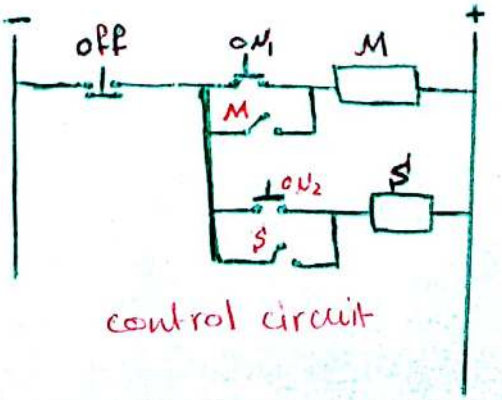




Case 4 (A) manual start of 3Φ synchronous motor



Power circuit



control circuit

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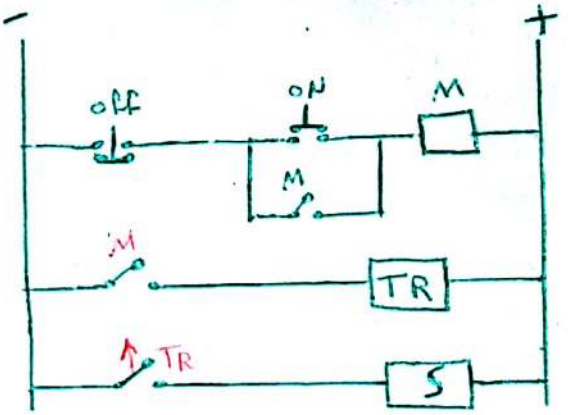
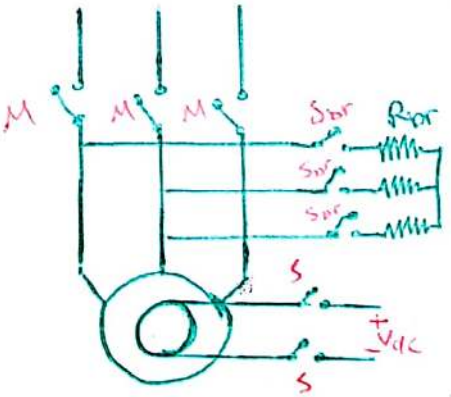
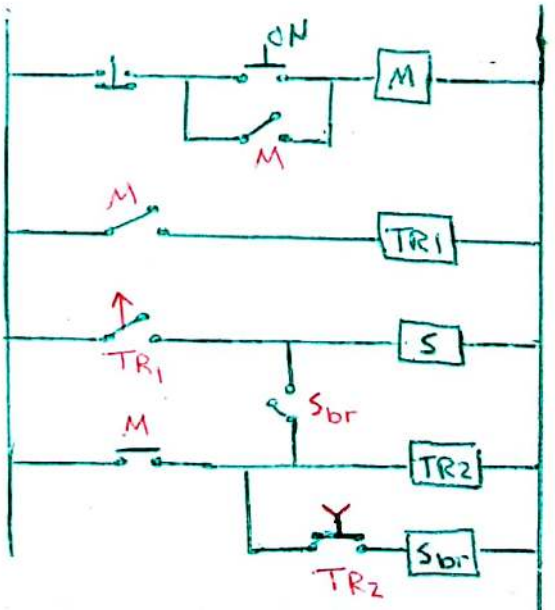
Case 4 (B) automatic start of syncronous motor



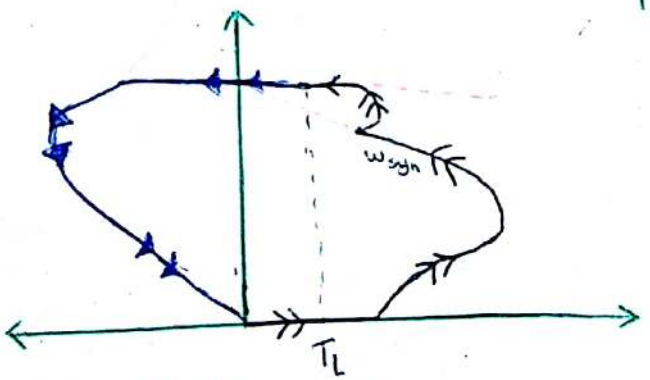
Case 5 Atuomatic start + Dynamic Braking syncronous motor.



Power circuit.



control circuit



Static char.

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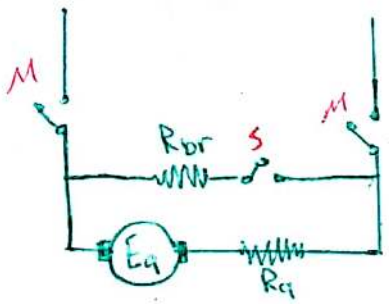
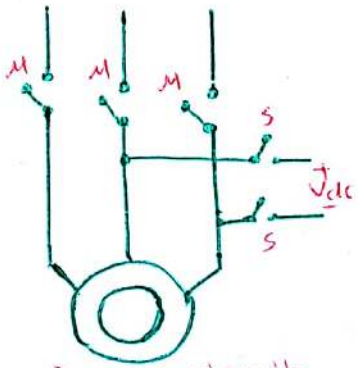
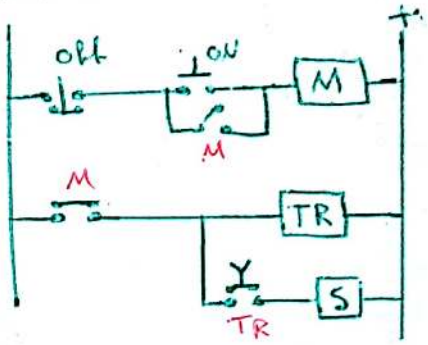
Case 6 Direct start and Dynamic Braking For DC & induction motor



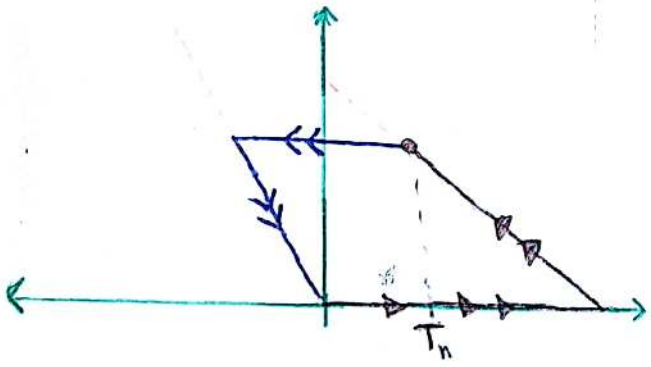
Power circuit of DC motor

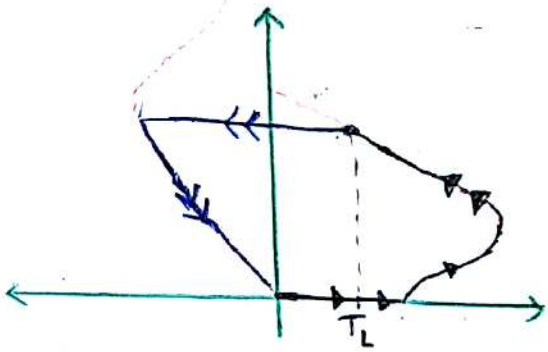


Power circuit of induction motor



control circuit





الى هنا "مادة الدرايف" تكون قد أكتملت

لا تنسونا من صالح دعائكم

✻ spark_team

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